

Causal State Binding: An Intervention-Based Evaluation Framework for Language Agents

Xiao Jia

School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen

xiaojia@link.cuhk.edu.cn

ORCID: <https://orcid.org/0009-0006-7711-4067>

Supplementary overview

This supplement documents controlled action responses, execution following agent-written state and software-localization prediction. Methods Tables M1–M4 define sampling units, intervention access, predictor features and execution inference; results tables retain their S-numbered identifiers. The opening sections describe memory-content verification, directional target scoring, independent prediction, baseline replication and the linear prediction comparison. Notes 1–15 give detailed analyses, annotation procedures and acquisition conditions. Figures S1–S14 show task-, corpus- and model-level results. Sections S1–S9 provide the source experiments, control tables and metric definitions. Protocol and export identifiers connect these results to their source records. Tables round reported values; accompanying CSV and JSON files retain full precision. Entropy, bootstrap and agreement statistics use the units and definitions specified below (Shannon, 1948; Efron and Tibshirani, 1993; Cohen, 1960).

Reading guide

Material	Page
Study structure, analysis units and intervention definitions	2
Memory-content development: 16 tasks, six conditions	5
Memory-content verification: 128 new tasks, eight conditions	8
Action direction under substituted content	9
Independent localization prediction: 800 issues	9
Complete baseline replication and observed-field coverage	15
Linear localization prediction	17
Notes 1–15: analyses, human evaluation and study records	18
Figures S1–S14: detailed behavioral and predictive results	30
Sections S1–S9: source experiments, control tables and definitions	46

Page numbers link to the corresponding material. PDF bookmarks provide individual section and note headings; figure references lead to the artwork. Longer panel descriptions are headed by the same figure number.

The figures present execution effects, prediction diagnostics, finite-action responses, output coupling, entropy and annotation agreement. Source data and input hashes accompany every figure. Task-level means use four continuations in the core execution experiment and two in the content verification; cross-validation folds are parts of a single grouped estimate. The supplementary specificity intervals and the content-separation experiment use their separately stated resampling and acquisition procedures, without changing the fixed localization predictors. The paired Brier histograms for the linear prediction analysis (Figure S13) use symmetric geometric bins from 10^{-4} to 1, including zero, with a symmetric-log threshold of 0.01. Captions specify the remaining transformations. Notes 1 and 13–15 distinguish reference-dependent prediction from the reference-free analyses.

Study structure and supplementary methods

The studies examine controlled action responses, execution following agent-written state and software-localization prediction. The tables specify the sampling units, information available to each intervention and the complete predictor features. Primary execution effects, supplementary contrasts and predictive comparisons have distinct analysis units and inferential roles.

Methods Table M1. Evidence roles and analysis units

Evidence	Sample and unit	Role and dependence
Finite-action assay	57,816 scored rows; 803 unique corpus/event identifiers	Seven source corpora with shared prompts. Corpus contrasts summarize this design; rows are not independent replications.
Open-weight controls	Four models; 126,000 rows each; 60 scaffold–task–family cells per model	Three scaffolds, 20 families; 108,000 generation and 18,000 offline-policy rows per model. Strict Qwen-32B deletion is an additional slice.
Agent-written execution	160 tasks per core model; 80 memory and 80 goal; four continuations/branch	Eight mechanisms, ten tasks per mechanism/component; 32 redundant tasks. The task is the paired unit. Astra uses the same 40-task subset.
Content validity controls	40 tasks per core model, 20/component	Paraphrase, replacement and history compensation. Undefined replacement targets are separately reported.
Reference-dependent localization	1,800 model–issue rows; 300 issues; six provider identifiers	Reference-dependent augmented features; grouped reanalysis is retrospective.
Reference-free bridge	1,200 model–issue rows; 300 issues; 297 issue/template groups	Four consistent core configurations; all copies of a group remain within one fold. Astra is a fixed 60-issue extension.
Reused holdout	400 model–issue rows; 100 issues	Disjoint from bridge fitting, but outcomes were known before the service-driven model replacement. Bootstrap conditions on frozen fits.
Independent prediction	3,200 model–issue records; 800 new issues from 12 repositories	Frozen predictors; outcome scoring released after all predictions. Paired issue resampling carries the four configurations together.

Evidence	Sample and unit	Role and dependence
Baseline replication	74,352 planned responses; 3,098 entries / 3,073 unique IDs	Seven source corpora; eight conditions and three repetitions. Membership, raw-field coverage and rule-completed compatibility output are provided separately.
Human output evaluation	1,200 items; three annotators; seven binary dimensions	25,200 individual scores; 8,400 item-dimension combinations. Agreement uses individuals, not majority labels.
Memory-content development	16 tasks per API configuration; six branches and two continuations	Two tasks per mechanism, shared by Sol and DeepSeek; 384 planned continuations. Development observations informed the subsequent fixed verification.
Memory-content verification	128 new tasks per API configuration; eight conditions and two continuations	Sixteen tasks per mechanism; 4,096 planned continuations. The paired unit is the task; task identities and analysis were fixed before acquisition.
Supplementary specificity	Same 640 core model-task records	Result-informed contrast of irrelevant perturbation versus targeted deletion; 5,000 stratified draws/cell and a separate eight-test Holm family.

Methods Table M2. State interventions and information access

Branch	Changed material	Fixed material and scoring
Sham	No saved component changed	Agent-written memory, goal and unrelated note; same environment and handoff.
Memory lesion	Entire saved memory removed	Goal, unrelated note, tool schema and current request retained. Identifiers in memory may also be removed.
Goal lesion	Entire saved goal removed	Memory, unrelated note, tool schema and current request retained. No evaluator-optimal policy inserted.
Irrelevant deletion	Unrelated note shortened toward the targeted serialized-character removal	Memory and goal retained. The available note can be shorter than the target; character matching does not imply token matching.
Rescue	Original agent-written component restored	Restore its actual content, including mistakes. Independent continuations, not replay of the same random draw.
Paraphrase	Semantically preserving re-expression of stored material	Same task, handoff and scorer. Twenty-task component subsets.
Replacement	One registered task-relevant saved field changed	Original environment scorer retained; replacement-target scorer computed separately only when definable. Targets can overlap.

Branch	Changed material	Fixed material and scoring
History compensa- tion	Allowed earlier history supplied with lesion	The same earlier information that preceded the agent’s write; no hidden answer added.
Redundancy stratum	Relevant information also present at handoff	Sixteen tasks/component; stratum was fixed before acquisition. This is a task-design stratum, not an extra model branch.

Identity-related memory fields are account (all mechanisms), participants (calendar), sku (inventory), customer code and currency (invoice), collection (merge), and report code (aggregation). The content-separation experiment partitions written memory using this classification. The original whole-memory lesion also removes an account value in 266 of 320 memory-task records. This records availability of the field; attribution of an execution failure to that field requires the separate intervention.

Methods Table M3. Linear reference-free feature dictionary

Let B_0, B_1, B_2 be normalized valid base-call file sets and $E_1, \dots, E_L$ the extra-call sets. Invalid outputs become empty sets. J is Jaccard similarity, assigned zero when a set is empty. The consensus C is the three highest-voted paths in the base calls, with the recorded deterministic tie order. Confidence is valid only for numeric values in $[0, 1]$; a missing confidence remains missing until training-fold imputation. Path-vote entropy uses natural logarithms and is zero when there are no votes. The normalized margin is the top path’s vote count minus the second path’s count, divided by the number of calls (zero when no votes exist).

Block	Feature	Exact definition
Shared numeric	Issue length	Character count of visible problem statement.
	Candidate count	Number of retrieved candidate implementation paths.
	Base agreement	Mean $J(B_i, B_j)$ over the three call pairs.
	Base vote margin	Normalized top-minus-second path-vote count over three base calls.
	Base entropy	Entropy of the base path-vote distribution, nats.
	Base invalid fraction	Number of invalid base outputs divided by three.
	Base confidence	Mean valid base-call confidence; missing if none valid.
Shared categorical	Model, repository	One-hot encoded within the training fold; unseen categories ignored.
State augmentation	1. Current agreement	$J(P_c, C)$.
	2. Minimal-state agreement	$J(P_c, P_m)$.
	3. Removal difference	$1 - J(P_c, P_{\text{removed}})$ when both outputs are valid; otherwise zero.
	4. Irrelevant agreement	$J(P_c, P_i)$.
	5. Mismatch difference	$1 - J(P_c, P_x)$ when both outputs are valid; otherwise zero.
	6–7. Validity/confidence	Invalid fraction and mean valid confidence over the five state probes.

Block	Feature	Exact definition
Generic augmentation	1. Consensus agreement	Mean $J(E_i, C)$ over extra calls.
	2. Extra agreement	Mean pairwise $J(E_i, E_j)$.
	3. Extra vote margin	Normalized top-minus-second path-vote count over L extra calls.
	4. Extra entropy	Entropy of the extra path-vote distribution, nats.
	5. Cross agreement	Mean $J(B_i, E_j)$ over every base/extra pair.
	6–7. Validity/confidence	Invalid fraction and mean valid confidence over extra calls.

The state sets $P_c, P_m, P_{\text{removed}}, P_i, P_x$ correspond to current, minimal, removed, irrelevant and mismatched saved working state. Mismatch candidates come from the same visible issue-specific pool and exclude the first output’s paths; no gold path or cross-fold donor is used. Both augmentations contain seven numeric features and use the same shared features, categories and estimator. Fixed generic diagnostics use five calls. Token funding can extend this to ten, governed by recorded usage alone.

Path normalization inspects at most the first three list entries, keeps strings, strips surrounding whitespace, replaces backslashes by forward slashes and removes a leading `./`. Empty paths, absolute paths, paths with a `..` component and duplicate normalized paths are discarded. A valid output additionally requires a list of at most three supplied files, a nonempty normalized set and membership of every retained path in the visible implementation-file inventory. These checks do not consult reference patch paths. For example, two valid sets sharing one of three total paths have $J = 1/3$ regardless of whether that path is correct; the scorer-only outcome is evaluated separately.

Supplementary specificity and replacement classification

The specificity estimator uses paired task differences, then equal mechanism means, as defined in the main paper. Each of eight core model/component cells contains 80 tasks. The bootstrap resamples ten tasks with replacement within each of eight mechanisms, recomputing the entire contrast for each of 5,000 draws. The 2.5th and 97.5th percentiles form a two-sided interval. Tests flip each task difference independently with probability one half, compare the absolute mean with its sign-flip reference and apply the plus-one correction over 50,000 draws. The eight supplementary tests share a separate Holm correction from the registered primary family. The all-zero Mistral memory sample produces a degenerate empirical interval and does not establish population equivalence.

Every one of the 640 replacement continuations is assigned exactly once. The replacement target is first checked for definability. Defined cases are partitioned by the original and replacement scorer: original-only, replacement-only, both, neither with a valid transaction, or neither with an invalid transaction. Thus overlapping targets do not count as two independent successes. Both targets are scored against the same saved environment and continuation action.

Memory content and transaction identifiers

The content-separation experiment specified 16 new memory tasks across eight mechanisms, shared by Sol and DeepSeek, before model acquisition. Its six conditions were full memory, identity fields only, content fields only, empty memory, semantic paraphrase and irrelevant-note perturbation. Each used two fresh continuations. Identity fields follow Supplementary Methods Table M2; the partition uses the agent’s actual memory without inserting missing or corrected values. The persistent goal, tool schema, execution context and initial environment are shared across conditions. This exploratory experiment

was informed by the execution results; its complete cohort and analysis were fixed before acquisition, with failures retained.

All 32 model-task records and 384 continuations were collected. Valid state writes numbered 15/16 for Sol and 16/16 for DeepSeek; the invalid write remains in the primary denominator. Sham success was 27/32 and 23/32. With identifiers retained but other memory content removed, success fell to 6/32 and 3/32. The paired content-loss contrasts were 0.656 [0.531, 0.781] and 0.625 [0.500, 0.750]. These descriptive 95% intervals use 5,000 paired-task bootstrap draws within each mechanism, seed 2026092602. With two tasks per mechanism and two continuations per branch, they describe variation within the tested set of mechanisms.

Removing identifiers while retaining content and removing all memory both yielded zero primary successes in each configuration. Thus execution depends on identity information as well as content. Sol’s paraphrase-minus-sham contrast was -0.094 [-0.156 , -0.031], while DeepSeek’s was 0.000 [-0.063 , 0.063], indicating different responses to paraphrase in the two configurations. Irrelevant-minus-sham contrasts were -0.063 [-0.188 , 0.063] and 0.000 [-0.063 , 0.063]. The table below gives the six development-condition counts. The independent content verification used 128 new tasks and eight conditions; its results appear in main Figure 5 and in Verification of memory-content dependence (p. 8).

Branch	Sol success/planned	DeepSeek success/planned
Sham	27/32	23/32
Identity fields only	6/32	3/32
Content fields only	0/32	0/32
Empty memory	0/32	0/32
Paraphrase	24/32	23/32
Irrelevant perturbation	25/32	23/32

Service and resource accounting. The first credential’s development batch returned 416 terminal insufficient-quota refusals and no valid model answer. That batch is retained as a service-access failure, outside the behavioral cohort. A separately registered second-credential batch used the same fixed development specification and completed 828 logical calls in 834 transport attempts. Final responses comprised 827 HTTP 200 and one terminal HTTP 404; six additional failed attempts and that terminal response lacked token usage. The saved reported attempt total was 1,194,825 tokens, with seven unmetered attempts. Two separate availability probes reported 4,445 tokens. Missing billing is not assigned zero. These records are separate from the earlier administrative upper bound of 42 potentially unlogged attempts. All model responses in this experiment were obtained through the API service.

Restoration from accessible history. A source-record baseline uses the same accessible structured history available to every candidate policy. On 192 fixed interface-validation cases (32 tasks and six state conditions), it exactly restored the declared memory and goal fields, without changing the 32 initially correct states or making a model call. Its implementation receives no task or evaluator argument. This establishes restoration feasibility at the interface, before evaluating downstream execution. The structured history supplies every declared field in these cases, making direct restoration the appropriate reference policy for this setting. The finite-action guard and software guard in the main text retain their measured outcomes and tradeoff.

Methods Table M4. Execution inference and reproducibility

Analysis	Resampling / test	Multiplicity and seed
Primary execution	5,000 within-mechanism task draws; 50,000 sign flips	Eight primary contrasts; shared seed 2026092411
Mechanism sensitivity	5,000 draws of eight mechanism means; leave one mechanism out	Descriptive; same ordered stream
Specificity supplement	5,000 task draws; 50,000 sign flips	Separate eight-test family; seed 2026092601
Content development	5,000 stratified task draws	Descriptive; seed 2026092602
Content verification	10,000 stratified task draws; 50,000 sign flips	Two primary contrasts; seed 2026092701

For each task, the four continuation outcomes are averaged within branch. The primary statistic is the mean of the eight mechanism-specific mean sham-minus-lesion differences. Five thousand bootstrap draws resample tasks within each mechanism. A separate sensitivity analysis resamples the eight mechanism means 5,000 times and reports leave-one-mechanism-out estimates. Two-sided sign-flip tests independently multiply each task difference by -1 or 1 with probability one half, then recompute the mechanism-equal mean. The reference assumes symmetry of task differences under the null. The p value is $(1 + \#\{|T^*| \geq |T| - 10^{-12}\})/50,001$. Holm’s monotone step-down procedure adjusts the eight primary model-component tests.

The analysis initializes NumPy’s default random generator once with seed 2026092411. Its ordered traversal is Sol, DeepSeek, Qwen, Mistral and Astra, with memory before goal; task files and mechanism keys are sorted. Bootstrap, mechanism sensitivity and sign-flip draws consume this shared stream in the archived analysis order, including secondary controls. Thus the seed reproduces the complete analysis sequence, not an independently restarted stream for each table cell. The Astra subset is outside the eight-test primary family. The separate specificity analysis uses seed 2026092601, 5,000 stratified draws and 50,000 sign flips; content development uses seed 2026092602 and 5,000 descriptive draws. The 128-task content verification uses seed 2026092701 and its separately fixed 10,000-draw analysis and two-test Holm family, described below.

Checkpoint encoding and tool access

Preparation exposes the previous request, remembered facts, persistent commitment and unrelated archived material through `inspect_history`. A subsequent `write_state` request must contain exactly three keys, `memory`, `goal` and `unrelated`, whose values are dictionaries. This validates the state-write structure; agreement with the observed facts is recorded separately. Two preparation calls are allowed, each with a maximum completion length of 1,536 tokens. A failed preparation retains its planned continuations and receives primary success zero.

The checkpoint stores the task, realized state, preparation calls, handoff messages and initial environment. Each branch deep-copies that environment. A lesion changes the selected dictionary to `{}` without removing its key; restoration reinstates its original content, including any error. Irrelevant editing shortens string values in the unrelated dictionary in sorted-key order, toward the number of serialized characters removed by the lesion. Continuation starts with fresh system and handoff messages, not the preparation conversation. It includes the stored state, tool schema and complementary component information; the preassigned redundant stratum also restates the targeted information. Only history compensation supplies the earlier history.

Each continuation allows two calls (maximum 1,024 completion tokens each): `inspect_environment`, then the designated transaction. The first call must have empty arguments. The transaction requires the exact argument-key set, registered outer value types and mechanism-specific feasibility checks; a failed call stops that continuation. The evaluator then

compares the realized transaction with the valid task solutions. The original 160-task experiment shuffles branch-repeat pairs within task using a Python random generator seeded by the first twelve hexadecimal digits of the canonical request-prefix hash. Environment reset fixes the starting state while repeats obtain new model draws.

Shared repository retrieval

At the pinned base commit, candidate implementation paths have extensions `.py`, `.pyx`, `.c`, `.cpp`, `.h`, `.js` or `.ts`. Paths are excluded when a lower-cased component equals `test`, `tests`, `testing`, `doc` or `docs`, or starts with `test_`. Camel-case boundaries are split before lower-casing; retrieval tokens match `[a-z][a-z0-9_]{2,}`. The visible problem statement supplies query counts q_t . For path-token frequency d_{pt} , document frequency df_t and N eligible paths, the score is

$$r(p) = \sum_{t \in p \cap q} \min(q_t, 3) d_{pt} \log \left(1 + \frac{N + 1}{df_t + 1} \right).$$

Paths are sorted by descending score and then lexicographic path; the first fifty are supplied. For each of the first eight paths, the source line maximizing the sum of $\min(q_t, 2)$ over its distinct matching tokens anchors an excerpt; ties select the earliest line. The excerpt begins up to eight lines before that anchor, includes at most 32 numbered lines and is truncated to 1,500 characters. The model-visible issue is truncated to 12,000 characters. All diagnostic arms use the same cached context. Feature tokenization is defined separately in *Reference-free response representations* and includes English stop-word removal; retrieval tokenization does not.

Verification of memory-content dependence

The content study used 128 newly generated tasks per API configuration: 16 instances of each of the eight transaction mechanisms. The tasks, identity-field partitions, conditions, repeats, scoring rule and analysis were fixed before acquisition. Their instance identifiers were disjoint from the 16-task development set. The agent first inspected the earlier interaction and wrote its own memory. Each saved checkpoint was then continued twice under each of eight conditions: the complete record, identity fields only, non-identity content only, empty memory, deterministic rewording, exact JSON text, irrelevant editing and restoration. This yielded 256 model-task checkpoints and 4,096 continuations. The same realized record was used across conditions; inaccurate agent-written values were retained.

Identity fields were account identifiers in allocation, routing and dependency tasks; account and participant identifiers in calendar tasks; account and item identifiers in inventory; account, customer and currency identifiers in invoice tasks; account and collection identifiers in record merging; and account and report identifiers in aggregation. Identity-only and content-only conditions formed disjoint partitions of the actual memory dictionary. The goal was held fixed. The exact-text condition used a lossless canonical JSON serialization of the dictionary. The deterministic rewording retained every original field value and used the same transformation as the development study. Comparing these two text conditions separated rewording from the change from a structured object to a string. Condition and repeat order were randomized within checkpoint using deterministic request-derived seeds.

The primary outcome required a valid state write and a successful final environment transaction. Raw environment success was also retained. The independent analysis unit was the task; its two continuation outcomes were averaged before comparison. Tasks received equal weight within each mechanism, and the eight mechanisms received equal weight. The two primary full-memory-minus-identity-only contrasts were evaluated with 10,000 stratified paired-task bootstrap draws and 50,000 two-sided sign-flip draws, with Holm adjustment across the two API configurations. The analysis seed was 2026092701. All registered secondary contrasts were retained. Reworded text versus exact JSON text

additionally had a prespecified, descriptive two-one-sided-test analysis with a ± 0.05 probability margin. Its standard error respected the eight equal-weight strata, with Welch–Satterthwaite degrees of freedom. This secondary equivalence analysis was not part of the family-adjusted superiority claim. Mechanism-resolved intervals in the figure are descriptive within-mechanism resamples with seed 2026092705.

Sol produced a valid state write for all 128 tasks, and DeepSeek for 127. Every planned continuation remained in the primary denominator. Sol used 4,352 logical requests and 4,353 recorded transport attempts; DeepSeek used 4,320 logical requests and attempts. Reported attempt usage was 12,204,784 and 3,746,841 tokens, respectively. One Sol transport failure and 30 DeepSeek HTTP 404 responses lacked token metering; their unknown costs were not assigned zero. No terminal scientific failure was resampled.

After the registered analysis, a separate service-failure sensitivity assigned each primary outcome affected by a terminal non-200 preparation or continuation request an arbitrary value in $[0, 1]$. A retry failure followed by a valid final response remained an observed outcome; malformed model content returned with HTTP 200 remained an observed failure. The resulting deterministic bounds were $[0.656, 0.656]$ for Sol and $[0.543, 0.582]$ for DeepSeek. These are worst-case missing-outcome bounds, not confidence intervals, and do not replace the registered all-planned estimates. The complete task records, intervention states, transactions, individual outcomes, attempts, source hashes and analysis outputs are retained in the research archive.

Direction of action under content replacement

The saved replacement experiments allow action direction to be distinguished from an undirected change in performance. For each task, we evaluate the four original-state continuations and the four replacement-state continuations against both the original and alternative targets. Let S_{ij} be the mean success of actions produced with state i , evaluated against target j , where 0 denotes the original and 1 the replacement. We summarize target alignment by

$$D = \frac{1}{2}\{(S_{00} - S_{01}) + (S_{11} - S_{10})\}. \quad (1)$$

The contrast ranges from -1 to 1 . A positive value indicates that actions under each state align more closely with that state’s corresponding target. Both target scores are retained when their accepted actions overlap. The evaluator uses the saved environment and continuation actions; no model response is generated during cross-scoring.

The analysis includes all 160 model–task records in the fixed replacement subset. An alternative target is defined for 130 records, supplying 1,040 cross-scored actions. Each table entry first averages the four continuations within task and then gives equal weight to the defined tasks. Mechanism-equal summaries and the redundant-information strata accompany the task-level source data. The alternative target is defined for all 20 memory and 20 goal tasks in each API model, 19 memory and 20 goal tasks in Qwen, and three memory and eight goal tasks in Mistral.

This supplementary cross-scoring was specified after inspecting the existing replacement results. It provides an explanatory comparison on those tasks. The original and replacement continuations were independently sampled; they are paired by task, without an assumption of common random numbers. Undefined targets are excluded from the conditional scores and retained in the planned denominator. Assigning their unknown task contrasts any value in $[-1, 1]$ yields all-planned bounds of $[-0.025, 0.075]$ for Qwen memory, $[-0.850, 0.850]$ for Mistral memory and $[-0.600, 0.600]$ for Mistral goals. These are sensitivity bounds, not confidence intervals.

Independent prediction of localization success

The independent study fixed 800 new SWE-bench test issues from twelve repositories after excluding the 423-identifier exclusion set, repository/commit duplicates and registered text/patch near-duplicates.

Table S4: Crossed scoring of original and replacement actions. Tasks gives defined/planned tasks; each defined task contributes four actions per state. Target scores can overlap. Qwen and Mistral refer to the fixed Qwen2.5-14B and Mistral-7B configurations.

Model	State	Tasks	S_{00}	S_{01}	S_{10}	S_{11}	D
Sol	Memory	20/20	0.938	0.050	0.238	0.763	0.706
Sol	Goal	20/20	0.938	0.050	0.188	0.738	0.719
DeepSeek	Memory	20/20	0.713	0.038	0.088	0.700	0.644
DeepSeek	Goal	20/20	0.800	0.050	0.088	0.613	0.638
Qwen	Memory	19/20	0.105	0.026	0.079	0.053	0.026
Qwen	Goal	20/20	0.163	0.163	0.125	0.125	0.000
Mistral	Memory	3/20	0.000	0.000	0.000	0.000	0.000
Mistral	Goal	8/20	0.000	0.000	0.000	0.000	0.000

The exclusion set is the disjoint union of 300 SWE-bench Lite test issues used for development, 100 SWE-bench Verified test issues used for reassessment, and the full 23-issue Lite development split. Twenty of those 23 development issues were used for implementation checks; all 23 were excluded to separate the development split from the independent sample. Issue-level membership is provided with the source data. There were 1,754 eligible unused issues. Selection was stratified by repository before outcomes, and the same issues were assigned to all four configurations. The fixed 800-issue size was a finite precision and acquisition choice: a development-conditional approximation required 658 groups for an AUC difference of 0.005 at 80% power, or 1,027 with 1.25-fold standard-deviation inflation. Sampling was not extended according to the new results. Membership, base commits and context hashes accompany the data.

The endpoint is success when at least two of three shared base calls each identify a reference implementation file among at most three returned paths. Gold paths are absent from generation and feature extraction. Three further current-state reevaluations, including the current-state probe and two ordinary repeats, combine with state removal and state mismatch to form five hybrid diagnostic calls. Pure state diagnosis uses the five original intervention conditions; ordinary self-consistency uses at least five calls. Token-funded generic sampling continues to the larger recorded state/hybrid budget, at most ten generic calls, based only on usage. Responses can be reused across strategies in joint acquisition, but deployment cost assigns every strategy its required calls.

The feature map uses model/repository identity, issue length and candidate count, base-answer agreement, validity, vote margin, entropy and confidence, then within-response confidence/validity, normalized retrieval ranks, lexical query agreement, state-answer agreement, pairwise file-set overlap and contrasts between query positions. Set overlap is Jaccard similarity with the frozen empty-set and invalid-output rules. Each arm has access to the original vector, its augmentation with ordered and aggregate features, its augmentation with aggregate features alone, and a compact relation-focused augmentation. Numeric inputs are represented consistently as float32 before the frozen transforms. Model/repository encoding ignores unseen categories. Training-fold median imputation retains entirely missing columns with zero fill and adds indicators for columns with missing training values; numeric features and indicators are standardized using training-fold statistics. Estimator random seeds are 17.

Development uses five outer issue/template-group folds and three inner folds. The common search has 60 configurations per arm: logistic $C \in \{0.1, 1, 10\}$; histogram gradient boosting with maximum leaves 7 or 15 and minimum leaf size 10 or 30; extra trees with depth 4 or 8 and minimum leaf size 5 or 15; and CatBoost with depth 4 or 6 and L2 penalty 3 or 10, crossed with the four representations. Histogram boosting uses 150 iterations, learning rate 0.05, L2 penalty 5 and no early stopping. Extra trees uses 160 trees and all features; CatBoost uses 160 iterations and learning rate 0.05. Logistic regression uses liblinear and at most 3,000 iterations. For each outer validation fold, inner AUC selects

one member of each learner family. Final configurations are then selected within each family by pooled five-fold out-of-fold AUC over the development cohort. Both selections use log loss and fixed candidate order to break ties. The four probabilities receive equal weight. The selected models are refitted on all development data and frozen before new acquisition; the 800 new issues do not enter either selection.

Table S5: Frozen main candidate and comparator configurations. Column counts precede one-hot expansion; each ensemble member has weight 0.25. Common settings are stated above.

Arm	Learner	Representation	Columns	Selected settings
Repetition-intervention	CatBoost	Aggregate	130	depth=4, l2=3
Repetition-intervention	Extra trees	Aggregate	130	depth=8, leaf=5
Repetition-intervention	Hist. boosting	Aggregate	130	leaves=15, leaf=10
Repetition-intervention	Logistic	Aggregate	130	C=0.1
State interventions	CatBoost	Aggregate	130	depth=6, l2=3
State interventions	Extra trees	Aggregate	130	depth=8, leaf=5
State interventions	Hist. boosting	Aggregate	130	leaves=15, leaf=10
State interventions	Logistic	Aggregate	130	C=0.1
Five generic queries	CatBoost	Aggregate	130	depth=6, l2=3
Five generic queries	Extra trees	Ordered + aggregate	279	depth=8, leaf=5
Five generic queries	Hist. boosting	Aggregate	130	leaves=15, leaf=10
Five generic queries	Logistic	Aggregate	130	C=0.1
Token-funded generic	CatBoost	Aggregate	130	depth=6, l2=10
Token-funded generic	Extra trees	Ordered + aggregate	279	depth=8, leaf=5
Token-funded generic	Hist. boosting	Aggregate	130	leaves=15, leaf=10
Token-funded generic	Logistic	Aggregate	130	C=0.1

Reference-free response representations

The five ordered diagnostic positions are fixed as follows. Superscripts index diagnostic responses, and subscripts index the three common base answers.

Position	Pure state diagnostic	Hybrid diagnostic	Generic diagnostic
0	Current record	Current record	Ordinary repeat 1
1	Minimal record	Ordinary repeat 1	Ordinary repeat 2
2	Removed record	Removed record	Ordinary repeat 3
3	Irrelevant edit	Ordinary repeat 2	Ordinary repeat 4
4	Mismatched record	Mismatched record	Ordinary repeat 5

Current, removed and mismatched responses are shared between pure-state and hybrid acquisition; the hybrid’s ordinary repeats reuse the first two generic responses. Deployment costs charge each strategy for all five required responses, even when joint acquisition reuses them. A funded generic arm can add positions beyond four. Let $P^{(i)}$ be the normalized ordered paths from diagnostic response i , B_j those from base response $j \in \{0, 1, 2\}$, and C the three most frequent paths across the base responses. Let T_i , U_i and T_0 denote the input, returned and original state’s candidate-file sets. Jaccard agreement is $J(A, B) = |A \cap B| / |A \cup B|$ when both sets are nonempty and zero otherwise. Path normalization, repository-membership validation and the at-most-three-path rule are identical for every diagnostic

arm. A response is parsed when a nonempty JSON object is recovered; validity additionally requires the registered file-output schema and valid implementation paths.

Each response supplies 21 quantities: confidence, validity, parsing success, path count, valid-path fraction; $J(P^{(i)}, C)$, $J(P^{(i)}, B_0)$, rank-weighted base votes and top-path agreement with C ; $J(P^{(i)}, T_i)$, $J(P^{(i)}, U_i)$, $J(P^{(i)}, T_0)$, the following shift $J(P^{(i)}, T_i) - J(B_0, T_i)$ and input-state distance $1 - J(T_i, T_0)$; retrieval coverage and best/mean retrieval rank; mean path–query lexical overlap, returned-state–query lexical overlap, serialized state length and a nonempty-returned-state indicator. An empty input state gives missing input-state distance. Rank-weighted base votes equal $\sum_{k=0}^{|P^{(i)}|-1} (3-k)v(P_k^{(i)})/18$, where v counts the three base outputs. Retrieval ranks are zero-based among the fifty visible candidates, with absent paths assigned rank 50; best and mean ranks are divided by 50, and both default to one for an empty output. Fractions use denominator $\max(|P^{(i)}|, 1)$. Lexical overlap applies J to lower-cased identifier tokens of more than two characters after English stop-word removal; returned state is serialized as JSON for this calculation. No reference implementation path enters these quantities.

The ordered representation records the first five response vectors, ten pairwise overlaps and their valid-output versions. The latter are missing unless both responses are valid. For positions $i = 1, \dots, 4$, it also records differences from position zero in confidence, base-consensus agreement, original-state agreement, following shift and input-state agreement. Its additional relative-disagreement term is

$$\{1 - J(P^{(0)}, P^{(i)})\} - \{1 - J(P^{(0)}, P^{(3)})\}.$$

For $i = 3$, this relative-disagreement term is identically zero; it is retained in the frozen ordered column list. It does not introduce an additional contrast. The aggregate representation summarizes every response quantity by finite-value mean, population standard deviation, minimum, maximum and missing fraction; it also summarizes all pairwise overlaps, the proportion of unique file sets, file-vote entropy in nats, the largest file-vote count divided by the number of calls, and that call count. Aggregates use all funded generic calls when more than five are available. These fields supplement the common base features and the seven original diagnostic features. Every arm can select the original feature vector, the original plus ordered and aggregate fields, the original plus aggregate fields, or a compact subset containing validity/parsing, state correspondence, following shifts, differences and pairs. The frozen column lists specify the compact representation exactly.

The auxiliary noise-relative representation compares altered responses with ordinary repetition. Neutral positions are $N = \{0, 1, 3\}$ and altered positions $A = \{2, 4\}$; in the hybrid arm these contain the current-state response and two ordinary repeats, then removal and mismatch. The corresponding positions are treated identically in the generic arm. The representation records the mean, population standard deviation and minimum overlap among neutral pairs. For each of confidence, base-consensus agreement, input/returned/original-state agreement, following shift, validity, parsing, best retrieval rank and retrieval coverage, it retains the neutral mean and standard deviation, plus each altered response’s difference from the neutral mean and that difference divided by $(SD + 0.05)$ and clipped to $[-10, 10]$. It also retains altered-to-neutral mean agreement, excess disagreement relative to neutral pairs, the difference between minimum neutral-pair agreement and maximum altered-to-neutral agreement, relative instability $(1 - \bar{J}_{i,N})/(1 - \bar{J}_N + 0.05)$, overlap between altered responses, and the mean and difference of their excess disagreements. These numeric contrasts are computed after the fixed float32 conversion. They receive their own equally specified candidate and generic fits and do not replace the frozen primary representation.

Independent fixed-weight inference used the pinned weights and llama.cpp runtime; the final Mistral configuration used four concurrent slots. Backend settings were checked on development inputs before use on the fixed evaluation cohort. Each logical request allowed at most three attempts, with completed responses retained and terminal failures included in the planned denominator.

The prespecified noise-contrast variant is an auxiliary candidate developed separately, with corresponding generic models receiving the same representation and search opportunities. Its comparison is

a prespecified secondary member of the three-comparison family.

All four 800-issue collections and their 3,200 predictions were completed before the separate gold file was released. The registered analysis retains failures and resamples issues 10,000 times within repository, carrying all four models together and conditioning on fixed predictors. Marginal intervals use the 2.5th and 97.5th percentiles. The three AUC comparisons use 0.8333rd and 99.1667th percentiles for Bonferroni family coverage. Brier score, log loss, ten fixed calibration bins and failure risk at 25/50/75/100% retained fractions are secondary measures; score ties are resolved by issue and model identifiers. Risk intervals condition on that observed selection. Calibration functions and retention rules remain fixed during evaluation.

Table S6: Prespecified independent AUC comparisons against the corresponding funded generic predictor, with marginal 95% and family-adjusted 98.333% intervals.

Candidate	Difference	Marginal 95%	Family 98.333%
Repetition–intervention	0.00382	[0.00023, 0.00743]	[-0.00055, 0.00821]
State interventions	0.00168	[-0.00247, 0.00589]	[-0.00345, 0.00677]
Noise contrast	0.00194	[-0.00214, 0.00608]	[-0.00305, 0.00697]

Table S7: Model-resolved independent discrimination. Each configuration has 800 issues; subgroup intervals are marginal and descriptive.

Model	Positive	Negative	Hybrid AUC	Generic AUC	Δ AUC [95% interval]
Sol	698	102	0.8343	0.8415	-0.00721 [-0.02631, 0.01151]
DeepSeek	564	236	0.9325	0.9282	0.00430 [-0.00401, 0.01287]
Qwen	531	269	0.7722	0.7352	0.03699 [0.01318, 0.06103]
Mistral	20	780	0.9879	0.9886	-0.00071 [-0.00496, 0.00327]

Reported joint acquisition usage was 241,102,491 tokens. There were 3,121 fully metered and sufficiently funded records out of 3,200; 81 attempts lacked metering. The full cohort is primary. On the complete-and-funded subset, the primary AUC difference is 0.00403 [0.00041, 0.00766]; the same-call-count comparison on all records is 0.00392 [0.00044, 0.00739]. Both are additional marginal descriptions. Costs include transport attempts and indivisible-call overshoot. The resource measure is provider-reported tokens; unmetered attempts form a separate accounting category. Requested and returned identifiers specify each API configuration.

The research archive includes the analysis specification, response files, frozen predictions, scoring-release record and bootstrap draws. Model-service interruptions in the development acquisitions are accounted for separately in Supplementary Notes 13–15.

Development ablations of diagnostic information

The 300-issue development cohort supports a targeted comparison of information blocks. All arms retain the same common base features. Quality fields describe confidence, parsing, path validity, retrieval compatibility and lexical/output properties. Response-relation fields describe agreement and contrasts across generated answers. State-correspondence fields describe input, returned and original state agreement, following shifts and input-state distance. We removed blocks and repeated the same nested search and learner-balanced ensemble construction within each diagnostic arm. Thus an ablation changes the available information while giving the reduced predictor the same opportunity to adapt.

The full hybrid ensemble has development AUC 0.94803, compared with 0.94331 using quality fields, 0.94870 with quality and answer relations, and 0.94254 with quality and state correspondence. The

corresponding pure-state values are 0.94756, 0.94071, 0.94690 and 0.94087. Relations among answers account for more of the observed increment than state correspondence alone. The auxiliary noise-relative hybrid attains 0.94469, below the original full hybrid, and remains a secondary candidate. The primary candidate and independent test population remained fixed throughout the ablation analysis.

Tables S8 and S9 report all ensembles and paired comparisons in the ablation analysis. Intervals use 10,000 paired issue/template-group resamples of the saved out-of-fold predictions (1,200 model–issue records in 297 groups), conditional on those fits and preceding development choices. These explanatory intervals resample predictions with the estimators held fixed; the independent evaluation is reported separately.

Table S8: Development performance for all retained information blocks. Hybrid combines repeated observations with state interventions; the two generic arms use five calls or the funded total.

Representation	AUC	Conditional 95%	Brier
State / Full	0.94756	[0.93447, 0.95997]	0.08499
State / Quality	0.94071	[0.92726, 0.95338]	0.09262
State / Quality + relations	0.94690	[0.93351, 0.95942]	0.08547
State / Quality + state	0.94087	[0.92724, 0.95379]	0.09248
Generic (five) / Full	0.94264	[0.92908, 0.95530]	0.09120
Generic (five) / Quality	0.93989	[0.92616, 0.95265]	0.09457
Generic (five) / Quality + relations	0.94233	[0.92882, 0.95488]	0.09200
Generic (five) / Quality + state	0.94018	[0.92671, 0.95273]	0.09456
Generic (funded) / Full	0.94334	[0.93008, 0.95563]	0.09128
Generic (funded) / Quality	0.93922	[0.92540, 0.95198]	0.09534
Generic (funded) / Quality + relations	0.94243	[0.92876, 0.95525]	0.09113
Generic (funded) / Quality + state	0.94022	[0.92680, 0.95282]	0.09442
Hybrid / Full	0.94803	[0.93490, 0.96042]	0.08433
Hybrid / Quality	0.94331	[0.93024, 0.95556]	0.09114
Hybrid / Quality + relations	0.94870	[0.93578, 0.96076]	0.08458
Hybrid / Quality + state	0.94254	[0.92922, 0.95503]	0.09112
Hybrid / Noise-relative	0.94469	[0.93113, 0.95737]	0.08668
Generic (five) / Noise-relative	0.94063	[0.92709, 0.95346]	0.09314
Generic (funded) / Noise-relative	0.94041	[0.92661, 0.95340]	0.09325

Table S9: Paired development AUC comparisons. Intervals resample issue groups in the saved out-of-fold predictions, conditional on the fitted estimators and development choices.

First representation	Second representation	Difference	Conditional 95%
State / Full	State / Quality	0.00684	[0.00384, 0.01002]
State / Full	State / Quality + relations	0.00065	[−0.00076, 0.00212]
State / Full	State / Quality + state	0.00669	[0.00364, 0.00993]
Generic (five) / Full	Generic (five) / Quality	0.00275	[−0.00013, 0.00571]
Generic (five) / Full	Generic (five) / Quality + relations	0.00031	[−0.00100, 0.00161]
Generic (five) / Full	Generic (five) / Quality + state	0.00246	[−0.00052, 0.00537]
Generic (funded) / Full	Generic (funded) / Quality	0.00412	[0.00132, 0.00700]
Generic (funded) / Full	Generic (funded) / Quality + relations	0.00092	[−0.00052, 0.00245]
Generic (funded) / Full	Generic (funded) / Quality + state	0.00313	[0.00001, 0.00639]

First representation	Second representation	Difference	Conditional 95%
Hybrid / Full	Hybrid / Quality	0.00472	[0.00091, 0.00872]
Hybrid / Full	Hybrid / Quality + relations	-0.00067	[-0.00246, 0.00101]
Hybrid / Full	Hybrid / Quality + state	0.00549	[0.00205, 0.00911]
State / Full	Generic (funded) / Full	0.00421	[-0.00124, 0.00964]
Hybrid / Full	Generic (funded) / Full	0.00469	[-0.00053, 0.00983]
State / Quality	Generic (funded) / Quality	0.00149	[-0.00375, 0.00686]
Hybrid / Quality	Generic (funded) / Quality	0.00409	[-0.00046, 0.00871]
State / Quality + relations	Generic (funded) / Quality + relations	0.00448	[-0.00104, 0.01011]
Hybrid / Quality + relations	Generic (funded) / Quality + relations	0.00627	[0.00107, 0.01154]
State / Quality + state	Generic (funded) / Quality + state	0.00065	[-0.00448, 0.00579]
Hybrid / Quality + state	Generic (funded) / Quality + state	0.00232	[-0.00209, 0.00674]
Hybrid / Noise-relative	Generic (funded) / Noise-relative	0.00428	[-0.00110, 0.00966]
Hybrid / Noise-relative	Hybrid / Full	-0.00334	[-0.00643, -0.00035]

Frozen auxiliary predictor settings

The auxiliary noise-relative ensemble uses the same four learner families and parameter grid. Its four representation choices are the original features, original plus noise contrasts, original/quality plus noise contrasts, and original plus the complete ordered/aggregate and contrast fields. The principal hybrid and funded-generic configurations used in the prespecified auxiliary comparison are listed below. Each member has probability weight 0.25; all other learner settings match the main ensemble.

Table S10: Frozen noise-relative candidate and corresponding funded-generic configurations.

Arm	Learner	Representation	Settings
Hybrid	Logistic	Full + contrasts	C=0.1
Hybrid	Hist. boosting	Full + contrasts	leaves=15, leaf=10
Hybrid	Extra trees	Full + contrasts	depth=8, leaf=5
Hybrid	CatBoost	Full + contrasts	depth=4, l2=3
Generic (funded)	Logistic	Original	C=0.1
Generic (funded)	Hist. boosting	Full + contrasts	leaves=7, leaf=10
Generic (funded)	Extra trees	Original	depth=8, leaf=5
Generic (funded)	CatBoost	Full + contrasts	depth=4, l2=3

Structured-state responses across source corpora

The seven-corpus structured-state experiment used a versioned sample drawn from the retained event pools. Sampling selected up to 500 entries per corpus; the available narrative pool contributed 98 entries. The resulting sample contained 3,098 entries and 3,073 unique corpus–event identifiers. Each entry was evaluated under the eight specified conditions in three repeated acquisitions, yielding all 74,352 planned records. The request route was `gpt-5.4-mini`, with a recorded output limit of 1,536 tokens, the specified condition-specific temperatures and memory initialization at each 50-entry block. Repetition indices do not imply service-side seed determinism.

The complete membership table identifies each source file, row, corpus, event and episode. Overlap

with the source-intake, comparison and finite-action samples is reported by event identifier. The two baseline acquisitions share a design and source pools; their response records are retained separately.

Table S11: Membership of the complete baseline acquisition. Duplicate event identifiers refer to distinct retained source rows and remain visible in the membership file.

Corpus	Available rows	Sample entries	Unique events
Narratives	98	98	84
ds000210	3,720	500	500
ds001618	3,636	500	500
ds001882	4,737	500	500
ds000212	3,584	500	500
ds004917	38,195	500	500
ds004042	4,017	500	489
Total		3,098	3,073

Table S12: Event overlap between the complete baseline sample and three related samples. Shared IDs are unique corpus–event identifiers; counts refer to the complete baseline sample.

Related sample	Entries	Unique IDs	Shared IDs
Initial source intake	224	224	81
Comparison pool	497	493	251
Finite-action sample	818	803	501

Raw model fields and outputs completed by the rule-based implementation were analyzed separately. The raw reason-response index is defined when a final action, an initial impulse and a reason graph with a node list are present; it equals one when the graph is nonempty and the final action differs from the initial impulse. Among paired scorable records, the structured condition exceeded reason removal in all seven corpora. The equal-corpus mean difference was 0.679, with a conditional 95% episode-resampling interval [0.657, 0.708]. These estimates describe the paired scorable subset. Coverage is retained in Table S13; unsupplied raw fields are not assigned zero.

Table S13: Structured-minus-reason-removal contrasts on paired raw fields. Planned pairs count entry–repetition pairs. Episodes are the source episode identifiers represented in the scorable subset.

Corpus	Planned pairs	Reason pairs	Episodes	Difference
Narratives	294	60	4	0.217
ds000210	1,500	387	4	0.765
ds001618	1,500	394	3	0.640
ds001882	1,500	371	8	0.571
ds000212	1,500	189	14	0.884
ds004917	1,500	508	1	0.927
ds004042	1,500	201	2	0.751

The primary field definitions also require numeric identity and continuity weights for the self-state index and a Boolean application field for the veto index. No paired self-state records satisfied those requirements. Only five pairs supplied the required veto fields on both sides, all with a zero contrast. The complete raw-field coverage table accompanies the source data. Outputs from the original field-completion rules are retained in a separate compatibility table, identifying their computational provenance.

The high-randomness condition had higher empirical final-action-string entropy than the structured condition in every corpus. Entropy is computed in bits on the same paired nonmissing actions within each corpus; it describes variation in the emitted strings, including their wording. It does not measure an open-ended action ontology or establish an entropy-matched comparison. Paired effects and entropy differences use the registered 2,000 resamples of source episode identifiers, carrying each selected episode’s conditions and repeats together. Corpora remain fixed in the equal-corpus summary. Where a corpus contributes one episode, resampling cannot estimate variation between episodes; the point estimate is retained without an inferential interval.

Table S14: Empirical action-string entropy in bits on matched nonmissing records. The paired denominator is shown for each corpus.

Corpus	Paired actions	Structured	Random	Difference
Narratives	233/294	7.090	7.769	0.678
ds000210	871/1500	9.173	9.415	0.242
ds001618	896/1500	7.973	9.038	1.064
ds001882	1015/1500	4.022	4.110	0.088
ds000212	363/1500	6.875	7.809	0.934
ds004917	1183/1500	7.625	9.846	2.221
ds004042	1444/1500	10.172	10.390	0.218

The acquisition saved 74,352 terminal logical requests and 74,353 attempts. Reported usage totalled 31,851,562 tokens; 3 attempts lacked usage. Those attempts and all terminal failures remain in the acquisition accounting. Missing usage contributes an unknown amount to the total cost.

Linear localization prediction

The six-model reference-dependent localization study found a substantial association between state diagnostics and correctness. Across 300 issues, 18,000 condition/repeat observations aggregate to 1,800 model–issue records, of which 1,425 meet the two-of-three localization criterion. Issue-grouped analysis gives AUC 0.737 for the ordinary baseline and 0.924 after adding the state features, a difference of 0.187 [0.127, 0.268] from 2,000 full-refit issue-bootstrap draws. Several augmented features use reference implementation files during scoring. The association therefore concerns reference-dependent diagnostics, even though reference files were excluded from generation prompts and the ordinary baseline.

The reference-free experiment tests predictive utility under information available before scoring. Sol, DeepSeek, Qwen and Mistral each evaluated the same 300 bridge issues. The common baseline was augmented with state probes, five generic probes, or generic acquisition funded to at least the state-probe token total when sufficient metering and calls were available. The resulting predictors use identical preprocessing and model complexity (Table S15).

Table S15: Reference-free localization reliability. Bridge estimates are grouped out-of-fold. Reused-holdout estimates condition on the bridge fits; these disjoint test outcomes had been viewed before the service-driven model amendment.

Feature set	Bridge AUC	Brier	Reused AUC	Brier
Shared baseline	0.920	0.108	0.950	0.074
Shared + state probes	0.923	0.106	0.954	0.072
Shared + five generic	0.924	0.105	0.957	0.072
Shared + funded generic	0.924	0.105	0.957	0.072

State-probe AUC was 0.923 on the bridge and 0.954 on the 100-issue reassessment. The relevant

incremental comparison is against the funded generic diagnostic: the paired AUC differences were -0.000972 $[-0.013172, 0.004733]$ and -0.003221 $[-0.010227, 0.002950]$, respectively. The reassessment is separate from model fitting, but its outcomes were observed before the amendment that replaced an unavailable API service; it is a conditional held-out evaluation.

Pooled discrimination combines within-model and between-model outcome variation. The bridge contains 1,200 model-issue records in 297 issue/template groups, with 678 positive outcomes. Within-model CSB/funded AUCs were 0.826/0.858 for Sol, 0.892/0.899 for DeepSeek, 0.674/0.651 for Qwen and 0.891/0.889 for Mistral. The reassessment contains 400 records and 258 positives; corresponding AUCs were 0.828/0.846, 0.916/0.924, 0.711/0.746 and 0.973/0.976. Sol has only four negative test outcomes and Mistral only three positives, making those estimates particularly sensitive to individual observations. Figure S12 presents the pooled and model-specific estimates with their support.

All four held-model point comparisons, and all four model-specific reassessment comparisons, favored funded generic diagnostics. Repository-held-out and token-prefix results were mixed. The conservative prior-interruption subset contains 693 bridge records and gives a saved out-of-fold AUC difference of 0.000495; its changed composition makes it a sensitivity analysis.

Recorded usage was complete and generic acquisition sufficiently funded in 1,151/1,200 bridge pairs; 68 attempts lacked usage. Figure S14 reports these accounting categories, retrospective budget comparisons and the upper bound of 42 potentially unlogged administrative attempts. Figure S13 complements discrimination with observed calibration, paired probability differences and selective localization risk. The figures summarize fixed predictions and the corresponding resampling distributions; model-resolved calibration is given in Supplementary Figure S11.

Supplementary analyses and research records

These notes detail the samples, controls and statistical analyses for controlled action responses, agent-written state and software localization, specifying the sampling units, available information and estimators.

Supplementary Note 1. Reference-dependent localization analysis

The localization endpoint requires at least two of three base calls each to retrieve a reference implementation file within their top three predictions. The 1,800 model-issue records contain 1,425 positive and 375 negative outcomes under this definition. The ordinary baseline uses reference-free features; the augmented predictor includes diagnostic components calculated with reference files.

Issue-grouped five-fold cross-validation with training-only preprocessing gives AUC 0.737216 for the baseline and 0.924486 after adding the reference-dependent CSB features. The difference is 0.187269, with a two-sided 95% full-refit issue-bootstrap interval $[0.126941, 0.267774]$; all 2,000 draws are estimable. Copies of a bootstrap-sampled issue share a fold. Numerical scaling is fitted separately from categorical encoding.

For held-group evaluation, each model or repository is excluded from training and predicted directly from the remaining groups. The archive also contains cross-validation on the retained groups as a separate sensitivity calculation. Direct held-group tests retain single-class groups as non-estimable for AUC. A held-model evaluation can include other models' records for the same issues in training, so its target is transfer across models conditional on those tasks.

Supplementary Note 2. Samples and generation accounting

Finite-action accounting layer	Count
Frozen intake: 32 source events in each of seven corpora	224
Sampled event entries for the 120-per-corpus request	818
Planned calls: $818 \times 6 \times 4 \times 3$	58,896
Acquisition log: recorded calls	58,836
Acquisition log: remote generations	58,835
Acquisition log: unrecovered generations	1
Merged scored source rows	61,056
Duplicate scored keys removed	3,240
Verified deduplicated scored rows	57,816
Distinct corpus/base-event pairs in scored rows	803

The accounting distinguishes planned calls, generation logs, merged exports and deduplicated scored observations. The difference $58,896 - 58,835$ spans these layers and is not a generation-failure count. The intake contains 32 source events per corpus; the finite-action evaluation requested 120 entries per corpus. All 803 unique identifiers in the finite-action list occur in the scored behavior set. The two lists share 73 identifiers, with 151 intake events absent from the finite-action sample and 730 finite-action events absent from the intake. This overlap specifies their shared source material.

The baseline lesion matrix uses a separate sample from the same seven-corpus pool. Its saved configuration sets a maximum of 500 events per corpus, eight conditions and three random seeds. The observed sample size depends on the available events in each corpus. The saved run-level metrics contain all 168 corpus-condition-seed cells, and the run summary records 74,352 calls, 74,336 remote generations and 16 unrecovered generations. These are run-level baseline totals, reported separately from the finite-action sample; the complete baseline replication provides entry-level membership in Tables S11–S12.

Sample	Observed entries or configured cap	Rows	Unique IDs	Overlap with 32	Overlap with 120
Frozen intake	32	224	224	—	73
Finite-action sample	Six corpora: 120; Narratives: 98	818	803	73	—
Comparison snapshot	71	497	493	74	237
Baseline lesion matrix	Configured cap: 500/corpus	—	—	—	—

Table note. Counts and intersections are computed directly from the retained Parquet files; unique IDs are corpus/event pairs. The 71-per-corpus comparison snapshot contains 497 entries and is distinct from the baseline execution cohort, which uses a cap of 500 per corpus. The 818-entry sample is stored in `full_sample_manifest.parquet`. The baseline row specifies its configured cap; entry-level membership and overlap are reported for the enumerated replication in Tables S11–S12. Dashes distinguish aggregate-only fields from observed sample counts. In the finite-action list, `ds004042` contributes 119 unique IDs among 120 entries and `Narratives` contributes 84 among 98; the other five corpora each contribute 120 unique IDs. Verified memberships and source hashes accompany the analysis records.

For each of the four controlled open-weight models, $3 \times 20 \times 100 \times 7 \times 3 = 126,000$ scored rows comprises 108,000 open-weight generation records and 18,000 offline-policy records, with no duplicate scoring keys. Recorded error rows are 689 (Qwen 14B), 1,523 (Qwen 7B), 212 (Qwen 32B) and 3,942 (Mistral 7B). The core seven variants include post-hoc/scrambled controls; strict decisive-field removal is a separate slice. Frozen plans identify the offline policy as the expected-action marginal within selected events, not a fitted empirical structured-output marginal.

Supplementary Note 3. Explicit rules and visible-output consistency

The deterministic baseline reconstructs the visible finite-action rule template and module settings without a hidden-target argument. It matches the target in all 14,454 structured-condition records. The result identifies explicit rule following as sufficient for performance in that experiment.

The output-only consistency rule excludes condition and provenance information and can score at least one dimension in 1.9075% of 26,946 traces. An exploratory semantic assessment uses 252 fixed-hash outputs, four per variant/corpus cell, with two judges and 504 logged calls. Sampling metadata is withheld. Each accepted judgment requires a quotation occurring exactly in the supplied text; absent or ambiguous evidence is unscorable. Coverage and agreement are reported separately because conditional agreement describes only the scorable subset. Equal-weight and four leave-one-component-out AFCI summaries assess sensitivity to the score definition; their weights were fixed without optimizing the contrasts.

Component	Jointly scorable / 252	Three-category κ	Agreement within jointly scorable
Reason	186	0.410	0.984
Memory	64	0.810	0.969
Persistent goal (judge field: self)	52	0.759	0.981
Veto	7	0.403	0.857

The three categories are unscorable, inconsistent and consistent; failed responses enter the unscorable category. Five of 504 terminal judgments returned HTTP 403, and the logs retain 506 total attempts. Coverage stratified by each corpus and variant is exported separately. Accepted quotations had to occur exactly in the supplied output; all were within the requested 160-character limit.

The equal-weight composites, including AFCI and the finite-action response summaries, are operational score definitions. Equal weights give each registered term the same arithmetic contribution while preserving its separate operational definition. Component-level values and the fixed leave-one-component-out analysis show how differences in coverage and numerical range affect the composite.

Supplementary Note 4. Patch application and repair outcomes

The official harness received 100 model–issue submissions from 50 issues and two models; 44 completed, one resolved and 56 returned harness errors. The application-and-repair experiment generated 100 nonempty initial patches and made 93 repair attempts. Eight final patches applied locally, spanning seven issues; all eight completed the official harness and none resolved its issue. Final apply categories are: 48 format errors, 25 context/hunk mismatches, 13 recorded empty repair patches, two missing-path errors, four other recorded failures and eight applicable patches. The 92 zero-length final exports represent filtered or rejected patches after initially nonempty generation.

Supplementary Note 5. Human evaluation and uncertainty reporting

Sample construction. The human-evaluation pool contains 532,800 candidate records from Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct and Mistral-7B-Instruct-v0.3 experiments and their accompanying controls. Stratified sampling selects 150 records from each of eight conditions: structured, full-context stochastic, state-field removal, context scrambling, goal deletion, post-hoc explanation/scrambling, distribution-matched and strict goal deletion. Within each condition, quotas are allocated across source blocks, with grouped sampling by prompt scaffold, task family and probe component. Both sampling and display-order randomization use seed 20260509. The resulting 1,200 items cover three prompt scaffolds and 20 task families: 1,050 are open-weight experiment records and 150 are offline-policy control records. Sampling and scoring programs are retained with the source records.

Masking and presentation. Each item receives a unique anonymous identifier. Model identity, condition, task source and reference target are stored in a separate hidden manifest. The rating form displays only the initial action tendency, candidate actions, reason graph, self-state, memory, veto status, final action and its rationale, in addition to the item identifier and display index. A predefined lexicon replaces model-, scaffold- and condition-related strings in the visible text with uniform placeholders; 223 records contain such replacements. All three annotators use the same item list and randomized display order.

Annotators and scoring instructions. Three graduate students in relevant university disciplines, including doctoral and master’s students, perform the evaluation using a common codebook and written instructions. They participate voluntarily, having been informed that the ratings will be used for this research. The seven dimensions assess consistency between the final action and stated reasons; adherence to relevant memory or prior commitments; appropriateness of veto; whether veto changes the initial action tendency; post-hoc characteristics of explanations; verbosity or templated expression without corresponding behavioral control; and inappropriate action changes accompanying irrelevant cues. The codebook defines each dimension, its binary judgment criteria and the handling of unjudgeable cases. These common criteria and written procedures standardize scoring across annotators. Instructions require independent judgments based only on visible item information, no discussion of individual items, no access to the hidden manifest, and no changes to item identifiers or output text.

Scores and aggregation. Each dimension is binary: 1 means its judgment statement holds and 0 means it does not. Because the statements have different directions, higher scores do not uniformly indicate better performance. Unjudgeable dimensions are marked missing and accompanied by an unscorable-item flag; forms also record three-level confidence, a metadata-exposure flag and optional notes. Each annotator completes all 1,200 items, with individual returns preserved separately. At each item and dimension, when all three scores are valid, the label agreed on by at least two annotators is the majority-vote result.

Agreement. Agreement is computed on individual scores before majority aggregation, for each of three annotator pairs and seven dimensions. Pairwise jointly valid observations are used if values are missing. All retained core ratings are valid binary values, yielding 25,200 individual scores. Of 8,400 item–dimension combinations, 8,250 are unanimous and 150 have a 2:1 split, affecting 68 items. The 21 Cohen’s κ values have mean 0.961, minimum individual pair-by-dimension value 0.860 and minimum dimension mean 0.897. The two-annotator evaluation used a separate 300-item sample.

In the control-summary tables, the column named “lower 95% bound” is the 2.5th percentile of the bootstrap distribution, corresponding to the lower endpoint of a two-sided 95% percentile interval. The controlled comparisons use seven corpus units; the descriptive Student t intervals in main Figure 2b and Supplementary Figure S4b are distinct from these percentile-bootstrap bounds. In the execution study, the task is the paired unit: the primary analysis resamples tasks within each of eight fixed mechanism strata and gives each mechanism equal weight. Resampling the eight mechanism clusters and leaving one mechanism out assess sensitivity to mechanism composition. Additional continuations improve measurement within a task without increasing the task count. Main Methods 5.1–5.2 specify these distinct uncertainty summaries. Methods 5.2 also collects the model and generation settings; Methods 5.3 defines the linear and richer predictive representations, budget comparisons and cohort roles.

Supplementary Note 6. Cohort membership and evaluation roles

The execution study contains 160 formal instances across eight mechanisms, including 32 predetermined redundancy cases and a 40-instance control subset; development instances are excluded. The linear

localization comparison separates 20 implementation-development issues, 300 bridge issues and 100 deduplicated reassessment issues. Its state features use no reference files. The 100-issue outcomes had been observed before the service-driven substitution of DeepSeek for Grok, so this cohort supports a conditional reassessment of the refitted four-model predictors. The independent prediction sample comprises 800 unused issues acquired after predictor selection, with all predictions frozen before outcome scoring. Supplementary Note 14 documents the model-service cohorts, their acquisition order and the service failures. Request records distinguish shared calls, diagnostic queries, retries and tokens, separating joint acquisition cost from the cost of each strategy.

Supplementary Note 7. Comparisons on shared model subsets

All five configurations are compared on the same 40 execution tasks and 60 bridge issues. The analysis uses DeepSeek together with the saved Sol, Qwen, Mistral and Astra observations. Supplementary Note 14 specifies the cohort composition and service-driven model substitution.

Table S16: Execution effects on the shared 40-task subset.

Model	State	Tasks	Effect	95% interval
Sol	memory	20	0.911	[0.854, 0.984]
Sol	goal	20	0.510	[0.344, 0.667]
DeepSeek route	memory	20	0.703	[0.635, 0.776]
DeepSeek route	goal	20	0.328	[0.208, 0.448]
Qwen 14B	memory	20	0.120	[0.042, 0.198]
Qwen 14B	goal	20	0.062	[0.010, 0.115]
Mistral 7B	memory	20	0.000	[0.000, 0.000]
Mistral 7B	goal	20	0.000	[0.000, 0.000]
Astra	memory	20	0.984	[0.969, 1.000]
Astra	goal	20	0.604	[0.458, 0.760]

Table S17: Prediction on the shared 60-issue subset.

Model	CSB AUC	Funded AUC	Δ AUC [conditional 95%]	Estimable
DeepSeek route	0.854	0.881	-0.02736 [-0.07446, 0.01563]	2000/2000
Sol	0.933	0.925	0.00721 [-0.03366, 0.04575]	2000/2000
Astra	0.973	0.964	0.00893 [0.00000, 0.03571]	1971/2000
Mistral 7B	0.977	0.953	0.02339 [0.00000, 0.10169]	1912/2000
Qwen 14B	0.619	0.620	-0.00118 [-0.10296, 0.09933]	2000/2000

Supplementary Note 8. Execution controls and sensitivity

Estimates use the planned task denominator and equal mechanism weights. Each cell reports a signed difference and a two-sided 95% within-mechanism task-bootstrap interval. The three core controls use all registered tasks: 80 per component for each core model and 20 per component for Astra. These intervals describe secondary contrasts. The eight-comparison multiplicity correction applies to the primary effects; no equivalence margin was specified for the controls.

Table S18: Restoration and irrelevant-deletion controls.

Model	State	Tasks	Rescue – lesion	Sham – irrelevant	Sham – rescue
Sol	memory	80	0.753 [0.662, 0.834]	0.006 [-0.016, 0.028]	-0.003 [-0.019, 0.013]
Sol	goal	80	0.366 [0.272, 0.462]	0.006 [-0.031, 0.044]	0.000 [-0.016, 0.016]
DeepSeek route	memory	80	0.519 [0.431, 0.600]	0.003 [-0.028, 0.034]	0.022 [-0.016, 0.059]

Table S18: Restoration and irrelevant-deletion controls. (continued)

Model	State	Tasks	Rescue – lesion	Sham – irrelevant	Sham – rescue
DeepSeek route	goal	80	0.238 [0.156, 0.319]	-0.013 [-0.047, 0.022]	0.022 [-0.013, 0.059]
Qwen 14B	memory	80	0.072 [0.025, 0.122]	0.003 [-0.037, 0.044]	0.009 [-0.009, 0.031]
Qwen 14B	goal	80	0.041 [0.013, 0.075]	0.003 [-0.022, 0.031]	0.003 [-0.009, 0.016]
Mistral 7B	memory	80	0.000 [0.000, 0.000]	0.003 [0.000, 0.009]	0.003 [0.000, 0.009]
Mistral 7B	goal	80	0.016 [0.000, 0.037]	0.000 [-0.009, 0.009]	-0.009 [-0.022, 0.000]
Astra	memory	20	1.000 [1.000, 1.000]	-0.016 [-0.031, 0.000]	-0.016 [-0.031, 0.000]
Astra	goal	20	0.604 [0.458, 0.760]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

The following validity contrasts use the fixed 40-task subset in each core model (20 per component). Fact replacement changes the stored fact or policy while leaving the environment unchanged. This contrast measures success against the original task target; following the replacement target is evaluated separately. Astra did not receive these additional branches. Supplementary Note 12 reports the corresponding replacement-target following rates.

Table S19: Fixed-subset paraphrase, replacement and history-compensation contrasts.

Model	State	Paraphrase – sham	Replacement – sham	Compensation – lesion
Sol	memory	-0.042 [-0.094, 0.000]	-0.693 [-0.844, -0.542]	0.563 [0.333, 0.771]
Sol	goal	0.016 [-0.031, 0.062]	-0.745 [-0.875, -0.615]	0.198 [0.062, 0.333]
DeepSeek route	memory	-0.016 [-0.078, 0.042]	-0.615 [-0.719, -0.495]	0.438 [0.250, 0.620]
DeepSeek route	goal	-0.021 [-0.104, 0.047]	-0.719 [-0.797, -0.641]	0.250 [0.125, 0.380]
Qwen 14B	memory	0.000 [0.000, 0.000]	-0.021 [-0.104, 0.042]	0.010 [0.000, 0.031]
Qwen 14B	goal	0.021 [-0.062, 0.104]	-0.031 [-0.094, 0.021]	0.000 [-0.042, 0.031]
Mistral 7B	memory	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Mistral 7B	goal	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

The mechanism sensitivity resamples the eight tested mechanisms, measuring dependence on their composition. Its precision is limited by those eight clusters. Holm adjustment is confined to the eight registered core primary contrasts.

Table S20: Mechanism-bootstrap sensitivity and adjusted primary tests.

Model	State	Primary effect	Mechanism-bootstrap 95% CI	Holm-adjusted p
Sol	memory	0.750	[0.656, 0.800]	< 0.001
Sol	goal	0.366	[0.269, 0.491]	< 0.001
DeepSeek route	memory	0.541	[0.344, 0.712]	< 0.001
DeepSeek route	goal	0.259	[0.147, 0.378]	< 0.001
Qwen 14B	memory	0.081	[0.016, 0.162]	0.031
Qwen 14B	goal	0.044	[0.009, 0.081]	0.137
Mistral 7B	memory	0.003	[0.000, 0.009]	1.000
Mistral 7B	goal	0.006	[0.000, 0.019]	1.000

Redundancy strata and the analysis restricted to a valid stored state are secondary. A valid write satisfies the tool/schema contract and need not contain correct facts. Selection on write validity changes the estimand; it does not replace the planned denominator. Each cell gives its actual task count.

Table S21: Redundancy and valid-write selected sensitivity.

Model	State	No redundant source	Redundant source	Valid writes only
Sol	memory	$n = 64$; 0.938 [0.898, 0.973]	$n = 16$; 0.000 [0.000, 0.000]	$n = 80$; 0.750 [0.659, 0.838]
Sol	goal	$n = 64$; 0.465 [0.359, 0.570]	$n = 16$; -0.031 [-0.062, 0.000]	$n = 80$; 0.366 [0.272, 0.459]
DeepSeek route	memory	$n = 64$; 0.680 [0.625, 0.730]	$n = 16$; -0.016 [-0.031, 0.000]	$n = 79$; 0.550 [0.465, 0.628]
DeepSeek route	goal	$n = 64$; 0.312 [0.230, 0.402]	$n = 16$; 0.047 [-0.016, 0.109]	$n = 78$; 0.264 [0.188, 0.345]
Qwen 14B	memory	$n = 64$; 0.090 [0.035, 0.148]	$n = 16$; 0.047 [0.000, 0.094]	$n = 76$; 0.081 [0.034, 0.134]
Qwen 14B	goal	$n = 64$; 0.055 [0.016, 0.102]	$n = 16$; 0.000 [-0.031, 0.031]	$n = 70$; 0.046 [0.009, 0.090]
Mistral 7B	memory	$n = 64$; 0.004 [0.000, 0.012]	$n = 16$; 0.000 [0.000, 0.000]	$n = 31$; 0.008 [0.000, 0.025]
Mistral 7B	goal	$n = 64$; 0.008 [0.000, 0.023]	$n = 16$; 0.000 [0.000, 0.000]	$n = 29$; 0.042 [0.000, 0.083]
Astra	memory	$n = 20$; 0.984 [0.969, 1.000]	Not estimable	$n = 20$; 0.984 [0.969, 1.000]
Astra	goal	$n = 20$; 0.604 [0.458, 0.750]	Not estimable	$n = 20$; 0.604 [0.458, 0.750]

Supplementary Note 9. Preparation, failures and execution cost

Valid/exact counts are tasks. The two sham columns are successful continuations over all planned sham continuations. “Raw” removes only the preparation-validity gate and is secondary. “Chars” is the number of tasks with equal serialized character counts after targeted and irrelevant deletion; this is not token or semantic equivalence.

Table S22: State preparation and sham-success denominators.

Model	State	N	Valid	Exact	Primary sham	Raw sham	Chars
Sol	memory	80	80	80	296/320	296/320	80/80
Sol	goal	80	80	70	300/320	300/320	80/80
DeepSeek route	memory	80	79	79	216/320	216/320	80/80
DeepSeek route	goal	80	78	78	218/320	218/320	80/80
Qwen 14B	memory	80	76	76	34/320	34/320	80/80
Qwen 14B	goal	80	70	69	40/320	40/320	79/80
Mistral 7B	memory	80	31	31	1/320	13/320	80/80
Mistral 7B	goal	80	29	29	4/320	20/320	80/80
Astra	memory	20	20	20	79/80	79/80	20/20
Astra	goal	20	20	18	80/80	80/80	20/20

The branch failure counts pool the two task components. Invalid/missing transactions and policy failures count continuations; invalid JSON and non-200 terminal HTTP responses count logical calls. Categories can overlap. The primary analysis includes all of these failed observations.

Table S23: Execution branch failures and retained calls.

Model	Branch	Contin.	Calls	Invalid transaction	Policy failure	Invalid JSON	HTTP failure
Sol	fact replace-ment	160	320	0	126	0	0

Table S23: Execution branch failures and retained calls. (continued)

Model	Branch		Contin.	Calls	Invalid transaction	Policy failure	Invalid JSON	HTTP failure
Sol	history	com-	160	320	0	14	0	0
	pensation							
Sol	irrelevant		640	1280	1	47	0	0
Sol	lesion		640	1280	200	201	0	0
Sol	paraphrase		160	320	0	13	0	0
Sol	rescue		640	1280	0	43	0	0
Sol	sham		640	1280	1	43	0	0
DeepSeek route	fact	replace-	160	318	40	106	40	1
	ment							
DeepSeek route	history	com-	160	319	31	12	31	3
	pensation							
DeepSeek route	irrelevant		640	1270	173	30	172	7
DeepSeek route	lesion		640	1274	312	150	311	11
DeepSeek route	paraphrase		160	318	29	14	29	4
DeepSeek route	rescue		640	1274	175	45	171	9
DeepSeek route	sham		640	1271	173	33	171	12
Qwen 14B	fact	replace-	160	320	39	105	0	0
	ment							
Qwen 14B	history	com-	160	320	22	122	0	0
	pensation							
Qwen 14B	irrelevant		640	1280	98	470	0	0
Qwen 14B	lesion		640	1280	180	426	0	0
Qwen 14B	paraphrase		160	320	31	106	0	0
Qwen 14B	rescue		640	1280	97	473	0	0
Qwen 14B	sham		640	1280	99	467	0	0
Mistral 7B	fact	replace-	160	304	51	98	0	0
	ment							
Mistral 7B	history	com-	160	303	64	88	0	0
	pensation							
Mistral 7B	irrelevant		640	1220	175	432	0	0
Mistral 7B	lesion		640	1239	189	425	0	0
Mistral 7B	paraphrase		160	301	47	102	0	0
Mistral 7B	rescue		640	1241	162	448	1	0
Mistral 7B	sham		640	1241	165	442	0	0
Astra	irrelevant		160	320	0	0	0	0
Astra	lesion		160	320	101	29	0	0
Astra	rescue		160	320	0	0	0	0
Astra	sham		160	320	0	1	0	0

Execution acquisition charges preparation once and includes all continuation branches and transport attempts. The totals quantify reported usage, with unmetered attempts counted separately. A failed first tool call terminates the continuation, reducing its logical call count. API and local token totals follow their respective tokenizers.

Table S24: Execution acquisition and missing token metering.

Model	Logical calls	Attempts	Reported tokens	Unmetered attempts
Sol	6400	6403	2,795,583	3
DeepSeek route	6363	6367	5,384,953	54
Qwen 14B	6400	6400	2,760,862	0
Mistral 7B	6071	6071	2,787,200	0
Astra	1360	1360	555,915	0

Supplementary Note 10. Prediction by model and cohort

Each row compares the state probes with the token-funded generic comparator using the same issues. Bridge predictions are pooled-model, grouped out-of-fold estimates; reused-holdout predictions use the four-model frozen bridge fit. Counts are model–issue rows and are not independent across models. AP denotes average precision. Brier scores quantify probability error; the calibration plots additionally show observed success against predicted probability.

Table S25: Prediction metrics by model, feature set and data role.

Model	Role	Probes	Positive	n predicted	AUC	AP	Brier
Sol	Bridge	State	270/300	300/300	0.826	0.963	0.071
Sol	Bridge	Generic	270/300	300/300	0.858	0.967	0.068
DeepSeek route	Bridge	State	212/300	300/300	0.892	0.935	0.104
DeepSeek route	Bridge	Generic	212/300	300/300	0.899	0.933	0.098
Qwen 14B	Bridge	State	186/300	300/300	0.674	0.770	0.225
Qwen 14B	Bridge	Generic	186/300	300/300	0.651	0.755	0.230
Mistral 7B	Bridge	State	10/300	300/300	0.891	0.604	0.023
Mistral 7B	Bridge	Generic	10/300	300/300	0.889	0.529	0.023
Sol	Reused	State	96/100	100/100	0.828	0.989	0.035
	test						
Sol	Reused	Generic	96/100	100/100	0.846	0.990	0.039
	test						
DeepSeek route	Reused	State	83/100	100/100	0.916	0.976	0.063
	test						
DeepSeek route	Reused	Generic	83/100	100/100	0.924	0.981	0.068
	test						
Qwen 14B	Reused	State	76/100	100/100	0.711	0.852	0.166
	test						
Qwen 14B	Reused	Generic	76/100	100/100	0.746	0.878	0.159
	test						
Mistral 7B	Reused	State	3/100	100/100	0.973	0.758	0.023
	test						
Mistral 7B	Reused	Generic	3/100	100/100	0.976	0.667	0.023
	test						

The completely metered and sufficiently funded bridge subset is selected solely from recorded metering and the generic token total. It is a sensitivity analysis, not a replacement of the full planned denominator. Its estimates use predictions from the full-cohort grouped fits.

Table S26: Prediction on the complete-and-funded metering subset.

Feature set	Rows	Positive	AUC	Brier
Shared baseline	1151	642	0.919	0.109
State probes	1151	642	0.923	0.106
Five generic	1151	642	0.923	0.106
Funded generic	1151	642	0.923	0.106

Joint bridge acquisition reports 76,484,918 tokens, counting shared calls once; 68 attempts have unreported usage. All 2,000 bridge full-refit bootstrap draws and all 5,000 fixed-predictor test-cluster draws were estimable for the primary differences. Single-class or otherwise non-estimable metrics remain NA in subgroup analyses. Repository- and model-held-out estimates and the attempt-level cost records accompany the numerical results. Held-model tests may contain the same issues under other models in training.

Supplementary Note 11. Retrospective token-budget sensitivity

The prefixes below use realized token counts without inspecting outcome labels. These retrospective thresholds summarize costs already incurred during acquisition, including the query that exceeded each threshold. Only pairs with sufficient metering enter this sensitivity; their denominator can vary by cap.

Table S27: Retrospective token-prefix sensitivity.

Cap	Pairs	Probes	AUC	Brier	Mean calls	Mean tokens
16,000	872	csb	0.882	0.126	0.396	14171.331
16,000	872	generic	0.884	0.125	0.390	14162.151
24,000	1160	csb	0.918	0.109	1.792	21617.003
24,000	1160	generic	0.918	0.109	1.778	21614.700
32,000	1162	csb	0.923	0.105	3.380	29184.468
32,000	1162	generic	0.923	0.106	3.361	29156.966
48,000	1151	csb	0.922	0.106	4.834	37017.755
48,000	1151	generic	0.922	0.106	4.996	37815.036
64,000	1151	csb	0.923	0.106	4.986	38087.260
64,000	1151	generic	0.923	0.106	5.260	39544.712
96,000	1151	csb	0.923	0.106	5.000	38211.056
96,000	1151	generic	0.923	0.106	5.288	39774.057

There were 553 excluded model–issue–cap cases, with reasons retained in the prefix exclusion ledger.

Supplementary Note 12. Replacement targets and their denominators

Replacement-target following is defined only when the changed record yields a definable target. Undefined cases remain in the planned task-success analysis but are not counted as failures to follow a nonexistent replacement target. These are continuation counts, with four continuations per validity task. Astra did not receive this branch.

Table S28: Replacement-target following with definable and planned denominators.

Model	State	Followed / definable	Undefined	Planned
Sol	memory	61/80	0	80
Sol	goal	59/80	0	80
DeepSeek route	memory	56/80	0	80
DeepSeek route	goal	49/80	0	80
Qwen 14B	memory	4/76	4	80
Qwen 14B	goal	10/80	0	80
Mistral 7B	memory	0/12	68	80
Mistral 7B	goal	0/32	48	80

Main Figures 3–4 and Supplementary Figures S1–S3 show the execution effects and controls. The task estimates use four continuations per branch within eight mechanisms; the secondary-control intervals describe differences without an equivalence test.

Supplementary Note 13. Held-group generalization and resource accounting

Each named group is excluded from training and evaluated directly. Repository tests withhold its issues across all four models. Model tests retain other models on the same issues and therefore do not simultaneously hold out tasks.

Table S29: Prediction in held-out repositories and models.

Excluded group	Train	Test	Positive	CSB AUC	Funded AUC	Brier	CSB / funded
repo: astropy	1176	24	16	0.977	0.977	0.048	/ 0.054
repo: django	744	456	253	0.944	0.948	0.099	/ 0.098
repo: matplotlib	1108	92	58	0.944	0.962	0.073	/ 0.070
repo: seaborn	1184	16	11	0.964	0.964	0.100	/ 0.100
repo: flask	1188	12	9	1.000	1.000	0.022	/ 0.021
repo: requests	1176	24	13	0.881	0.867	0.135	/ 0.140
repo: xarray	1180	20	14	0.952	0.917	0.089	/ 0.109
repo: pylint	1176	24	11	0.832	0.790	0.175	/ 0.182
repo: pytest	1132	68	40	0.929	0.929	0.089	/ 0.091
repo: scikit-learn	1108	92	64	0.941	0.930	0.073	/ 0.076
repo: sphinx	1136	64	32	0.913	0.905	0.121	/ 0.122
repo: sympy	892	308	157	0.880	0.902	0.150	/ 0.132
model: DeepSeek route	900	300	212	0.908	0.911	0.112	/ 0.108
model: Sol	900	300	270	0.863	0.901	0.074	/ 0.072
model: Mistral 7B	900	300	10	0.677	0.680	0.038	/ 0.040
model: Qwen 14B	900	300	186	0.624	0.644	0.257	/ 0.254

Table S30: Bridge service failures and request-level metering.

Bridge model	Logical calls	Attempts	Reported tokens	Unknown	Issues with failure	HTTP
Sol	3900	3910	15,533,921	11	1/300	
DeepSeek route	4058	4061	20,647,973	37	33/300	
Qwen 14B	3909	3909	17,112,923	0	0/300	
Mistral 7B	4095	4109	23,190,101	20	3/300	
Astra	780	781	3,102,504	1	0/60	

The four-model development cohort reports 76,484,918 tokens with 68 unmetered attempts. The conservative prior-interruption sensitivity retains 693 of 1,200 pairs; its CSB/funded AUC is 0.8313/0.8308. The administrative-interruption bound covers up to 42 additional attempts with indeterminate usage; it is reported separately from observed failures and token totals.

Supplementary Note 14. Service configurations and cohort composition

The Grok service cohort contained 604 terminal HTTP failures in 4,221 logical calls, affecting 101/300 issues (92 base, 211 state and 301 generic failures). The Sol and Grok service cohorts each had 1,300 HTTP403 quota refusals and no valid model answer. Service-access refusals are classified separately from generated task responses. The Sol/Grok/Qwen/Mistral bridge CSB/funded AUCs were 0.944870/0.944280, with difference 0.000590 and full-refit interval [-0.008658, 0.005812]. Its 100-issue comparison difference was 0.005166, with conditional interval [-0.002629, 0.013293], and was confounded by the unavailable API rows.

Sol responses were recovered through a second credential on the same 100 issues. DeepSeek was initially evaluated on those issues with the first predictor’s unknown-category encoding, without DeepSeek bridge training. The Sol recovery CSB/funded AUCs were 0.833333/0.848958, a difference of -0.015625 with interval [-0.061224, 0.012249]. The DeepSeek transfer values were 0.922750/0.915663, a difference of 0.007087 with interval [-0.025514, 0.036864]. These estimates condition on the Sol/Grok/Qwen/Mistral fits; Supplementary Note 15 reports the Sol/DeepSeek/Qwen/Mistral predictors refitted on the consistent model cohort.

The amendment collected DeepSeek’s 160 execution tasks and 300 bridge issues and refitted the four reliability predictors on the consistent model cohort. Sol, Qwen, Mistral, Astra and all 100-issue responses were reused. The change followed service failure and observation of the preceding results. It consequently defines an amended analysis, with no additional independent issue cohort.

The API configurations are specified by their requested and returned third-party aliases. DeepSeek’s request configuration and second-credential route match the preceding transfer evaluation. The provider’s V4-Pro alias-migration notice documents the naming transition (DeepSeek, 2026). Eight preceding connectivity calls report 4,980 tokens and four unmetered attempts, counted once outside the formal denominators.

Supplementary Note 15. Prediction on the reused holdout

Estimates use the same 100 held-out issues with saved Sol, DeepSeek, Qwen and Mistral responses and the refitted bridge predictors. These issues remain excluded from bridge training, while their outcomes were known before the amendment. The intervals therefore describe a post-amendment reassessment conditional on the fixed fits.

Table S31: Reused-holdout results for the consistent-model predictors.

Model	Features	Positive /100	AUC	AP	Brier
Sol	Base	96	0.8255	0.9894	0.0395
Sol	state probes	96	0.8281	0.9894	0.0349
Sol	five generic	96	0.8490	0.9907	0.0380
Sol	funded generic	96	0.8464	0.9905	0.0387
DeepSeek route	Base	83	0.9227	0.9796	0.0647
DeepSeek route	state probes	83	0.9157	0.9759	0.0627
DeepSeek route	five generic	83	0.9235	0.9809	0.0677
DeepSeek route	funded generic	83	0.9242	0.9810	0.0676
Qwen 14B	Base	76	0.6787	0.8294	0.1698
Qwen 14B	state probes	76	0.7111	0.8524	0.1664
Qwen 14B	five generic	76	0.7429	0.8778	0.1594
Qwen 14B	funded generic	76	0.7456	0.8779	0.1585
Mistral 7B	Base	3	0.9828	0.7917	0.0225
Mistral 7B	state probes	3	0.9725	0.7576	0.0226
Mistral 7B	five generic	3	0.9759	0.6667	0.0222
Mistral 7B	funded generic	3	0.9759	0.6667	0.0229

Table S32: Reused-holdout paired comparisons and conditional uncertainty.

CSB minus	AUC difference	Estimable /5000	Conditional 95% interval
Shared + funded generic	-0.003221	5000	[-0.010227, 0.002950]
Shared + five generic	-0.003194	5000	[-0.010034, 0.002680]
Shared baseline	0.003931	5000	[-0.001138, 0.009401]

Sol has four negative outcomes and Mistral three positives, limiting the resolution of their model-specific AUCs. Terminal and invalid responses remain in the planned denominator. The recovered Sol records contain six terminal HTTP 524 responses across five issues; the DeepSeek records contain ten terminal HTTP 404 responses across nine issues and two retried HTTP 429 responses. Original quota attempts, these failures and unreported costs are preserved. Reanalysis of the saved responses incurred no further model-query cost.

Supplementary figures

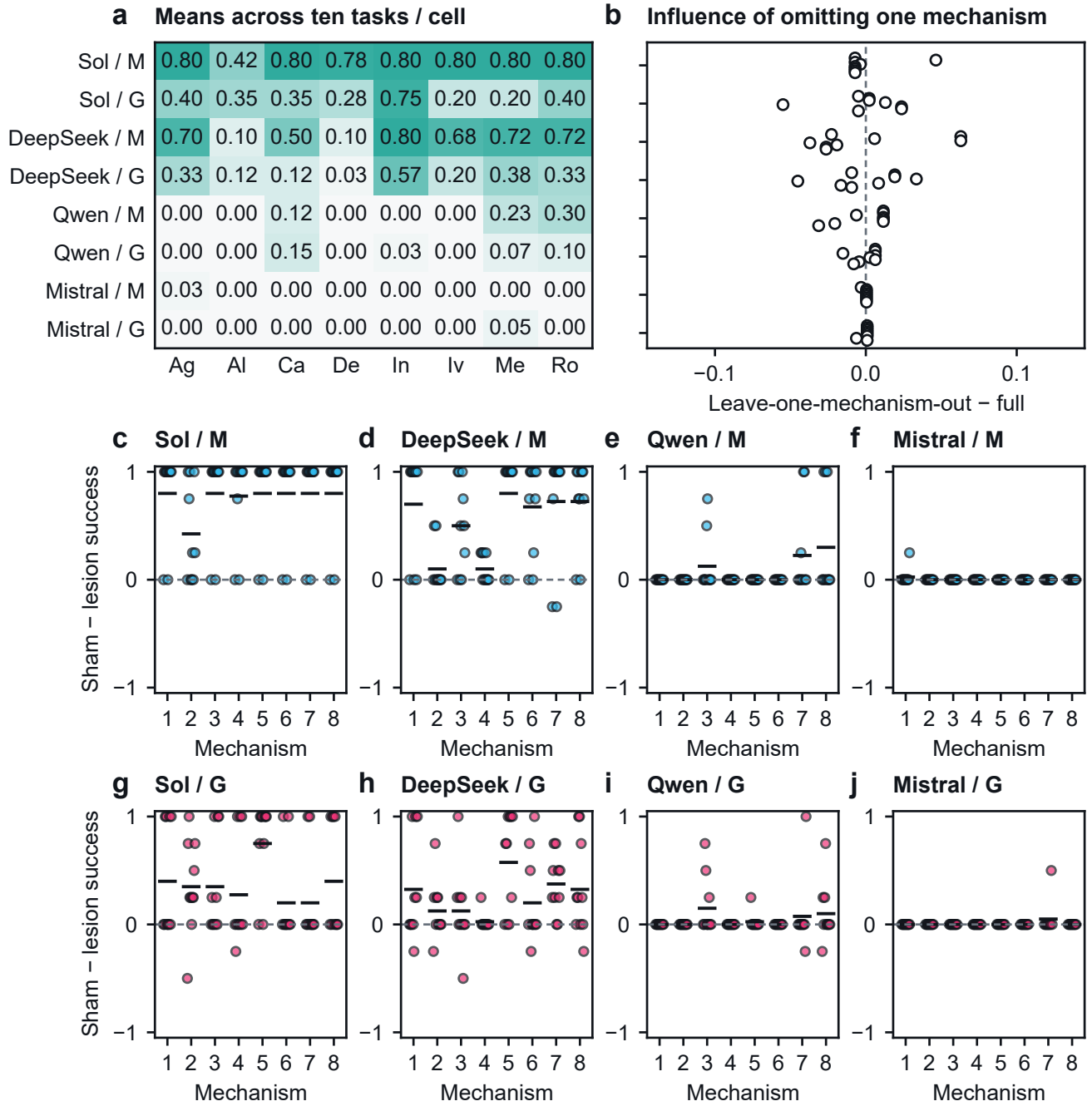


Figure S1: Mechanism heterogeneity and task variation. **a**, The 64 model/component-by-mechanism means, fixed diverging scale $[-1, 1]$. **b**, Each leave-one-mechanism-out estimate minus its corresponding full estimate, with the same row order as **a**; zero is no change, not a significance boundary. **c–j**, All ten task differences within each of eight mechanisms for each model/component cell, using four continuations per branch. Horizontal jitter is deterministic by task ID and is not uncertainty; black horizontal marks show each mechanism mean, with no connecting trend line between categorical mechanisms. Mechanism IDs 1–8 (Ag, Al, Ca, De, In, Iv, Me, Ro in **a**) denote aggregation, allocation, calendar, dependencies, inventory, invoice, merge and routing. The two rows are memory and goal; all 640 task differences are retained. Mechanism-bootstrap intervals are reported in Table S20 and the numerical results.

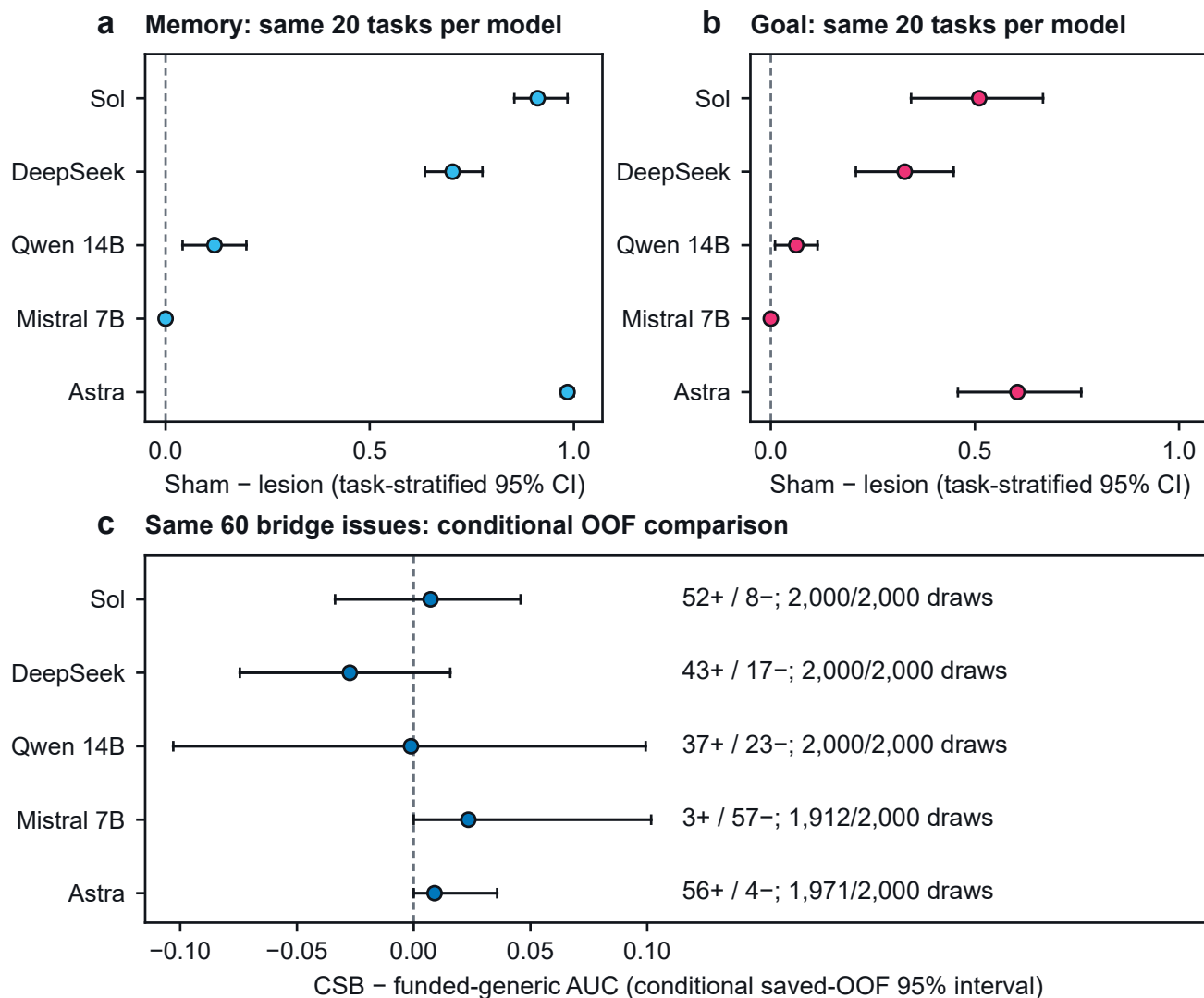


Figure S2: Fixed common subsets with Astra. **a,b**, All five models on the same 40 execution tasks, 20 per component, with equal-mechanism effects and task-stratified 95% intervals. The subset's weighting can differ from a raw continuation average. **c**, CSB minus funded-generic AUC on the same 60 bridge issues, with 95% intervals conditional on the out-of-fold (OOF) predictions. Printed positive/negative counts and estimable/2,000 draw counts retain single-class bootstrap failures. Intervals condition on the five-model subset fits and resample its issues; they do not include training variation.

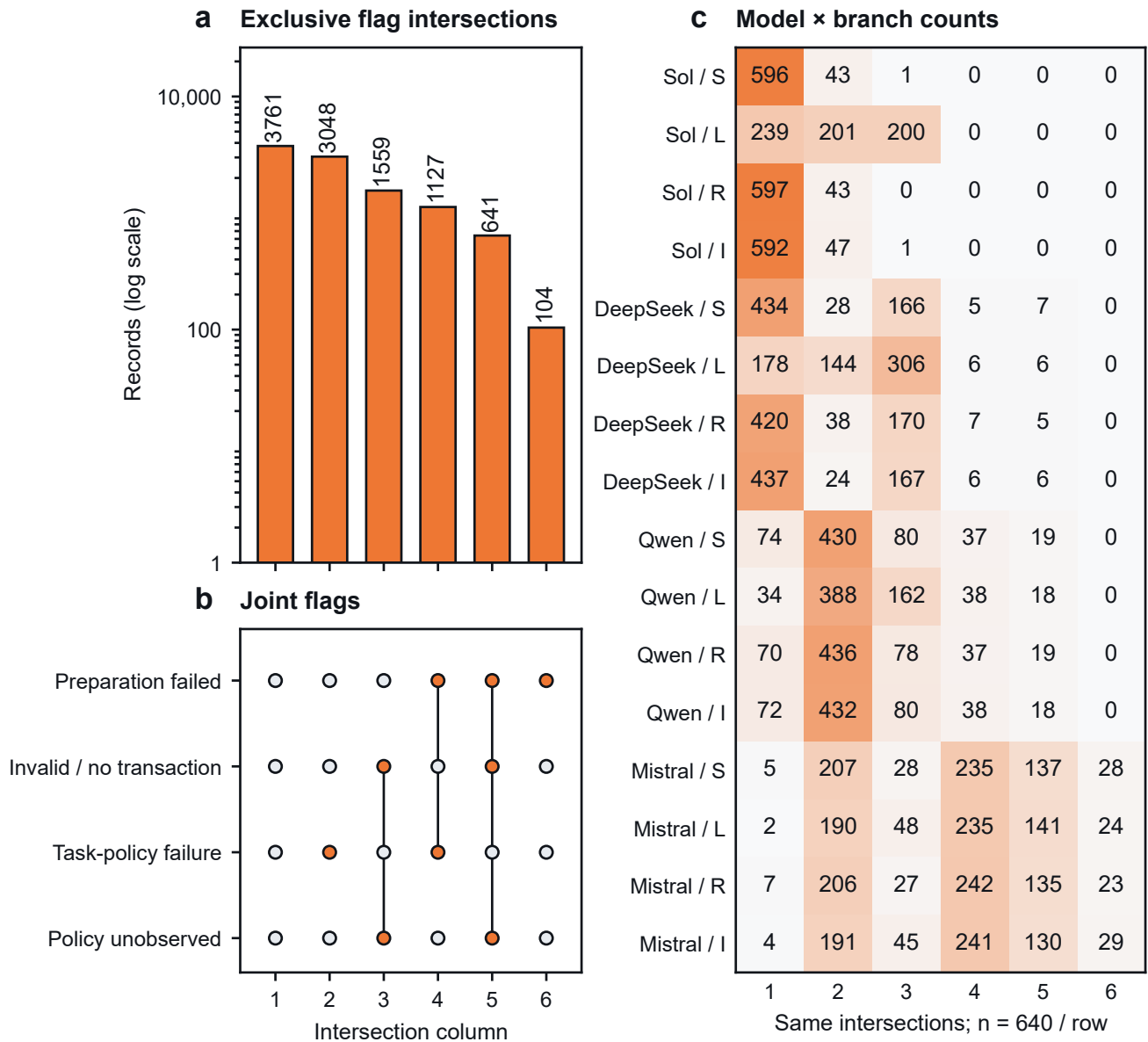


Figure S3: Failure flags without double counting. **a,b**, Mutually exclusive intersections of the three recorded failure flags plus an explicit unobserved-policy flag. Intersection 1 has no recorded failure flags. Log-count bars and incidence dots share the same intersection-column order. All 10,240 core continuations across sham, lesion, rescue and irrelevant branches are counted exactly once; 1,768 records have multiple failure flags and 2,200 lack an observed task-policy flag. **c**, The same intersections resolved by model and branch (S, sham; L, lesion; R, rescue; I, irrelevant), with exactly 640 records per row and a 0–640 sequential color scale. Unobserved policy outcomes form a separate category. The intersections describe the observed execution outcomes and response validity.

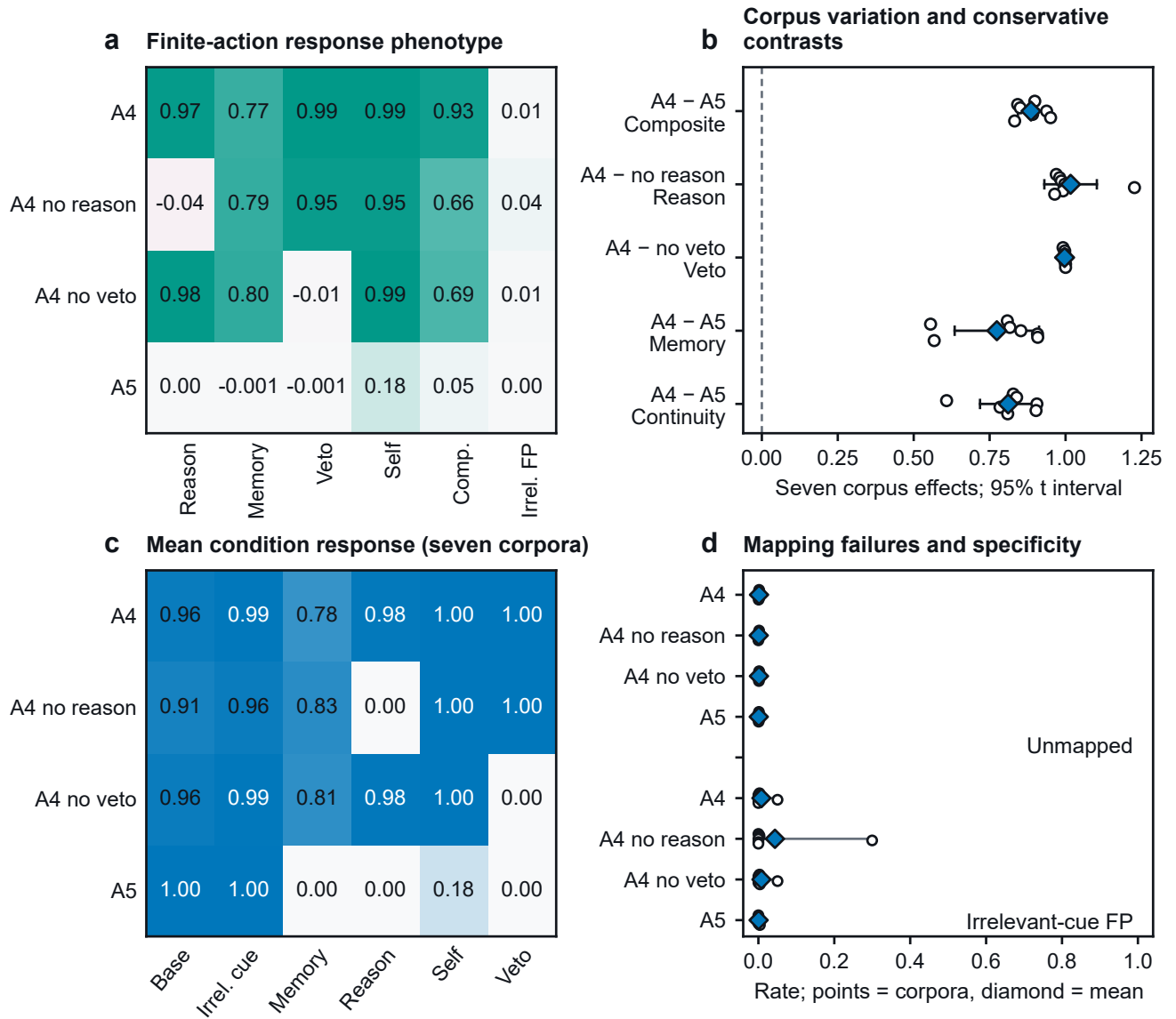


Figure S4: Explicit-rule finite-action responses. **a**, Mean behavioral response profiles, including reason, memory, veto, self, composite and irrelevant-cue false-positive measures; the symmetric diverging scale makes signed responses visible. **b**, Seven corpus effects (hollow points), means (diamonds) and descriptive two-sided 95% Student t intervals (6 degrees of freedom) for five registered contrasts. Table S45 separately retains the original percentile-bootstrap criterion. **c**, Condition-level mean response over seven corpora and three retained replicate indices; detailed corpus cells are shown in Figure S5. **d**, Unmapped-action and irrelevant-cue false-positive rates, with individual corpus observations, ranges and means. A4 denotes the structured condition, A5 the high-stochasticity comparator, and the two ablations remove reason or veto. Visible rules permit deterministic reconstruction of all 14,454 structured targets. The assay measures responses to supplied information across corpora with shared templates.

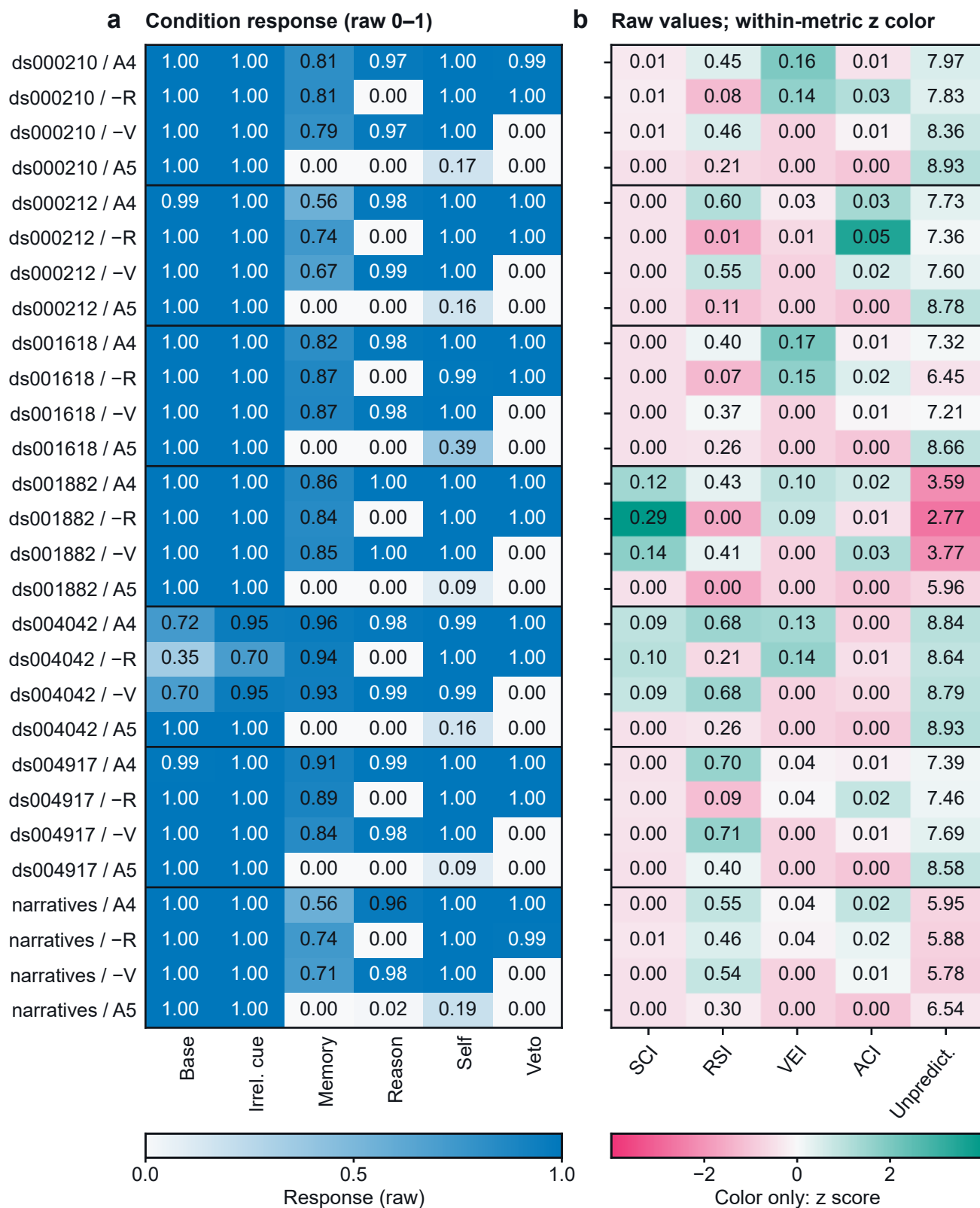


Figure S5: Corpus-resolved action and trace profiles. Cells average three replicate indices from separate finite-action (a) and structured-state trace (b) cohorts; matching corpus-variant names do not identify paired tasks. **a**, Raw condition responses (0–1). **b**, Raw metric values, colored by within-metric z scores across 28 corpus-variant means with symmetric limits. SCI, RSI, VEI and ACI denote self-continuity, reason-sensitivity, veto-effect and action-continuity indices. Rules separate corpora; -R/-V remove reason/veto. Source data retain 420 replicate metric values and 504 condition/replicate rows.

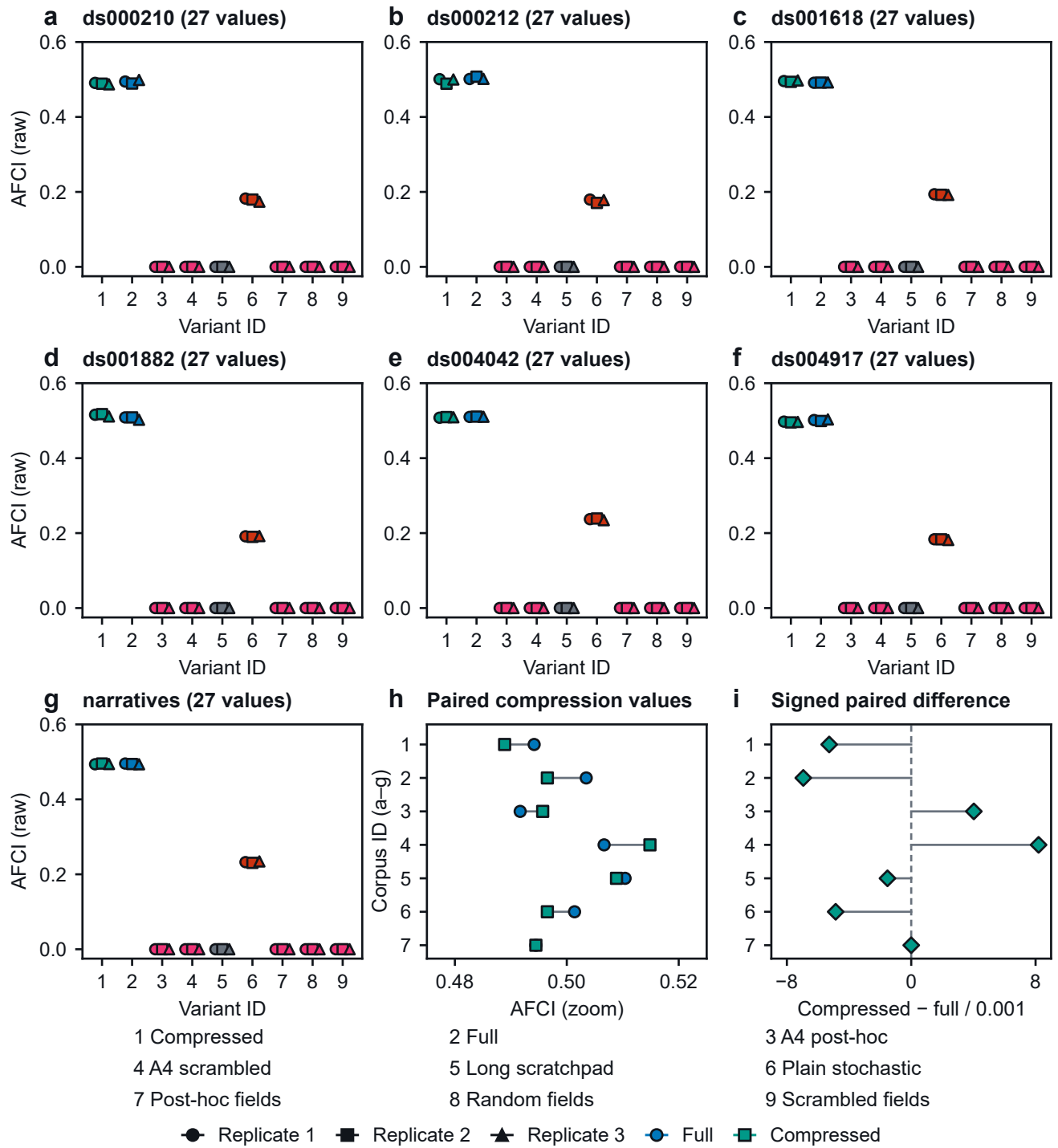


Figure S6: Action-field coupling by corpus and reporting condition. **a–g**, Seven corpus panels display all 189 original observations: nine variants times three retained replicate indices per corpus. The common displayed AFCI scale is 0–0.6 and contains all observations; the possible score range is 0–1; small horizontal offsets and circle/square/triangle shapes identify replicate indices 1/2/3, not uncertainty. The in-figure key maps variant IDs 1–9 to the original variants. Colors distinguish compressed, full, plain-stochastic, long-scratchpad and field-control families. AFCI uses both reported fields and their source information. **h**, Full and compressed structured-condition means paired within each corpus, with an explicitly magnified AFCI axis. **i**, Their signed compressed-minus-full differences in units of 0.001; **a–g** provide the full-scale context. Corpus IDs 1–7 in **h,i** follow panels **a–g**. Compression is a positive sensitivity case outside the predefined negative comparator set. Source data give both signed contrast directions; the points summarize the observed corpus means.

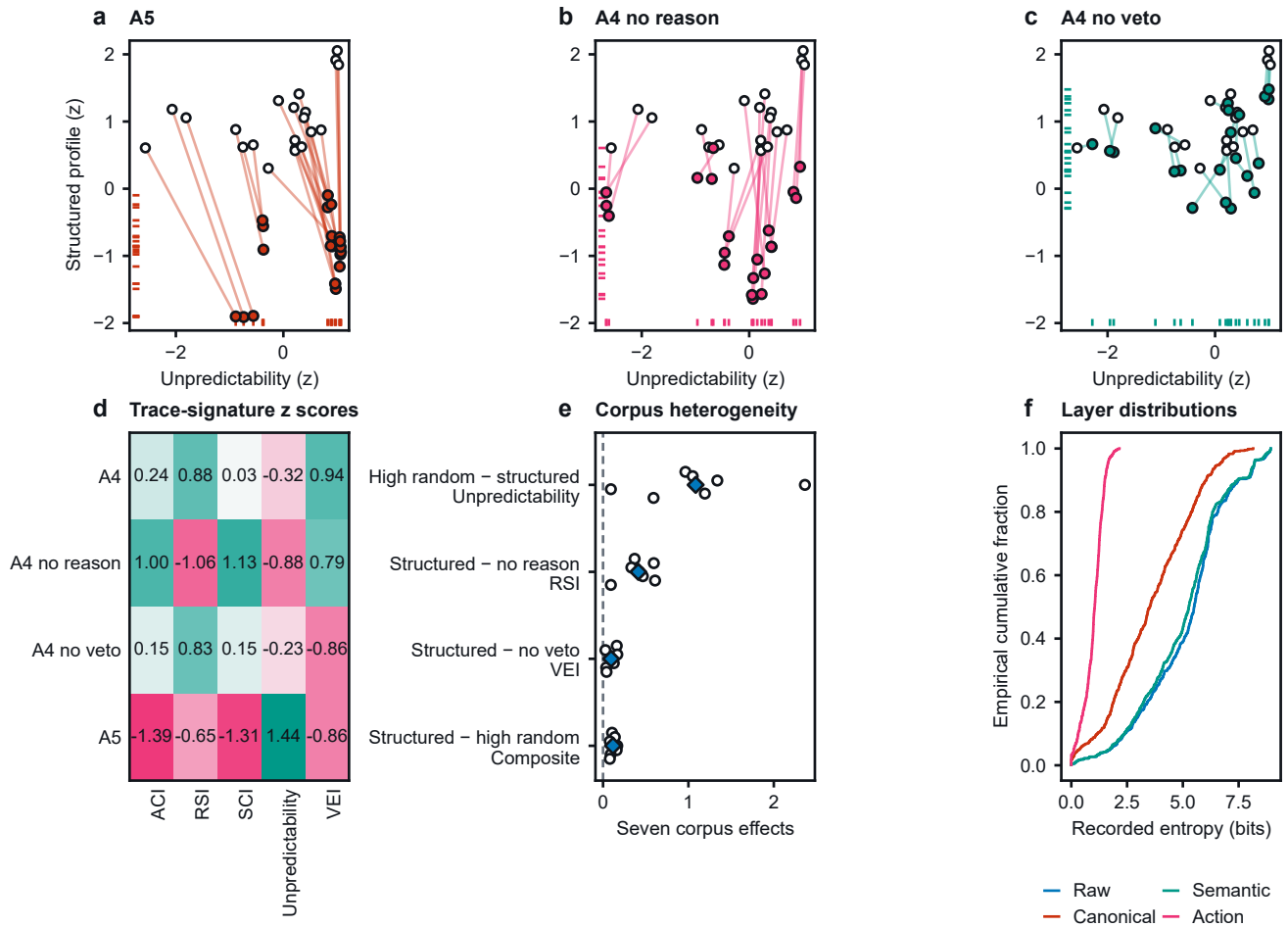


Figure S7: Paired trace geometry and entropy profiles. **a–c**, All 21 corpus–replicate displacements from A4 (open markers) to the indicated variant (colored markers), on identical standardized axes. Marginal rugs show the variant endpoints, and lines connect paired observations. **d**, Standardized trace signatures with the original metric names and exact values; symmetric color limits span the displayed data. **e**, Seven corpus effects for each indicated contrast (open points) with the corpus mean (diamond). These are descriptive contrasts. U denotes unpredictability and C the structured composite. **f**, Empirical distributions of recorded entropy by layer, retaining all rows. Entropy is in bits from the original base-2 implementation; these layers differ in representation and are not independent datasets. The standardized coordinates describe relationships among the recorded trace metrics.

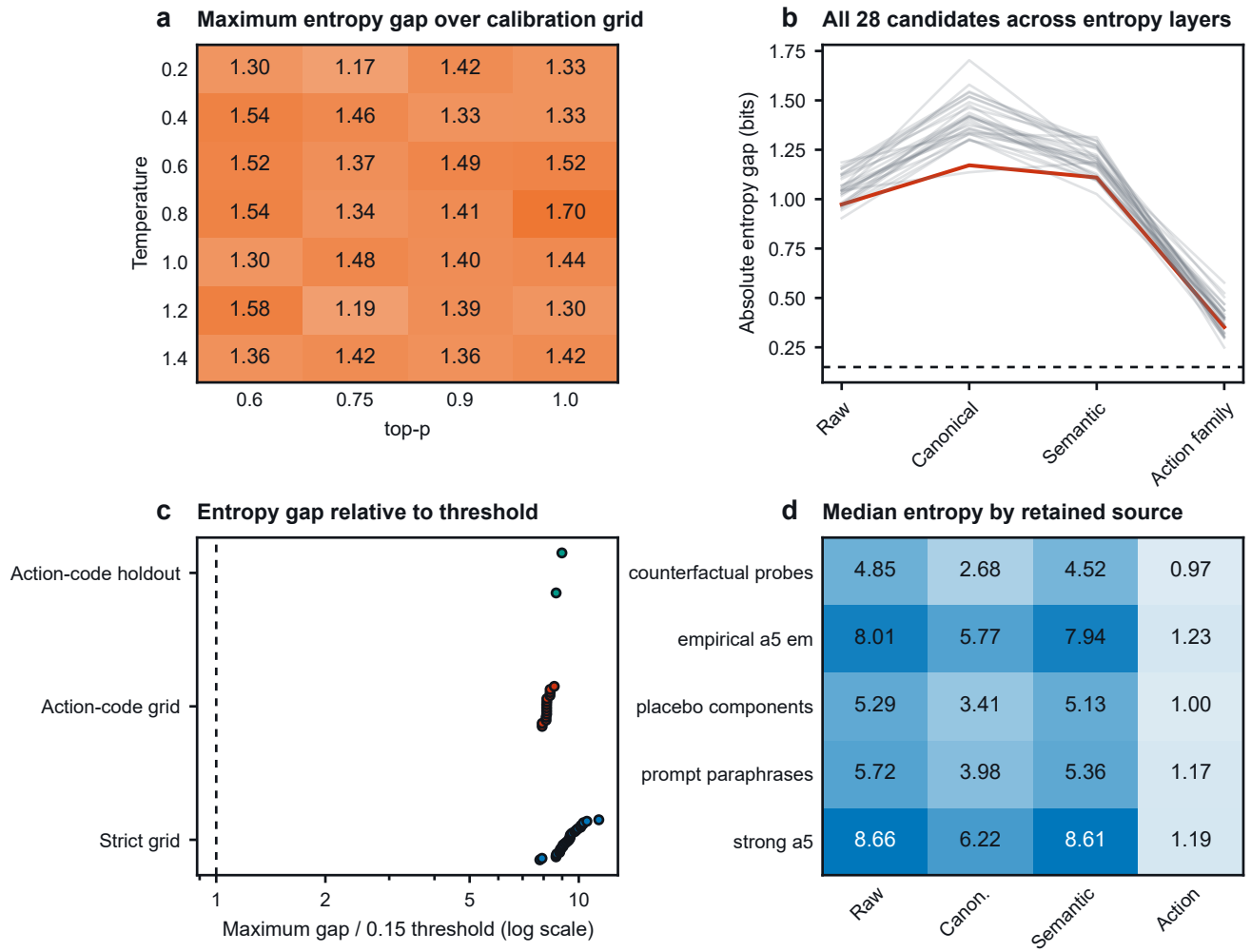


Figure S8: Entropy differences across the calibration search. **a**, Maximum entropy-layer gap for every cell of the 28-candidate strict temperature/top- p grid (bits). **b**, The same candidates' four entropy gaps; the candidate with the smallest gap is red and the dashed line is the 0.15-bit threshold. **c**, All 44 retained candidates across calibration/holdout sources, shown by maximum-gap/threshold ratio on a logarithmic axis. Every ratio exceeds 1; source names identify the calibration and holdout samples; all displayed candidates belong to the documented search. **d**, Median entropy by source and layer (bits), with calibration and holdout sources shown separately. The minimum maximum gap is 1.171 bits, compared with the 0.15-bit threshold.

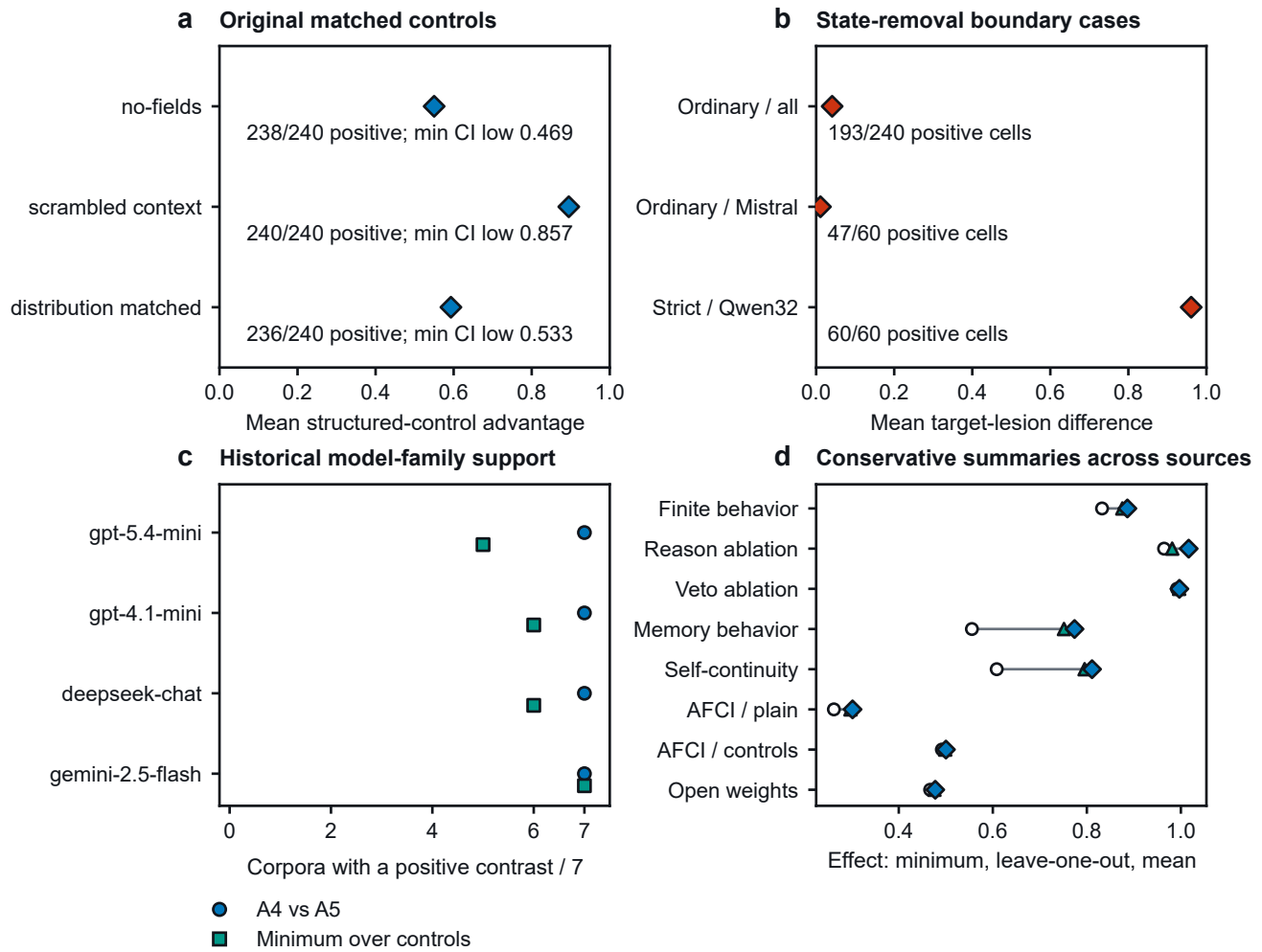


Figure S9: Open-weight control contrasts and corpus sensitivity. **a**, Three original matched controls in 240 model-scaffold-family cells, with mean accuracy differences, positive-cell counts and the minimum lower endpoint across scaffold, model, task-family and model-scaffold-family bootstrap clusters. These endpoints summarize lower limits, not complete intervals. **b**, Ordinary and stricter target-lesion contrasts with their original denominators. **c**, Provider identifiers and positive-corpus counts against A5 and the least favorable control; these are different configurations from the core model cohort. **d**, Source-specific minimum corpus effect (hollow circle), minimum leave-one-corpus-out mean (triangle) and mean (diamond). Each comparison retains its own analysis unit and score scale; values are summarized within their source study.

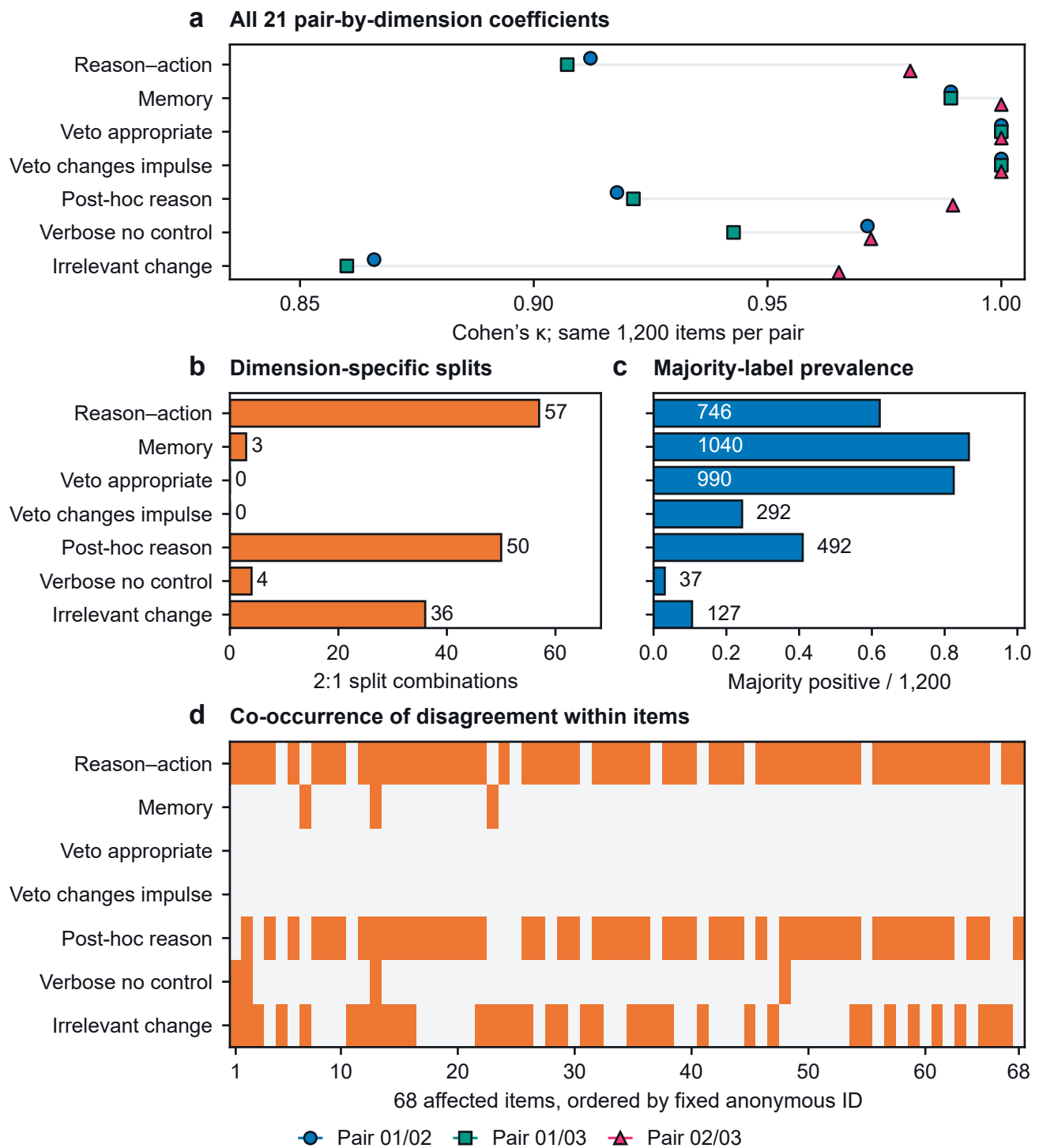


Figure S10: Annotation agreement, prevalence and item-level disagreements. **a**, All 21 pair-by-dimension Cohen's κ values, each based on 1,200 jointly valid binary ratings. Pairs 01/02, 01/03 and 02/03 identify the anonymous annotators; the light range segments are descriptive, not uncertainty intervals. **b**, Two-to-one split counts by dimension. **c**, Majority-positive prevalence, with actual positive counts printed and denominator 1,200. **d**, All 68 affected items in fixed anonymous-ID order; colored cells identify the 150 split combinations among 8,400 item-dimension combinations. The other 1,132 items have no disagreement and remain in the agreement and prevalence denominators. Dimensions differ in direction and are not combined into a quality score. The same annotators and items are reused across dimensions. Supplementary Note 5 specifies sampling, masking and independent scoring.

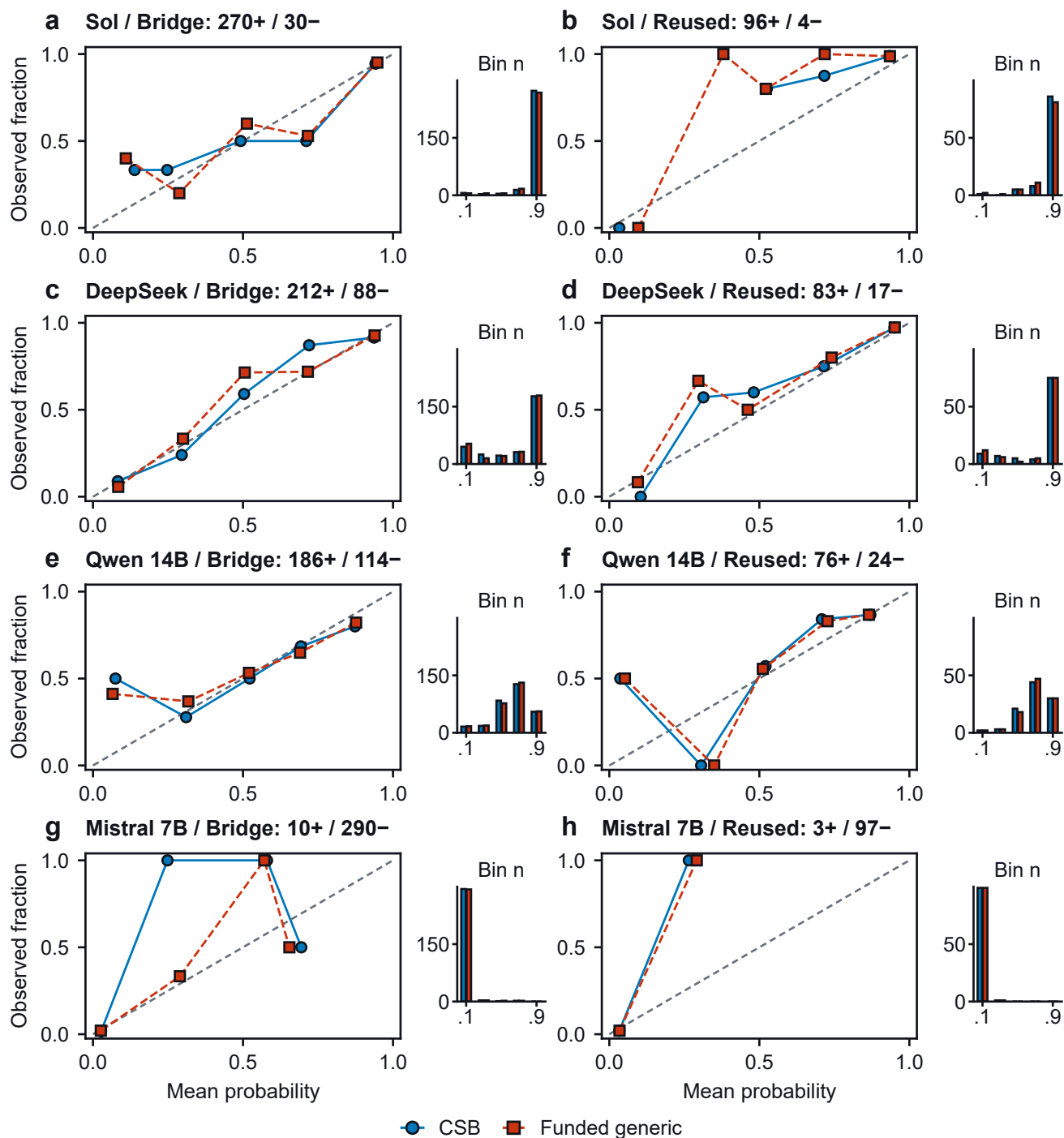


Figure S11: Calibration and support resolved by model. Rows are the four core models; columns are bridge and reused holdout. Blue circles are CSB and red squares funded generic. Calibration uses the same five fixed bins as Supplementary Figure S13; empty bins stay disconnected. The adjacent marginal bars show sample counts in all five bins, left to right from low to high probability, with separate bars per strategy and total denominators 300 or 100. Titles retain positive/negative counts, including Sol's four holdout negatives and Mistral's three positives. The source data include every bin, including empty bins. The plots are descriptive: no calibration curve was fitted, and model records sharing an issue remain statistically dependent.

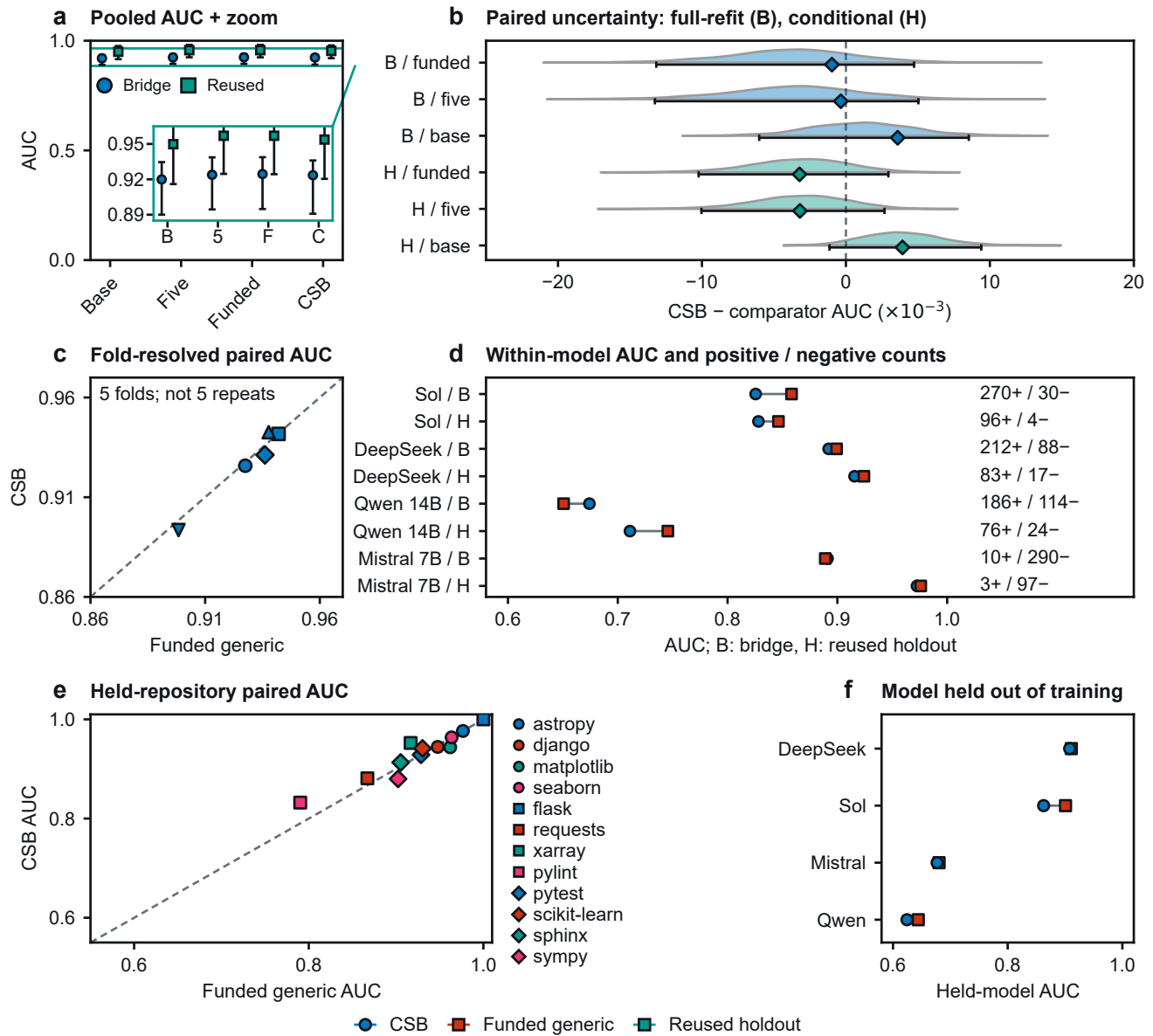


Figure S12: Predictive information from pooled performance to paired and grouped comparisons. **a**, Four strategies on a full AUC scale with an explicitly bounded magnification (B, baseline; 5, five generic probes; F, funded generic; C, CSB). Blue circles denote bridge and teal squares reused holdout in a–b. Intervals in a are marginal percentile intervals from grouped bootstrap draws. **b**, Paired CSB-minus-comparator differences: observed diamonds, two-sided 95% intervals and Scott-bandwidth half-densities of the same draws, with independently normalized heights. Bridge B uses 2,000 full-refit grouped draws and holdout H 5,000 conditional test-group draws. Axis units are 0.001 AUC. **c**, Five cross-validation (CV) fold AUC pairs; folds share training data and are not five independent replications. **d**, Model-resolved AUC and positive/negative counts for both stages. **e**, **f**, Repository- and model-held-out AUC pairs; in e marker size is constant and shape/color identify the held repository. Group counts are supplied in source data. Model exclusion retains other models on the same issues. In c–f blue denotes CSB and red funded generic, except the repository key in e. These are descriptive point estimates. The reused holdout is conditional on the model-cohort amendment.

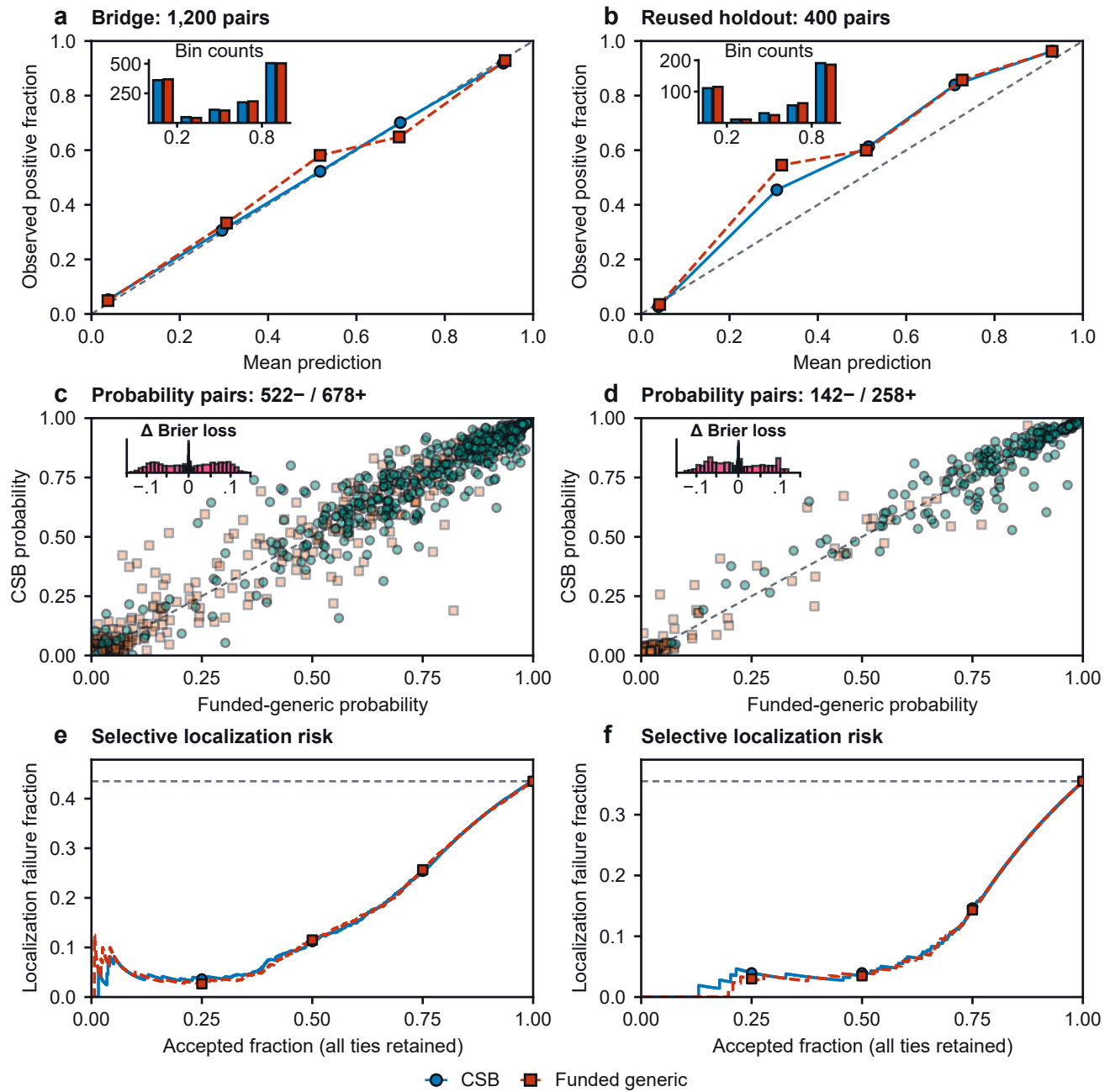


Figure S13: Probability calibration, paired prediction disagreement and selective risk. Columns are the 1,200 bridge model-issue pairs and 400 reused holdout pairs. **a,b**, Empirical positive fraction versus mean probability in five fixed bins, $[0, 0.2), \dots, [0.8, 1]$, with bin-count insets and ideal-calibration diagonals. **c,d**, All matched CSB and funded-generic probabilities, with scorer-side outcome classes shown as orange squares (negative) and teal circles (positive); titles report class counts. Insets show the complete distribution of paired Brier-loss differences, CSB minus funded generic, using signed geometric bins and a symmetric-log axis with linear threshold 0.01. Negative loss differences favor CSB. **e,f**, Descriptive localization failure fraction among predictions retained at every unique saved probability threshold; ties are retained together. Symbols locate the closest available 25/50/75/100% coverage values, and dashed horizontal lines mark all-pair failure fractions. The curves use fixed predictions without recalibration or confidence bands and were not used to select an operating threshold. Model records share issues. SI Figure S11 gives model-specific calibration and the observation counts supporting each bin.

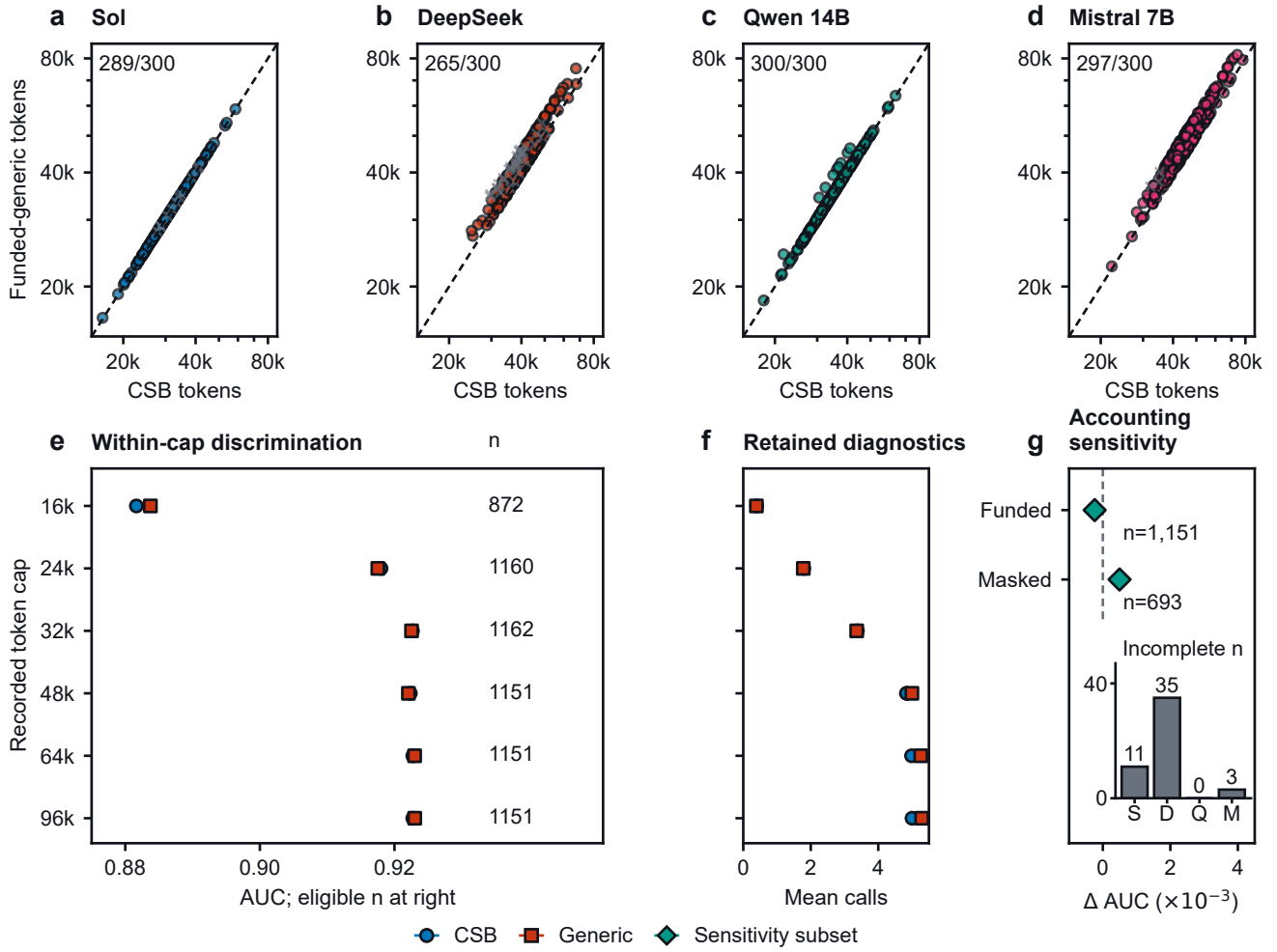


Figure S14: Reported resource use and budget comparisons. **a–d**, All 300 reported-token pairs per model on common log-log axes, with equality diagonals. Colored circles have complete recorded usage and funded-generic tokens at least equal to CSB; gray crosses retain other pairs. Printed fractions give complete-and-funded counts. All plotted pairs have positive reported totals; the flags identify pairs with incomplete metering. **e,f**, Within-cap retrospective AUC comparisons and actual mean retained diagnostic calls. Segments pair strategies within a cap only; the eligible population changes (872, 1,160, 1,162, then 1,151 pairs), so these are not common-cohort scaling curves. Actual overshoot calls remain paid and the specified missing-prefix rules apply. **g**, Selected-subset AUC differences for complete-and-funded accounting ($n = 1,151$) and the conservative prior-interruption mask ($n = 693$), shown descriptively without refitted intervals. Funded and Masked label the complete-and-funded and conservative prior-interruption subsets, respectively. The inset retains all 49 incomplete/unfunded pairs by model (S, Sol; D, DeepSeek; Q, Qwen; M, Mistral); these are accounting categories, not 49 model failures. Resource use is measured in provider-reported tokens. Unmetered usage and the upper bound of 42 potentially unlogged administrative attempts are accounted for separately in Supplementary Note 13.

Table S33: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports.

Canonical variant	Used in	Definition	Export aliases or descriptive names	Role
A4	baseline, behavior, strong controls	structured variant with branching, self-state, memory, reason graph and veto enabled	none	primary structured variant
A5	baseline, behavior, strong controls	stochastic comparator with branching and high-temperature sampling while self/memory/reason/veto are disabled	none	primary stochastic comparator
A4_no_reason	baseline, behavior, robustness	A4 with reason component removed	none	reason-ablation comparator
A4_no_veto	baseline, behavior, robustness	A4 with veto component removed	none	veto-ablation comparator
A4_full	matched-interface AFCI	full structured matched-interface variant	none	positive matched-interface variant verbosity sensitivity
A4_compressed	matched-interface AFCI	compressed positive sensitivity variant excluded from the predefined control-comparator set	none	predefined control comparator
A5_plain	matched-interface AFCI	plain stochastic matched-interface comparator	none	predefined control comparator
A5_schema_random_fields	matched-interface AFCI	A5 matched-interface fields generated as random or template fields	random-field schema A5	predefined control comparator
A5_schema_posthoc_fields	matched-interface AFCI and open-weight reference model	A5 final action generated before post-hoc fields	A5_schema posthoc in post-hoc schema A5; posthoc open-weight runtime exports	predefined control comparator
A5_schema_scrambled_fields	matched-interface AFCI	A5 fields scrambled from another event/action source	scrambled-field schema A5	predefined control comparator
A4_posthoc_fields	matched-interface AFCI	A4 final action retained while fields are generated after final-action selection	A4 post-hoc fields	predefined control comparator
A4_scrambled_fields	matched-interface AFCI	A4 final action retained while fields are paired with wrong event/action information	A4 scrambled fields	predefined control comparator
A5_long_scratchpad	matched-interface AFCI	A5 stochastic comparator with extended scratchpad/field budget	long-scratchpad A5	predefined control comparator
A5_entropy_attempt	strong stochastic controls, entropy diagnostics	lower-temperature stochastic attempt	lower-temperature stochastic-attempt export alias	predefined entropy-closeness criterion unsatisfied
A5_empirical_entropy_attempt	entropy-calibration diagnostic	selected temperature/top-p candidate from empirical calibration diagnostic	calibration candidate aliases A5_attempt_t02_p07 A5_attempt_t020_p075	predefined entropy-closeness criterion unsatisfied

Continued on next page

Table S33: Variant protocol-identifier registry. Canonical labels are raw protocol identifiers retained for reproducibility; alias notes identify export names or descriptive names that appear in derived result exports. (continued)

Canonical variant	Used in	Definition	Export aliases or descriptive names	Role
A5_compute matched	strong stochastic controls	compute-control attempt under the implemented protocol; raw identifier does not indicate strict token/compute matching	none	strong stochastic comparator
A5_self consistency	strong stochastic controls	self-consistency stochastic control under the implemented protocol	none	strong stochastic comparator

Table S34: Mechanistic claim components, operational tests and contrary results.

Table role	Claim component	Operational test	Contrary result	Interpretive boundary
mechanism_map	stochastic contrast	The high-stochasticity comparator increases action unpredictability while disabling self, memory, reason and veto modules.	A strong stochastic control recovers structured-control metrics.	Operational stochastic-substitution test; no claim about consciousness, free will or neural homology.
mechanism_map	matched-interface action-field coupling	All controls use the same reason/memory/self/veto reporting format and are scored by AFCI.	Post-hoc, scrambled, random-field or verbosity controls match the structured matched-interface AFCI.	Tests reporting-format availability under a shared interface.
mechanism_map	finite-action-code behavior	Scorer-side records define hidden expected labels and score only canonical final action codes.	The structured-control variant does not pass behavioral criteria or ablations preserve the full structured-control profile.	Fixed action ontology; does not test unconstrained natural-language decision making.

S1. Agent configurations and source corpora

The implemented family is defined by explicit module toggles and sampling settings. The table reports the operational variant definitions used by the lesion tests.

Table S35: Implemented agent-family variants and operational roles.

Variant	Branch	Self-state	Memory	Reason graph	Veto	Temperature	Role
A0	no	no	no	no	no	0.1	minimal non-structured reference
A1	yes	no	no	no	no	0.2	branching-only reference
A2	yes	yes	no	yes	no	0.2	self plus reason without memory/veto
A3	yes	yes	yes	yes	no	0.2	memory-enabled structured variant without veto
A4	yes	yes	yes	yes	yes	0.2	primary structured variant
A4_no_reason	yes	yes	yes	no	yes	0.2	reason-graph lesion
A4_no_veto	yes	yes	yes	yes	no	0.2	veto lesion
A5	yes	no	no	no	no	0.9	primary stochastic comparator
A5_entropy attempt	yes	no	no	no	no	0.2	lower-temperature stochastic attempt
compute-control attempt	yes	no	no	no	no	0.9	compute-control attempt; raw identifier retained in Table S33
A5_self consistency	yes	no	no	no	no	0.9	multi-sample stochastic control
A5_empirical entropy_attempt	yes	no	no	no	no	1	empirical entropy-attempt comparator

Table S36: Dataset namespaces used as prompt/event corpora.

Dataset namespace	Prompt-corpus role	Construct pressure
narratives_formal	narrative prompt corpus	agency and self-narrative cues
ds000210_self_loop	self/prospection prompt corpus	self-continuity
ds001618_self_loop	autobiographical/self prompt corpus	self-memory loop
ds001882_choice	choice prompt corpus	reason/value conflict
ds000212_choice	choice prompt corpus	reason, veto and agency
ds004917		
uncertainty	uncertainty/risk prompt corpus	stochastic contrast
ds004042_recall	recall/memory prompt corpus	memory constraint

S2. Baseline intervention matrix

The baseline lesion matrix evaluates whether output dispersion and structured-control traces separate under the implemented structured-control/high-stochasticity lesion design. Raw protocol identifiers are retained in the tables for exact export matching. The criterion summary records the realized call count and dataset-direction outcomes.

Table S37: Baseline lesion-matrix criterion summary.

Criterion	Field	Value
baseline lesion matrix	status	pass
baseline lesion matrix	profile	full
baseline lesion matrix	calls	74352
baseline lesion matrix	remote calls	74336
baseline lesion matrix	unrecovered generations	16
baseline lesion matrix	unrecovered-generation rate	0.000215193
baseline lesion matrix	model	gpt-5.4-mini
baseline lesion matrix	fallback model	qwen3-30b-a3b
baseline lesion matrix	structured-control SCI > high-stochasticity datasets	5
baseline lesion matrix	high-stochasticity unpredictability > structured-control datasets	7
baseline lesion matrix	reason-lesion drop datasets	7
baseline lesion matrix	veto-lesion drop datasets	7

Table S38: Baseline trace metrics summarized by variant.

Variant	SCI mean	RSI mean	VEI mean	ACI mean	Unpredictability mean	Distance to structured mean	Rows
A0	0	0.14931	0	9.524e-05	8.073	1.59321	21
A1	0	0.18819	0	0	8.00038	0.476597	21
A2	0.00423593	0.788509	9.524e-05	0	7.99426	0.320232	21
A3	0.0378864	0.537366	0	0.00956706	7.15695	0.172573	21
A4	0.0328108	0.544336	0.0944023	0.014518	6.9708	0	21
A4_no_reason	0.0596636	0.131598	0.0868688	0.0212342	6.62649	0.474022	21
A4_no_veto	0.0355957	0.533555	0	0.0137792	7.0287	0.181687	21
A5	0	0.219036	0	0.000190476	8.05561	0.447575	21

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index.

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds000210_self_loop	1	A0	0	0.328	0	0	8.76825	1.35526
ds000210_self_loop	1	A1	0	0.144	0	0	8.89627	0.442842
ds000210_self_loop	1	A2	0.001	0.648	0	0	8.94578	0.245967
ds000210_self_loop	1	A3	0.0144	0.462	0	0.0106667	8.16901	0.162433
ds000210_self_loop	1	A4	0.007	0.47	0.162	0.006	8.03157	0
ds000210_self_loop	1	A4_no_reason	0.02	0.096	0.166	0.0193333	7.77433	0.393241
ds000210_self_loop	1	A4_no_veto	0.0114	0.412	0	0.00933333	8.38953	0.212166
ds000210_self_loop	1	A5	0	0.218	0	0	8.93627	0.315964
ds000210_self_loop	2	A0	0	0.344	0	0	8.76751	1.21486
ds000210_self_loop	2	A1	0	0.14	0	0	8.87448	0.346922
ds000210_self_loop	2	A2	0.001	0.66	0	0	8.94178	0.356704
ds000210_self_loop	2	A3	0.014	0.466	0	0.006	8.51228	0.249926
ds000210_self_loop	2	A4	0.015	0.416	0.144	0.0106667	7.53976	0
ds000210_self_loop	2	A4_no_reason	0.014	0.064	0.142	0.024	7.86176	0.367291
ds000210_self_loop	2	A4_no_veto	0.01	0.47	0	0.0113333	8.16759	0.213498
ds000210_self_loop	2	A5	0	0.226	0	0	8.95378	0.292231
ds000210_self_loop	3	A0	0	0.36	0	0	8.83731	1.32969
ds000210_self_loop	3	A1	0	0.19	0	0	8.88506	0.397445
ds000210_self_loop	3	A2	0.001	0.688	0	0	8.91476	0.330203
ds000210_self_loop	3	A3	0.013	0.468	0	0.0113333	8.46181	0.179912
ds000210_self_loop	3	A4	0.009	0.462	0.176	0.00466667	8.33933	0
ds000210_self_loop	3	A4_no_reason	0.011	0.078	0.122	0.034	7.85439	0.408794
ds000210_self_loop	3	A4_no_veto	0.016	0.508	0	0.0113333	8.52001	0.18609
ds000210_self_loop	3	A5	0	0.176	0	0	8.91476	0.35543
ds000212_choice	1	A0	0	0.296	0	0	8.72665	1.15236
ds000212_choice	1	A1	0	0.102	0	0	8.7556	0.657512
ds000212_choice	1	A2	0	0.876	0	0	8.71947	0.242358
ds000212_choice	1	A3	0.0054	0.434	0	0.011	7.07074	0.245605
ds000212_choice	1	A4	0.00252	0.656	0.028	0.0266667	7.85523	0
ds000212_choice	1	A4_no_reason	0.00108	0.012	0.006	0.0606667	7.55452	0.680466
ds000212_choice	1	A4_no_veto	0.00396	0.526	0	0.0233333	7.81526	0.146769
ds000212_choice	1	A5	0	0.116	0	0	8.77971	0.541392
ds000212_choice	2	A0	0	0.246	0	0	8.7878	1.18602
ds000212_choice	2	A1	0	0.086	0	0	8.69956	0.724426
ds000212_choice	2	A2	0.00036	0.884	0	0	8.76596	0.313144
ds000212_choice	2	A3	0.00216	0.492	0	0.018	7.55397	0.172383

Continued on next page

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds000212_choice	2	A4	0.00216	0.634	0.032	0.026	7.80838	0
ds000212_choice	2	A4_no_reason	0.0036	0.01	0.012	0.038	7.2857	0.671147
ds000212_choice	2	A4_no_veto	0.00252	0.498	0	0.0124	7.31826	0.166749
ds000212_choice	2	A5	0	0.098	0	0	8.79493	0.53777
ds000212_choice	3	A0	0	0.296	0	0	8.83555	1.0744
ds000212_choice	3	A1	0	0.09	0	0	8.6902	0.591411
ds000212_choice	3	A2	0	0.868	0	0	8.75269	0.397617
ds000212_choice	3	A3	0.00144	0.63	0	0.0253333	7.79394	0.119062
ds000212_choice	3	A4	0.00288	0.52	0.024	0.0326667	7.53217	0
ds000212_choice	3	A4_no_reason	0.00144	0.004	0.006	0.0653333	7.25292	0.541525
ds000212_choice	3	A4_no_veto	0.0036	0.618	0	0.0213333	7.65724	0.168122
ds000212_choice	3	A5	0	0.116	0	0	8.77164	0.406437
ds001618_self_loop	1	A0	0	0.112	0	0	8.47286	1.35064
ds001618_self_loop	1	A1	0	0.19	0	0	8.61063	0.428248
ds001618_self_loop	1	A2	0	0.72	0.002	0	8.77191	0.320151
ds001618_self_loop	1	A3	0	0.362	0	0.005	7.81575	0.274838
ds001618_self_loop	1	A4	0	0.444	0.154	0.0173333	7.52728	0
ds001618_self_loop	1	A4_no_reason	0	0.064	0.204	0.0133333	6.53898	0.383927
ds001618_self_loop	1	A4_no_veto	0	0.366	0	0.012	7.66101	0.317151
ds001618_self_loop	1	A5	0	0.246	0	0	8.67665	0.271338
ds001618_self_loop	2	A0	0	0.144	0	0	8.52178	1.36285
ds001618_self_loop	2	A1	0	0.2	0	0	8.56596	0.487458
ds001618_self_loop	2	A2	0	0.694	0	0	8.77412	0.33743
ds001618_self_loop	2	A3	0	0.368	0	0.0085	7.72458	0.17737
ds001618_self_loop	2	A4	0	0.444	0.14	0.00866667	7.73685	0
ds001618_self_loop	2	A4_no_reason	0	0.076	0.122	0.026	6.41047	0.402855
ds001618_self_loop	2	A4_no_veto	0	0.368	0	0.012	6.46868	0.166994
ds001618_self_loop	2	A5	0	0.248	0	0	8.63753	0.298441
ds001618_self_loop	3	A0	0	0.106	0	0	8.43722	1.43727
ds001618_self_loop	3	A1	0	0.184	0	0	8.45131	0.580472
ds001618_self_loop	3	A2	0	0.596	0	0	8.76532	0.413091
ds001618_self_loop	3	A3	0	0.396	0	0.00333333	7.68315	0.226955
ds001618_self_loop	3	A4	0	0.306	0.208	0.004	6.70393	0
ds001618_self_loop	3	A4_no_reason	0	0.058	0.11	0.014	6.39787	0.334682
ds001618_self_loop	3	A4_no_veto	0	0.388	0	0.0106667	7.50181	0.226884
ds001618_self_loop	3	A5	0	0.282	0	0	8.66935	0.263545
ds001882_choice	1	A0	0	0.004	0	0	6.52926	1.94798
ds001882_choice	1	A1	0	0.004	0	0	5.99445	0.479681
ds001882_choice	1	A2	0.008	0.75	0	0	5.55912	0.361726
ds001882_choice	1	A3	0.162	0.396	0	0.008	3.81626	0.118239
ds001882_choice	1	A4	0.152	0.446	0.106	0.0193333	3.72608	0
ds001882_choice	1	A4_no_reason	0.27	0	0.08	0.002	2.82488	0.462926
ds001882_choice	1	A4_no_veto	0.12	0.424	0	0.03	4.03051	0.113551
ds001882_choice	1	A5	0	0	0	0	5.9395	0.998752
ds001882_choice	2	A0	0	0.002	0	0	6.40637	1.95812
ds001882_choice	2	A1	0	0.002	0	0	5.64805	0.476378
ds001882_choice	2	A2	0.007	0.768	0	0	5.77716	0.363858
ds001882_choice	2	A3	0.141	0.468	0	0.015	4.70427	0.125849
ds001882_choice	2	A4	0.104	0.448	0.118	0.024	4.15845	0
ds001882_choice	2	A4_no_reason	0.312	0.002	0.114	0.006	2.74187	0.492869
ds001882_choice	2	A4_no_veto	0.151	0.404	0	0.0233333	3.91573	0.13466
ds001882_choice	2	A5	0	0.004	0	0	6.24003	0.696129
ds001882_choice	3	A0	0	0	0	0	6.52098	1.92553
ds001882_choice	3	A1	0	0.006	0	0	5.86191	0.406223
ds001882_choice	3	A2	0.02	0.756	0	0	5.22555	0.402404

Continued on next page

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds001882_choice	3	A3	0.145	0.45	0	0.012	3.74179	0.108793
ds001882_choice	3	A4	0.109	0.384	0.078	0.018	2.89757	0
ds001882_choice	3	A4_no_reason	0.294	0	0.082	0.0113333	2.74725	0.426429
ds001882_choice	3	A4_no_veto	0.162	0.414	0	0.026	3.36946	0.0994636
ds001882_choice	3	A5	0	0.002	0	0	5.69531	0.945267
ds004042_recall	1	A0	0	0.03	0	0	8.91725	1.92048
ds004042_recall	1	A1	0	0.192	0	0	8.93057	0.551628
ds004042_recall	1	A2	0.012952	0.902	0	0	8.95378	0.277931
ds004042_recall	1	A3	0.100937	0.65	0	0	8.3718	0.14564
ds004042_recall	1	A4	0.103603	0.682	0.14	0.002	8.83922	0
ds004042_recall	1	A4_no_reason	0.0956766	0.194	0.14	0.006	8.54148	0.496309
ds004042_recall	1	A4_no_veto	0.0962956	0.66	0	0.002	8.82551	0.16193
ds004042_recall	1	A5	0	0.244	0	0	8.94578	0.471786
ds004042_recall	2	A0	0	0.026	0	0.002	8.90648	1.95812
ds004042_recall	2	A1	0	0.218	0	0	8.95227	0.562195
ds004042_recall	2	A2	0.0092848	0.87	0	0	8.95778	0.227454
ds004042_recall	2	A3	0.091722	0.72	0	0	8.88925	0.131892
ds004042_recall	2	A4	0.0748846	0.698	0.116	0.006	8.79395	0
ds004042_recall	2	A4_no_reason	0.118549	0.232	0.17	0.00266667	8.74416	0.477901
ds004042_recall	2	A4_no_veto	0.0741564	0.688	0	0.006	8.71487	0.121295
ds004042_recall	2	A5	0	0.276	0	0.002	8.91306	0.446837
ds004042_recall	3	A0	0	0.032	0	0	8.91048	1.941
ds004042_recall	3	A1	0	0.226	0	0	8.94027	0.492784
ds004042_recall	3	A2	0.0276378	0.904	0	0	8.94427	0.287693
ds004042_recall	3	A3	0.0847924	0.724	0	0.01	8.82833	0.144852
ds004042_recall	3	A4	0.0965382	0.66	0.122	0	8.87975	0
ds004042_recall	3	A4_no_reason	0.0786418	0.216	0.11	0.01	8.62266	0.458524
ds004042_recall	3	A4_no_veto	0.0886568	0.698	0	0.006	8.83318	0.133679
ds004042_recall	3	A5	0	0.262	0	0.002	8.92978	0.428004
uncertainty ds004917	1	A0	0	0.018	0	0	8.62299	2.0015
uncertainty ds004917	1	A1	0	0.338	0	0	8.63624	0.400878
uncertainty ds004917	1	A2	0.00036	0.912	0	0	8.43571	0.494227
uncertainty ds004917	1	A3	0.00324	0.682	0	0.013	8.09108	0.161046
uncertainty ds004917	1	A4	0.00432	0.692	0.046	0.0113333	7.01934	0
uncertainty ds004917	1	A4_no_reason	0.00288	0.076	0.048	0.0246667	7.65113	0.616474
uncertainty ds004917	1	A4_no_veto	0.00324	0.692	0	0.0088	7.91809	0.0461257
uncertainty ds004917	1	A5	0	0.382	0	0	8.54235	0.455262
uncertainty ds004917	2	A0	0	0.012	0	0	8.51087	2.13843
uncertainty ds004917	2	A1	0	0.336	0	0	8.40745	0.387633
uncertainty ds004917	2	A2	0.00036	0.912	0	0	8.47606	0.382314
uncertainty ds004917	2	A3	0.00468	0.69	0	0.006	7.58568	0.0960863
uncertainty ds004917	2	A4	0.00432	0.712	0.048	0.0126667	7.65424	0
uncertainty ds004917	2	A4_no_reason	0.00324	0.086	0.024	0.0226667	7.29729	0.667423
uncertainty ds004917	2	A4_no_veto	0.00252	0.73	0	0.012	7.56667	0.117761

Continued on next page

Table S39: Baseline trace metrics by dataset namespace, variant and replicate index. (continued)

Dataset	Replicate index	Variant	SCI	RSI	VEI	ACI	Unpredictability	Distance to structured
ds004917								
uncertainty	2	A5	0	0.392	0	0	8.65413	0.391514
ds004917								
uncertainty	3	A0	0	0.004	0	0	8.60747	2.05311
ds004917								
uncertainty	3	A1	0	0.304	0	0	8.61613	0.414245
ds004917								
uncertainty	3	A2	0	0.916	0	0	8.47654	0.331396
ds004917								
uncertainty	3	A3	0.00324	0.692	0	0.0173333	7.85269	0.0245462
ds004917								
uncertainty	3	A4	0.0018	0.704	0.018	0.006	7.49391	0
ds004917								
uncertainty	3	A4_no_reason	0.00828	0.118	0.054	0.02	7.4201	0.64978
ds004917								
uncertainty	3	A4_no_veto	0.00216	0.708	0	0.01	7.58817	0.151182
ds004917								
uncertainty	3	A5	0	0.424	0	0	8.54861	0.447065
narratives_formal	1	A0	0	0.265306	0	0	6.49726	1.28397
narratives_formal	1	A1	0	0.346939	0	0	6.53308	0.387361
narratives_formal	1	A2	0	0.72449	0	0	6.5943	0.221234
narratives_formal	1	A3	0.00742347	0.530612	0	0.0102041	6.02814	0.220534
narratives_formal	1	A4	0	0.530612	0.0510204	0.0178571	6.23594	0
narratives_formal	1	A4_no_reason	0	0.418367	0.0510204	0.0153061	5.56579	0.33566
narratives_formal	1	A4_no_veto	0	0.632653	0	0.0244898	5.32241	0.317218
narratives_formal	1	A5	0	0.316327	0	0	6.54578	0.247869
narratives_formal	2	A0	0	0.214286	0	0	6.46415	1.22598
narratives_formal	2	A1	0	0.387755	0	0	6.48456	0.349778
narratives_formal	2	A2	0	0.785714	0	0	6.5943	0.260554
narratives_formal	2	A3	0	0.581633	0	0	5.92067	0.167048
narratives_formal	2	A4	0	0.530612	0.0306122	0.0306122	5.91899	0
narratives_formal	2	A4_no_reason	0.00707041	0.540816	0.0408163	0	6.05625	0.137085
narratives_formal	2	A4_no_veto	0	0.510204	0	0	6.10536	0.269589
narratives_formal	2	A5	0	0.336735	0	0	6.52537	0.218853
narratives_formal	3	A0	0	0.295918	0	0	6.48456	1.63981
narratives_formal	3	A1	0	0.265306	0	0	6.57389	0.443026
narratives_formal	3	A2	0	0.72449	0	0	6.53308	0.157421
narratives_formal	3	A3	0.00117857	0.622449	0	0.0102041	5.68077	0.371015
narratives_formal	3	A4	0	0.591837	0.0408163	0.0204082	5.69492	0
narratives_formal	3	A4_no_reason	0.0114776	0.418367	0.0204082	0.0306122	6.01247	0.549153
narratives_formal	3	A4_no_veto	0	0.489796	0	0.0170068	5.91339	0.344546
narratives_formal	3	A5	0	0.234694	0	0	6.55349	0.369185

S3. Action-field coupling under a common reporting interface

Matched-interface controls required comparator variants to use the same reporting format and then scored action-field coupling. AFCI is an interface-level analysis that uses reported content, final action and predefined field-source information; the finite-action-code benchmark supplies the reporting-format-independent behavioral test. The separate output-only analyses in Supplementary Note 3 assess coverage and consistency using visible content. The compressed structured-control sensitivity variant (**A4_compressed**) is reported as a positive sensitivity variant excluded from the seven predefined control comparators: **A5_plain**, **A5_schema_random_fields**, **A5_schema_posthoc_fields**, **A5_schema_scrambled_fields**, **A4_posthoc_fields**, **A4_scrambled_fields** and **A5_long_scratchpad**. The self-action AFCI term is sparse in the matched-interface trace scorer. The finite-action-code B_{SCI} score is defined separately over final action-code stability under the predefined self-continuity probes.

Table S40: Matched-interface control criterion summary.

Criterion	Field	Value
matched-interface controls	status	pass
matched-interface controls	dataset_count	7
matched-interface controls	control_count	7
matched-interface controls	min_a4_afci_gt_control_dataset_count	7
matched-interface controls	cluster_bootstrap_afci_delta_low95	0.451911
matched-interface controls	call_count	26946
matched-interface controls	remote calls	26946
matched-interface controls	unrecovered_count	0
matched-interface controls	unrecovered_rate	0
matched-interface controls	planned calls	26946
matched-interface controls	model	gpt-5.4-mini

Table S41: Predefined control-comparator recovery status for matched-interface AFCI.

Comparator variant	Datasets	Structured AFCI > comparator datasets	Comparator recovered structured datasets
A5_plain	7	7	0
A5_schema_random_fields	7	7	0
A5_schema_posthoc_fields	7	7	0
A5_schema_scrambled_fields	7	7	0
A4_posthoc_fields	7	7	0
A4_scrambled_fields	7	7	0
A5_long_scratchpad	7	7	0

Table S42: AFCI component means by matched-interface variant. A4_compressed is a compressed positive sensitivity variant excluded from the seven predefined control comparators.

Variant	Reason alignment mean	Memory alignment mean	Veto alignment mean	Self alignment mean	Irrelevant suppression mean	AFCI mean	AFCI + suppression mean	Rows
A4_compressed	0.491286	0.502222	1	0.00412698	1	0.499409	0.599527	21
A4_full	0.494461	0.503968	1	0.00285714	1	0.500322	0.600257	21
A4_posthoc								
fields	0	0	0	0	0	0	0	21
A4_scrambled								
fields	0	0	0	0	0	0	0	21
A5_long								
scratchpad	0	0	0	0	0	0	0	21
A5_plain	0.069306	0.225474	0.5	0	0	0.198695	0.158956	21
A5_schema								
posthoc_fields	0	0	0	0	0	0	0	21
A5_schema								
random_fields	0	0	0	0	0	0	0	21
A5_schema								
scrambled_fields	0	0	0	0	0	0	0	21

Table S43: Dataset-level AFCI contrasts against the structured matched-interface variant (A4_full).

Dataset	Comparator variant	Comparator AFCI	Structured AFCI	Structured minus comparator
ds000210_self_loop	A4_compressed	0.488889	0.494167	0.00527778
ds000210_self_loop	A4_posthoc			
ds000210_self_loop	fields	0	0.494167	0.494167
ds000210_self_loop	A4_scrambled			
ds000210_self_loop	fields	0	0.494167	0.494167
ds000210_self_loop	A5_long			
ds000210_self_loop	scratchpad	0	0.494167	0.494167
ds000210_self_loop	A5_plain	0.178611	0.494167	0.315556
ds000210_self_loop	A5_schema			
ds000210_self_loop	posthoc_fields	0	0.494167	0.494167
ds000210_self_loop	A5_schema			
ds000210_self_loop	random_fields	0	0.494167	0.494167
ds000210_self_loop	A5_schema			
ds000210_self_loop	scrambled_fields	0	0.494167	0.494167
ds000212_choice	A4_compressed	0.496528	0.503472	0.00694444
ds000212_choice	A4_posthoc			
ds000212_choice	fields	0	0.503472	0.503472
ds000212_choice	A4_scrambled			
ds000212_choice	fields	0	0.503472	0.503472
ds000212_choice	A5_long			
ds000212_choice	scratchpad	0	0.503472	0.503472
ds000212_choice	A5_plain	0.176111	0.503472	0.327361
ds000212_choice	A5_schema			
ds000212_choice	posthoc_fields	0	0.503472	0.503472
ds000212_choice	A5_schema			
ds000212_choice	random_fields	0	0.503472	0.503472
ds000212_choice	A5_schema			
ds000212_choice	scrambled_fields	0	0.503472	0.503472
ds001618_self_loop	A4_compressed	0.495694	0.491667	-0.00402778
ds001618_self_loop	A4_posthoc			
ds001618_self_loop	fields	0	0.491667	0.491667
ds001618_self_loop	A4_scrambled			
ds001618_self_loop	fields	0	0.491667	0.491667
ds001618_self_loop	A5_long			
ds001618_self_loop	scratchpad	0	0.491667	0.491667
ds001618_self_loop	A5_plain	0.192708	0.491667	0.298958
ds001618_self_loop	A5_schema			
ds001618_self_loop	posthoc_fields	0	0.491667	0.491667

Continued on next page

Table S43: Dataset-level AFCI contrasts against the structured matched-interface variant (A4_full). (continued)

Dataset	Comparator variant	Comparator AFCI	Structured AFCI	Structured minus comparator
ds001618_self_loop	A5_schema	0	0.491667	0.491667
ds001618_self_loop	random_fields	0	0.491667	0.491667
ds001882_choice	A5_schema	0	0.491667	0.491667
ds001882_choice	scrambled_fields	0.514861	0.506667	-0.00819444
ds001882_choice	A4_compressed	0	0.506667	0.506667
ds001882_choice	A4_posthoc	0	0.506667	0.506667
ds001882_choice	fields	0	0.506667	0.506667
ds001882_choice	A4_scrambled	0	0.506667	0.506667
ds001882_choice	fields	0	0.506667	0.506667
ds001882_choice	A5_long	0	0.506667	0.506667
ds001882_choice	scratchpad	0	0.506667	0.506667
ds001882_choice	A5_plain	0.191181	0.506667	0.315486
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	posthoc_fields	0	0.506667	0.506667
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	random_fields	0	0.506667	0.506667
ds001882_choice	A5_schema	0	0.506667	0.506667
ds001882_choice	scrambled_fields	0	0.506667	0.506667
ds004042_recall	A4_compressed	0.508889	0.510417	0.00152778
ds004042_recall	A4_posthoc	0	0.510417	0.510417
ds004042_recall	fields	0	0.510417	0.510417
ds004042_recall	A4_scrambled	0	0.510417	0.510417
ds004042_recall	fields	0	0.510417	0.510417
ds004042_recall	A5_long	0	0.510417	0.510417
ds004042_recall	scratchpad	0	0.510417	0.510417
ds004042_recall	A5_plain	0.236944	0.510417	0.273472
ds004042_recall	A5_schema	0	0.510417	0.510417
ds004042_recall	posthoc_fields	0	0.510417	0.510417
ds004042_recall	A5_schema	0	0.510417	0.510417
ds004042_recall	random_fields	0	0.510417	0.510417
ds004042_recall	A5_schema	0	0.510417	0.510417
ds004042_recall	scrambled_fields	0	0.510417	0.510417
ds004917	A4_compressed	0.496528	0.501389	0.00486111
uncertainty	A4_posthoc	0	0.501389	0.501389
ds004917	fields	0	0.501389	0.501389
uncertainty	A4_scrambled	0	0.501389	0.501389
ds004917	fields	0	0.501389	0.501389
uncertainty	A5_long	0	0.501389	0.501389
ds004917	scratchpad	0	0.501389	0.501389
uncertainty	A5_plain	0.182847	0.501389	0.318542
ds004917	A5_schema	0	0.501389	0.501389
uncertainty	posthoc_fields	0	0.501389	0.501389
ds004917	A5_schema	0	0.501389	0.501389
uncertainty	random_fields	0	0.501389	0.501389
ds004917	A5_schema	0	0.501389	0.501389
uncertainty	scrambled_fields	0	0.501389	0.501389
narratives_formal	A4_compressed	0.494473	0.494473	0
narratives_formal	A4_posthoc	0	0.494473	0.494473
narratives_formal	fields	0	0.494473	0.494473
narratives_formal	A4_scrambled	0	0.494473	0.494473
narratives_formal	fields	0	0.494473	0.494473
narratives_formal	A5_long	0	0.494473	0.494473
narratives_formal	scratchpad	0	0.494473	0.494473
narratives_formal	A5_plain	0.232462	0.494473	0.262011
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	posthoc_fields	0	0.494473	0.494473
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	random_fields	0	0.494473	0.494473
narratives_formal	A5_schema	0	0.494473	0.494473
narratives_formal	scrambled_fields	0	0.494473	0.494473

S4. Finite-action responses

The finite-action-code benchmark scores canonical final-action behavior under the finite action alphabet. Expected labels are scorer-side rule annotations derived from the condition map; the model-visible prompt specifies the task context, manipulation, module-lesion rule and allowed action ontology but not the hidden expected label used for scoring.

Table S44: Finite-action-code ontology and scorer-side expected labels.

Probe condition	Valid action codes	Expected before	Expected after	Interpretation
baseline	ACTION_A; ACTION_B	ACTION_A	ACTION_A	No specialized component is required; correct behavior is to keep ACTION_A.
reason_flip	ACTION_A; ACTION_B	ACTION_A	ACTION_B	The decisive reason is moved from A to B; reason-enabled variants should change.
memory_conflict	ACTION_A; RECALL_PRIOR; DEFER	ACTION_A	RECALL_PRIOR	A prior commitment conflicts with the immediate action impulse.
veto_cue	ACTION_A; VETO; DEFER	ACTION_A	VETO	A hard stop or constraint cue invalidates the first impulse.
self_continuity	ACTION_A; ACTION_B; DEFER	ACTION_A	ACTION_A	A persistent self-state or long-term commitment supports continuity.
irrelevant_cue	ACTION_A; ACTION_B	ACTION_A	ACTION_A	An irrelevant cue should not change the action.

Table S45: Finite-action-code behavioral benchmark criterion summary.

Criterion	Field	Value
finite-action-code behavioral benchmark	status	pass
finite-action-code behavioral benchmark	dataset_count	7
finite-action-code behavioral benchmark	scored records	57816
finite-action-code behavioral benchmark	source_scored records	61056
finite-action-code behavioral benchmark	planned calls	58896
finite-action-code behavioral benchmark	remote calls	58835
finite-action-code behavioral benchmark	unrecovered generations	1
finite-action-code behavioral benchmark	parse-error rate	1.700e-05
finite-action-code behavioral benchmark	unmapped action rate	0.00129722
finite-action-code behavioral benchmark	structured-control irrelevant false-positive rate	0.0083022
finite-action-code behavioral benchmark	structured-control behavioral composite > high-stochasticity datasets	7
finite-action-code behavioral benchmark	structured-control B_RSI > reason-lesion datasets	7
finite-action-code behavioral benchmark	structured-control B_VEI > veto-lesion datasets	7
finite-action-code behavioral benchmark	structured-control B_MCI > high-stochasticity datasets	7
finite-action-code behavioral benchmark	structured-control B_SCI > high-stochasticity datasets	7
finite-action-code behavioral benchmark	dataset-cluster bootstrap lower endpoint of the two-sided 95% interval	0.855732

Table S46: Behavioral component scores by variant; means and standard deviations.

Variant	Statistic	RSI	MCI	VEI	SCI	Cue	Composite	FP	n
A4	Mean	0.973	0.773	0.989	0.990	0.931	0.931	0.008	7
A4	SD	0.023	0.151	0.019	0.021	0.037	0.037	0.019	7
A4_no_reason	Mean	-0.044	0.788	0.954	0.954	0.663	0.663	0.044	7
A4_no_reason	SD	0.113	0.089	0.112	0.113	0.102	0.102	0.113	7
A4_no_veto	Mean	0.977	0.801	-0.008	0.990	0.690	0.690	0.009	7
A4_no_veto	SD	0.019	0.081	0.019	0.021	0.020	0.020	0.019	7
A5	Mean	0.002	-0.001	-0.001	0.179	0.045	0.045	0.001	7
A5	SD	0.004	0.001	0.001	0.101	0.025	0.025	0.001	7

Table S47: Behavioral component scores by dataset namespace and variant.

Dataset	Variant	B_RSI	B_MCI	B_VEI	B_SCI	Cue	Composite	FP
ds000210_self_loop	A4	0.969444	0.808333	0.988889	0.994444	0.940278	0.940278	0.00277778
ds000210_self_loop	A4_no_reason	0	0.813889	0.997222	0.997222	0.702083	0.702083	0
ds000210_self_loop	A4_no_veto	0.969444	0.788889	-0.00277778	0.994444	0.6875	0.6875	0.00277778
ds000210_self_loop	A5	0	0	0	0.166667	0.0416667	0.0416667	0
ds000212_choice	A4	0.977778	0.555556	0.997222	1	0.882639	0.882639	0
ds000212_choice	A4_no_reason	-0.00277778	0.738889	0.994444	0.997222	0.681944	0.681944	0.00277778
ds000212_choice	A4_no_veto	0.988889	0.672222	0	1	0.665278	0.665278	0
ds000212_choice	A5	0	0	0	0.161111	0.0402778	0.0402778	0
ds001618_self_loop	A4	0.983333	0.816667	0.997222	1	0.949306	0.949306	0
ds001618_self_loop	A4_no_reason	-0.00277778	0.863889	0.997222	0.988889	0.711806	0.711806	0.00277778
ds001618_self_loop	A4_no_veto	0.983333	0.869444	0	1	0.713194	0.713194	0
ds001618_self_loop	A5	0.00277778	0	0	0.391667	0.0986111	0.0986111	0
ds001882_choice	A4	0.997222	0.852778	0.994444	0.997222	0.960417	0.960417	0.00277778
ds001882_choice	A4_no_reason	0	0.836111	1	1	0.709028	0.709028	0
ds001882_choice	A4_no_veto	1	0.847222	0	1	0.711806	0.711806	0
ds001882_choice	A5	0	0	0	0.0916667	0.0229167	0.0229167	0
ds004042_recall	A4	0.927171	0.907563	0.946779	0.943978	0.931373	0.931373	0.0504202
ds004042_recall	A4_no_reason	-0.29972	0.635854	0.70028	0.697479	0.433473	0.433473	0.29972
ds004042_recall	A4_no_veto	0.938375	0.879552	-0.0504202	0.943978	0.677871	0.677871	0.0504202
ds004042_recall	A5	0	0	0	0.159664	0.039916	0.039916	0
ds004917_uncertainty	A4	0.991667	0.908333	1	0.997222	0.974306	0.974306	0
ds004917_uncertainty	A4_no_reason	0	0.891667	1	1	0.722917	0.722917	0

Continued on next page

Table S47: Behavioral component scores by dataset namespace and variant.
(continued)

Dataset	Variant	B_RSI	B_MCI	B_VEI	B_SCI	Cue	Composite	FP
ds004917 uncertainty	A4_no_veto	0.980556	0.838889	0	0.997222	0.704167	0.704167	0
ds004917 uncertainty	A5	0	0	0	0.0944444	0.0236111	0.0236111	0
narratives formal	A4	0.964286	0.563492	0.996032	1	0.880952	0.880952	0
narratives formal	A4_no_reason	0	0.738095	0.988095	1	0.681548	0.681548	0
narratives formal	A4_no_veto	0.97619	0.710317	-0.00396825	0.996032	0.669643	0.669643	0.00396825
narratives formal	A5	0.0119048	-0.00396825	-0.00396825	0.190476	0.0486111	0.0486111	0.00396825

Table S48: Predefined dataset-direction behavioral criterion contrasts.

Dataset	structured comp.	reason-lesion comp.	veto-lesion comp.	HS comp.	structured- HS comp.	reason contrast	veto contrast	memory contrast	self contrast
ds000210_self_loop	0.940278	0.702083	0.6875	0.0416667	0.898611	0.969444	0.991667	0.808333	0.827778
ds000212_choice	0.882639	0.681944	0.665278	0.0402778	0.842361	0.980556	0.997222	0.555556	0.838889
ds001618_self_loop	0.949306	0.711806	0.713194	0.0986111	0.850694	0.986111	0.997222	0.816667	0.608333
ds001882_choice	0.960417	0.709028	0.711806	0.0229167	0.9375	0.997222	0.994444	0.852778	0.905556
ds004042_recall	0.931373	0.433473	0.677871	0.039916	0.891457	1.22689	0.997199	0.907563	0.784314
ds004917 uncertainty	0.974	0.723	0.704	0.024	0.951	0.992	1	0.908	0.903
narratives formal	0.880952	0.681548	0.669643	0.0486111	0.832341	0.964286	1	0.56746	0.809524

S5. Stochastic controls and entropy

Strong stochastic controls test whether alternative stochastic implementations recover the structured-control profile. Entropy diagnostics summarize action variability under raw strings, rule-canonical actions, semantic clusters and action-family mappings. The calibration tables report the maximum entropy gaps alongside the predefined matching threshold.

Table S49: Proxy token/compute accounting summary. Provider-side prompt-token, completion-token and total-token usage were not uniformly retained; this table reports availability and proxy measurements for length and structure.

Audit item	Measurement or proxy	Availability or value	Evidence source	Interpretation
provider token usage	prompt_tokens, completion_tokens, total_tokens	not uniformly retained	token_verbosity_audit criterion summary reports provider token usage unavailable	provider-token accounting incomplete
prompt length	model-visible prompt character or token count	not uniformly retained	no complete prompt-length column in retained token audit summary	prompt-token matching unavailable
completion length proxy	json_trace_chars, final_action_chars, first_impulse_chars	84,516 trace rows; 552 source/dataset/variant summaries	retained token-verbosity audit table	field-length/verbosity proxy only
trace structure proxy	reason_node_count, branch_count, memory_item_count	84,516 trace rows; 552 summaries	retained token-verbosity audit table	structure proxy for compute-risk assessment
schema field count	matched-interface emitted JSON schema	nine external fields in matched-interface protocol	matched-interface protocol schema	controls schema availability under the matched interface
retry count	provider retry attempts	not uniformly retained	unrecovered generations are reported separately by experiment	retry-cost inference unavailable
latency proxy	latency_ms	available where retained in token_verbosity_audit	wall-clock proxy affected by provider and network conditions	indirect model-compute/cost proxy
compressed structured-control sensitivity	AFCI_mean	A4_full=0.500; A4_compressed=0.499	matched-interface AFCI table	coupling persisted under compressed trace expression
A5_long scratchpad control	AFCI_mean	A5_long scratchpad=0.000	matched-interface AFCI table	compute matching not established

Table S50: Robustness and stochastic-control criterion summary. Prompt-paraphrase, expanded counterfactual and placebo rows are auxiliary robustness blocks. They are distinct from the formal counterfactual-transform quality-control screen and the consistency-passing transform reported in Supplementary Table S65.

Experiment	Status	Calls	Remote	Unrec.	Unrec. rate	Min ds	HS rec.	AFCI ds	Parse n	Parse r	Kappa	Leak
strong stochastic controls	pass	15490	15490	0	0	7	0	–	–	–	–	–
A5_empirical	predefined	17082	17082	0	0	7	0	–	–	–	–	–
entropy_attempt	closeness not satisfied											
prompt paraphrases	pass	11400	11400	0	0	17	0	–	–	–	–	–
expanded counterfactual probes	auxiliary	30584	30584	0	0	61	0	–	–	–	–	–
placebo components	pass	9960	9960	0	0	24	0	–	–	–	–	–
cross-model subset	pass	–	–	–	–	–	–	–	–	–	–	–
open-weight	pass	–	–	–	–	–	–	7	8	0.000762	–	–
reference model												
returned-score	pass	–	–	–	–	–	–	–	–	–	0.961	0
audit												

Table S51: Entropy diagnostics summarized by experiment source and variant.

Source	Variant	Raw H mean	Rule H mean	Semantic H mean	Family H mean	Raw H SD	Rule H SD	Semantic H SD	Family H SD	Aggregate rows
counterfactual probes	A4	4.180	2.647	3.962	0.863	1.703	1.490	1.625	0.461	70
counterfactual probes	A4_no reason	4.013	2.549	3.851	0.883	1.753	1.543	1.683	0.512	70
counterfactual probes	A4_no veto	4.179	2.591	3.969	0.913	1.734	1.508	1.709	0.490	70
counterfactual probes	A5	5.519	3.362	5.445	1.086	1.410	1.738	1.432	0.452	70
A5_emp.										
entropy	A4	6.596	5.106	6.360	1.070	1.292	1.541	1.301	0.411	21
A5_emp.										
entropy	A5	7.569	5.600	7.514	1.241	0.956	1.390	0.979	0.131	21
A5_emp.	A5_emp.									
entropy	entropy	7.584	5.477	7.529	1.236	0.924	1.432	0.951	0.134	21
placebo_components	A4	4.445	3.443	4.312	0.822	1.594	1.628	1.546	0.393	33
placebo_components	A4_no reason	4.156	3.232	4.027	0.879	1.622	1.472	1.596	0.500	33
placebo_components	A4_no veto	4.698	3.546	4.551	0.865	1.508	1.678	1.460	0.432	33
placebo_components	A5	5.364	4.048	5.330	1.091	1.442	1.677	1.450	0.378	33
prompt_paraphrases	A4	5.232	3.848	5.000	1.159	1.409	1.501	1.341	0.466	21
prompt_paraphrases	A4_no reason	4.895	3.419	4.720	1.103	1.503	1.495	1.458	0.514	21
prompt_paraphrases	A4_no veto	5.464	4.010	5.267	1.002	1.356	1.515	1.326	0.410	21
prompt_paraphrases	A5	6.302	4.653	6.183	1.204	0.995	1.340	1.000	0.251	21
strong stochastic controls										
strong stochastic controls	A4	7.009	5.462	6.777	1.090	1.849	1.757	1.815	0.415	7
strong stochastic controls										
strong stochastic controls	A5	7.998	5.914	7.943	1.204	1.377	1.545	1.401	0.228	7
strong stochastic controls	compute									
strong stochastic controls	control	7.994	5.915	7.936	1.238	1.422	1.569	1.452	0.151	7
strong stochastic controls	A5_entropy									
strong stochastic controls	attempt	7.941	5.896	7.896	1.161	1.465	1.754	1.463	0.211	7
strong stochastic controls	A5_self									
strong stochastic controls	consistency	7.994	5.926	7.928	1.173	1.382	1.640	1.414	0.176	7

Table S52: Predefined empirical entropy-attempt calibration criterion summary.

Criterion	Field	Value
strict_a5_em_calibration	calibration_events_per_dataset	50
strict_a5_em_calibration	closeness_threshold	0.15
strict_a5_em_calibration	grid_coverage_status	complete
strict_a5_em_calibration	max_selected_strict_entropy_gap	1.17146
strict_a5_em_calibration	note	maximum gap across the four entropy layers exceeds threshold
strict_a5_em_calibration	observed_required_setting_count	28
strict_a5_em_calibration	required_setting_count	28
strict_a5_em_calibration	selected_temperature	0.2
strict_a5_em_calibration	selected_top_p	0.75
strict_a5_em_calibration	selected_variant	A5_attempt_t020_p075
strict_a5_em_calibration	status	predefined closeness not satisfied
strict_a5_em_calibration	strict_entropy_close_enough	False

Table S53: Predefined empirical entropy-attempt calibration grid.

Variant	Temp.	Top p	Raw H gap	Rule H gap	Semantic H gap	Family H gap	Strict score	Max strict gap
A5_attempt								
t020_p060	0.2	0.6	0.944	1.300	1.096	0.302	0.910	1.300
A5_attempt								
t020_p075	0.2	0.75	0.973	1.171	1.110	0.353	0.902	1.171
A5_attempt								
t020_p090	0.2	0.9	1.102	1.420	1.224	0.404	1.037	1.420
A5_attempt								
t020_p100	0.2	1	0.955	1.331	1.094	0.365	0.936	1.331
A5_attempt								
t040_p060	0.4	0.6	1.125	1.545	1.271	0.388	1.082	1.545
A5_attempt								
t040_p075	0.4	0.75	1.154	1.462	1.295	0.249	1.040	1.462
A5_attempt								
t040_p090	0.4	0.9	1.070	1.328	1.217	0.362	0.994	1.328
A5_attempt								
t040_p100	0.4	1	1.186	1.334	1.314	0.317	1.038	1.334
A5_attempt								
t060_p060	0.6	0.6	1.129	1.518	1.264	0.296	1.052	1.518
A5_attempt								
t060_p075	0.6	0.75	1.123	1.373	1.260	0.403	1.040	1.373
A5_attempt								
t060_p090	0.6	0.9	1.114	1.492	1.253	0.504	1.091	1.492
A5_attempt								
t060_p100	0.6	1	1.155	1.520	1.273	0.303	1.063	1.520
A5_attempt								
t080_p060	0.8	0.6	1.167	1.539	1.295	0.371	1.093	1.539
A5_attempt								
t080_p075	0.8	0.75	1.043	1.344	1.178	0.322	0.972	1.344
A5_attempt								
t080_p090	0.8	0.9	1.068	1.414	1.207	0.324	1.003	1.414
A5_attempt								
t080_p100	0.8	1	1.058	1.704	1.205	0.307	1.069	1.704
A5_attempt								
t100_p060	1	0.6	0.904	1.305	1.026	0.418	0.913	1.305
A5_attempt								
t100_p075	1	0.75	0.973	1.476	1.106	0.469	1.006	1.476
A5_attempt								
t100_p090	1	0.9	0.960	1.403	1.099	0.404	0.966	1.403
A5_attempt								
t100_p100	1	1	1.017	1.438	1.164	0.394	1.003	1.438
A5_attempt								
t120_p060	1.2	0.6	1.045	1.579	1.184	0.435	1.061	1.579
A5_attempt								
t120_p075	1.2	0.75	1.043	1.136	1.188	0.577	0.986	1.188
A5_attempt								
t120_p090	1.2	0.9	0.984	1.389	1.128	0.395	0.974	1.389
A5_attempt								
t120_p100	1.2	1	1.029	1.298	1.170	0.469	0.991	1.298
A5_attempt								
t140_p060	1.4	0.6	1.041	1.357	1.179	0.525	1.025	1.357
A5_attempt								
t140_p075	1.4	0.75	1.028	1.419	1.157	0.397	1.000	1.419
A5_attempt								
t140_p090	1.4	0.9	1.051	1.357	1.198	0.437	1.011	1.357
A5_attempt								
t140_p100	1.4	1	0.966	1.424	1.119	0.439	0.987	1.424

Table S54: Finite-action-code entropy-calibration criterion summary.

Criterion	Field	Value
finite-action entropy calibration	structured-control behavioral composite > selected high-stochasticity datasets	7
finite-action entropy calibration	behavior pass threshold	6
finite-action entropy calibration	calibration events per dataset	14
finite-action entropy calibration	calibration maximum predefined entropy gap	1.22474
finite-action entropy calibration	calibration mean predefined entropy gap	1.13035
finite-action entropy calibration	calls	11088
finite-action entropy calibration	closeness threshold	0.15
finite-action entropy calibration	dataset count	7
finite-action entropy calibration	dry run	False
finite-action entropy calibration	experiment	finite-action entropy calibration
finite-action entropy calibration	unrecovered-generation rate	0
finite-action entropy calibration	grid setting count	14
finite-action entropy calibration	maximum model-generation budget	12000
finite-action entropy calibration	holdout events per dataset	18
finite-action entropy calibration	holdout maximum predefined entropy gap	1.34765
finite-action entropy calibration	holdout mean predefined entropy gap	1.14711
finite-action entropy calibration	holdout predefined entropy close enough	False
finite-action entropy calibration	holdout_unmapped action rate	0
finite-action entropy calibration	planned calibration calls	8820
finite-action entropy calibration	planned calls	11088
finite-action entropy calibration	planned holdout calls	2268
finite-action entropy calibration	remote calls	11088
finite-action entropy calibration	selected high-stochasticity candidate did not recover structured-control behavior	True
finite-action entropy calibration	selected behavior bootstrap lower endpoint of the two-sided 95% interval	0.863095
finite-action entropy calibration	selected temperature	0.2
finite-action entropy calibration	selected top-p	0.7
finite-action entropy calibration	selected variant	A5_attempt_t02_p07
finite-action entropy calibration	status	predefined closeness not satisfied
finite-action entropy calibration	unrecovered generations	0

Table S55: Finite-action-code entropy-calibration grid.

Variant	Sets	Rows	Mean gap	Max gap	Unmap	Raw H	Code H	Family H	Temp.	Top p
A5_attempt										
t01_p07	7	588	1.138	1.189	0	1.138	1.138	1.138	0.1	0.7
A5_attempt										
t01_p10	7	588	1.171	1.249	0.002	1.171	1.171	1.171	0.1	1
A5_attempt										
t02_p07	7	588	1.130	1.225	0.002	1.130	1.130	1.130	0.2	0.7
A5_attempt										
t02_p10	7	588	1.142	1.250	0	1.142	1.142	1.142	0.2	1
A5_attempt										
t04_p07	7	588	1.139	1.189	0	1.139	1.139	1.139	0.4	0.7
A5_attempt										
t04_p10	7	588	1.164	1.219	0	1.164	1.164	1.164	0.4	1
A5_attempt										
t06_p07	7	588	1.162	1.250	0.002	1.162	1.162	1.162	0.6	0.7
A5_attempt										
t06_p10	7	588	1.161	1.223	0	1.161	1.161	1.161	0.6	1
A5_attempt										
t08_p07	7	588	1.157	1.225	0.002	1.157	1.157	1.157	0.8	0.7
A5_attempt										
t08_p10	7	588	1.156	1.223	0	1.156	1.156	1.156	0.8	1
A5_attempt										
t10_p07	7	588	1.153	1.223	0	1.153	1.153	1.153	1	0.7
A5_attempt										
t10_p10	7	588	1.153	1.284	0.003	1.153	1.153	1.153	1	1
A5_attempt										
t12_p07	7	588	1.152	1.223	0	1.152	1.152	1.152	1.2	0.7
A5_attempt										
t12_p10	7	588	1.142	1.219	0.002	1.142	1.142	1.142	1.2	1

Table S56: Finite-action-code entropy-calibration holdout grid.

Variant	Sets	Rows	Mean gap	Max gap	Unmap	Raw H	Code H	Family H	Temp.	Top p
A5	7	756	1.165	1.299	0	1.165	1.165	1.165	–	–
A5_attempt										
t02_p07	7	756	1.147	1.348	0	1.147	1.147	1.147	0.2	0.7

S6. Model sensitivity and human agreement

Robustness analyses compare the tested instructions, provider identifiers and a local open-weight reference model. The annotation tables report agreement in returned individual scores, using Cohen’s κ as a chance-adjusted nominal-agreement statistic (Cohen, 1960). The sampling and scoring procedures are described in Supplementary Note 5.

Table S57: Cross-model subset criterion summary.

Model	Status	Exp.	Profile	Calls	Remote	Unrecov.	Unrec. rate	Vars.	Ctrls.	Min ctrl ds	Strong HS ds	Fallback	SC>HS ds
gpt-5.4 mini	pass	cross subset	full	1995	1995	0	0	5	4	5	0	qwen3 30b-a3b	7
gpt-4.1 mini	pass	cross subset	full	1995	1995	0	0	5	4	6	0	qwen3 30b-a3b	7
deepseek-chat	pass	cross subset	full	1995	1995	0	0	5	4	6	0	qwen3 30b-a3b	7
gemini-2.5 flash	pass	cross subset	full	1995	1995	2	0.001	5	4	7	0	qwen3 30b-a3b	7

Table S58: Open-weight Qwen2.5-14B-Instruct AFCI components by variant.

Variant	Statistic	Reason	Memory	Veto	Self	Support	AFCI	AFCI+	n
A4	Mean	1	0.5	1	1	1	0.875	0.9	21
A4	SD	0	0	0	0	0	0	0	21
A4_no reason	Mean	0.5	0.5	1	1	1	0.75	0.8	21
A4_no reason	SD	0	0	0	0	0	0	0	21
A4_no veto	Mean	1	0.5	0.5	1	1	0.75	0.8	21
A4_no veto	SD	0	0	0	0	0	0	0	21
A5	Mean	0.446	0.25	0.5	0.393	0	0.397	0.318	21
A5	SD	0.019	0	0	0.031	0	0.008	0.006	21
A5_schema posthoc_fields	Mean	0	0	0	0	0	0	0	21
A5_schema posthoc_fields	SD	0	0	0	0	0	0	0	21

Table S59: Agreement in returned annotations by dimension.

Dimension	Annotator A	Annotator B	Overlap items	Agreement	Cohen kappa
final action reason-consistent	annotator_01	annotator_02	1200	0.958333	0.912127
final action reason-consistent	annotator_01	annotator_03	1200	0.955833	0.907158
final action reason-consistent	annotator_02	annotator_03	1200	0.990833	0.980522
action respects relevant memory	annotator_01	annotator_02	1200	0.997500	0.989211
action respects relevant memory	annotator_01	annotator_03	1200	0.997500	0.989211
action respects relevant memory	annotator_02	annotator_03	1200	1.000000	1.000000
veto_was_appropriate	annotator_01	annotator_02	1200	1.000000	1.000000
veto_was_appropriate	annotator_01	annotator_03	1200	1.000000	1.000000
veto_was_appropriate	annotator_02	annotator_03	1200	1.000000	1.000000
veto changed first impulse	annotator_01	annotator_02	1200	1.000000	1.000000
veto changed first impulse	annotator_01	annotator_03	1200	1.000000	1.000000
veto changed first impulse	annotator_02	annotator_03	1200	1.000000	1.000000
reason_looks_posthoc	annotator_01	annotator_02	1200	0.960000	0.917801
reason_looks_posthoc	annotator_01	annotator_03	1200	0.961667	0.921303
reason_looks_posthoc	annotator_02	annotator_03	1200	0.995000	0.989652
verbose/template-like without control	annotator_01	annotator_02	1200	0.998333	0.971364
verbose/template-like without control	annotator_01	annotator_03	1200	0.996667	0.942728
verbose/template-like without control	annotator_02	annotator_03	1200	0.998333	0.972113
irrelevant cue caused inappropriate change	annotator_01	annotator_02	1200	0.974167	0.865842
irrelevant cue caused inappropriate change	annotator_01	annotator_03	1200	0.972500	0.860015
irrelevant cue caused inappropriate change	annotator_02	annotator_03	1200	0.993333	0.965260

Annotation methods: three graduate students in relevant university disciplines, including doctoral and master’s students, each rated the same 1,200 items using a common seven-dimension codebook. Supplementary Note 5 specifies sampling, metadata masking, randomized order, written instructions and majority aggregation. All 25,200 individual core ratings are valid binary values; 8,250 item–dimension combinations are unanimous and 150 have a 2:1 split, affecting 68 items. The table uses individual ratings before aggregation: mean pairwise Cohen’s kappa is 0.961, the minimum individual value is 0.860 and the minimum dimension mean is 0.897.

S7. Source records and reproducibility

Numbered manuscript-table locations inside the archival map refer to the original source release. The analyses in Supplementary Notes 1–15 and the current S-numbered tables provide the corresponding interpretation and results.

The reproducibility map links reported claims to tables, result exports, decision criteria, analysis code and checksums. Full SHA-256 values are retained in the CSV export; the table prints 12-character prefixes. Export paths identify the original source records. Figure S9 displays the open-weight controls, and Supplementary Note 1 reports the grouped reference-dependent localization analysis. “Legacy export ID” identifies an archived filename; current supplementary tables use the S prefix.

Table S60: Validation source-data and control-construction map.

Item	Role	Source data, config or script path	Checksum or provenance
Main validation panel a source data	Open-weight core no-context controls	manuscript/figures/source_data/figure5a_open_weight_core_controls.csv	12ef8d181136
Main validation panel b source data	Open-weight state-boundary lesion	manuscript/figures/source_data/figure5b_open_weight_state_boundary_lesion.csv	803140cd02ce
Main validation panel c source data	SWE-bench Lite full 300-record incremental AUC	manuscript/figures/source_data/figure5c_swebench_incremental_auc.csv	261436e42398
Main validation panel d source data	SWE-bench Lite single-predictor AUC comparison	manuscript/figures/source_data/figure5d_swebench_single_predictor_auc.csv	8897f8c61c79
Main validation panel e source data	SWE-bench Lite sensitivity and binding-guard summary	manuscript/figures/source_data/figure5e_swebench_sensitivity_guard.csv	4dcd90ac380e
Main validation rendering and export QA	Figure rendering and export-QA script	manuscript/figures/scripts/build_validation_figure5.R; audit/figure5_primary_validation_export_qc_20260530.csv	61d69f9d8264 / 80f8d170bb37
SWE-bench reliability calibration source data	Issue-grouped five-fold reliability calibration for the full 300-record issue-to-file endpoint	manuscript/figures/source_data/extended_data_figure9_swebench_reliability_calibration.csv; manuscript/figures/source_data/swebench_cv_reliability_predictions.csv	archived manifest
SWE-bench threshold analysis source data	Thresholded localization-guard analysis over predicted hit@3 probabilities	manuscript/figures/source_data/extended_data_figure10_swebench_threshold_analysis.csv; manuscript/figures/scripts/build_swebench_reliability_figures.R	archived manifest
Submitted main Table 1 source data	Claim ledger with supported claims and excluded inferences	manuscript/tables/table1_validation_evidence_interpretation.csv	archived manifest
Supplementary Table 33 source data	Validation controls and interpretation status	manuscript/tables/supplementary_table_33_validation_controls.csv	7a88ca507fb7
Decisive-field positive-control summaries	Two-model decisive-field and wrapper result tables	manuscript/tables/table8_decisive_field_positive_control_condition_summary.csv; manuscript/tables/table8_control_binding_wrapper_scrambled_summary.csv	archived manifest
Decisive-field positive-control leakage audit	Prompt/scorer leakage and scope checks	manuscript/tables/table8_decisive_field_positive_control_leakage_audit.csv	e1fdfc5005ce
Local strict matching summaries	Local Qwen2.5-1.5B forced-budget entropy/token audit	manuscript/tables/table16_local_strict_matching_condition_summary.csv; manuscript/tables/table16_local_strict_matching_gate_summary.csv	archived manifest

Continued on next page

Table S60: Validation source-data and control-construction map. (continued)

Item	Role	Source data, config or script path	Checksum or provenance
SWE-bench Lite issue-to-file summaries	Complete 300-record issue-localization summaries and execution-slice preparation status	results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/gate_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_environment_status.json; results/paper1_revision/real_task_predictive_validity/swebench_v2_full300_api_20260529_merged/official_execution_slice_analysis_v3_apply_repair/official_execution_slice_gate_status.json	archived manifest
Open-weight matrix config path	Model, scaffold, task-family and variant manifest	results/paper1_revision/remote_5090_run3_20260508/nmi_formal_execution_manifest/execution_manifest.json	11e3390109fd
Formal API validation config path	API validation run manifest	results/paper1_revision/remote_5090_run3_20260508/formal_api_validation/formal_api_full_20260510_0030/run_manifest.json	8712dbea8b1b
Distribution-matched control construction	Offline/no-context action-prior policy and prompt construction	src/freewill/structured_control_validation.py	eba9d1039034
Strict-lesion construction	Removal of decisive intervention-carrying field with non-decisive context retained	src/freewill/structured_control_validation.py	eba9d1039034
Counterfactual transform QC and consistency-passing transform	Transform definitions and formal holdout launcher	src/freewill/structured_control_validation.py; scripts/run/run_formal_open_weight_validation.py	eba9d1039034 / 0f9cfd41e1af
Token/compute criterion script	Prompt-token, total-token and latency-proxy criterion analysis	scripts/run/analyze_formal_validation_results.py; scripts/run/audit_open_weight_token_entropy.py	404f5edb4132 / f037f6424513
Entropy-prior behavioral-control script	Calibration-split action-prior selection and entropy-prior no-field behavioral control	scripts/run/select_formal_entropy_settings.py; src/freewill/structured_control_validation.py	f29b12ebe627 / eba9d1039034
Local strict matching script	Local tokenizer padding, fixed-generation and forced-budget audit implementation	scripts/run/run_strict_matching_local.py; src/freewill/strict_matching_local.py	archived manifest
SWE-bench Lite localization scripts	Dataset fetch, oracle-free issue-to-file prompts, shard merge, readiness audit, execution-slice patch generation and v3 apply-check/repair filtering	scripts/run/run_swebench_file_localization_v2_api.py; scripts/run/merge_swebench_fileloc_v2_shards.py; scripts/run/run_swebench_execution_slice_patch_generation_api.py; scripts/run/run_swebench_patch_generation_v3_apply_repair.py; src/freewill/real_task_predictive_validity.py	archived manifest

Table S61: Source evidence and export map. Export IDs identify archived source files.

ID	Evidence	Original export	Analysis	Checksum prefix
H01	baseline lesion matrix separates unpredictability and trace structure	Export 2	Baseline lesion matrix	5f2fae5d2ab5
H02	matched-interface AFCI rules out field availability alone	Export 3	Matched-interface controls	2fdaffdbfdb
H03	finite-action-code behavior supports action-level component coupling	Export 4	Finite-action behavior	55820262e805
H04	strong stochastic controls do not recover structured-control profile	Export 9	Stochastic controls	1f1de7d4d18d

Table S61: Source evidence and export map. Export IDs identify archived source files. (continued)

ID	Evidence	Original export	Analysis	Checksum prefix
H05	predefined entropy closeness was not satisfied	Export 6 / Export 12	Entropy diagnostics and calibration	b6150257
H06	cross-model tested subset preserves structured-control greater-than high-stochasticity direction	Export 9	Cross-model subset	c3ae
H07	open-weight reference model preserves structured-control greater-than high-stochasticity AFCI direction	Export 9 / Export 10	Open-weight anchor	2f8b5ff04901
H08	decisive-field positive control recovers the finite-action binding readout	Export 8	Decisive-field control	d76affe4ffe8
H09	local wrapper reduces scrambled-field binding	Export 8	Decisive-field control	archived manifest
H10	bounded local forced-budget entropy/token audit preserves structured-state advantage	Export 16	Strict-matching local audit	archived manifest
H11	SWE-bench Lite historical association; the augmented features depend on gold	Export 17	SWE localization v2	0206bfa8d6db
H12	Individual ratings and majority labels document inter-annotator agreement	Export 9	Blind-rating export	archived manifest
				bffe01d6d1fd

The complete source identifiers and analysis mappings are provided in `supplement_tables/historical_claim_artifact_map.csv`, supplied with the editable source and research package.

02

Interface-diagnostic rows are summarized separately from effect estimates. The taxonomy distinguishes final-criterion-compatible interface diagnostics, non-final provider-interface attempts and entropy-calibration diagnostics. The export-coverage table lists the table exports underlying the manuscript and this supplement.

Table S62: Interface-diagnostic taxonomy and interpretation.

Source	Experiment	Model	Diagnostic type	Rows	Interpretation
Cross-model robustness	Gemini full run	gemini-2.5-flash	payload not object	2	Final criterion met; retained as interface diagnostic.
Provider-interface compatibility check	Qwen direct-provider check	qwen3-30b-a3b	HTTP 400	100	Compatibility rows outside the final-evidence criteria.
Entropy-calibration diagnostic	entropy-attempt calibration	qwen3-30b-a3b fallback	HTTP 500 / fallback metadata	594	Diagnostic calibration rows; predefined closeness was not satisfied.
Baseline lesion matrix	aggregate baseline accounting	gpt-5.4-mini	aggregate only	16	Aggregate unrecovered-output count; no trace-level taxonomy.

Table S63: Table-export coverage. Full-precision values are retained in the CSV exports.
Data rows exclude the header row and any empty trailing lines.

Export label	Data rows	Columns	Leading columns
Variant name registry	17	5	canonical variant, used in, definition, aliases, role
Mechanistic claim map	3	5	claim component, operational test, contrary result, interpretive boundary
Baseline lesion-matrix criterion summary	12	3	criterion, field, value
Baseline trace metrics by dataset, variant and replicate index	168	19	variant, VBI, SCI, VEI, RSI, ACI, CSVI, unpredictability ...
Baseline trace metrics by variant	8	18	variant, VBI mean, SCI mean, VEI mean, RSI mean, ACI mean, CSVI mean, unpredictability mean ...
Matched-interface AFCI by dataset, variant and replicate index	189	10	dataset namespace, variant, replicate index, reason alignment, memory alignment, veto alignment ...
Matched-interface AFCI by variant	9	16	variant, alignment means, AFCI mean, AFCI with suppression mean, SD fields ...
Matched-interface dataset contrasts	56	5	dataset namespace, variant, AFCI, structured full AFCI, paired contrast
Matched-interface comparator criterion status	7	4	control variant, dataset count, structured greater-than-control dataset count, recovery count
Matched-interface criterion summary	11	3	criterion, field, value
Finite-action condition rates	504	7	dataset namespace, variant, replicate index, condition, correct rate, unmapped rate, row count
Finite-action dataset criterion contrasts	7	10	dataset namespace, structured variant, ablation comparators, high-stochasticity comparator, component contrasts ...
Finite-action criterion summary	16	3	criterion, field, value
Finite-action metrics by dataset, variant and replicate index	84	16	dataset namespace, variant, replicate index, condition rates and component scores ...
Finite-action summary by dataset and variant	28	9	dataset namespace, variant, B scores, cue specificity, behavioral structured action control ...
Finite-action summary by variant	4	16	variant, component-score means, behavioral composite mean, false-positive and row-count summaries ...
Entropy diagnostics by source and variant	20	11	source, variant, raw, canonical, semantic and action-family entropy means and SDs ...
Full entropy diagnostic units	594	14	source, output namespace, variant, replicate index, row count, unique action counts ...
Strict entropy-attempt calibration grid	28	12	variant, temperature, top-p, paired counts, absolute entropy gaps ...
Strict entropy-attempt criterion summary	13	3	criterion, field, value
Finite-action entropy-calibration grid	14	11	variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...
Finite-action entropy-calibration criterion summary	30	3	criterion, field, value
Finite-action entropy-calibration holdout grid	2	11	variant, dataset count, row count, predefined entropy gap, unmapped action rate, entropy gaps ...
Blinded-audit reliability by dimension	7	6	dimension, annotator labels, overlap items, agreement, Cohen's kappa
Cross-model criterion summary	4	14	model, status, experiment profile, calls, interface diagnostics, directional dataset counts ...
Open-weight AFCI by dataset, variant and replicate index	105	10	dataset namespace, variant, replicate index, alignment terms and AFCI

Continued on next page

Table S63: Table-export coverage. Full-precision values are retained in the CSV exports.
Data rows exclude the header row and any empty trailing lines. (continued)

Export label	Data rows	Columns	Leading columns
Open-weight AFCI by variant	5	16	variant, alignment means, AFCI means, SD fields and rows
Robustness criterion summary	8	13	experiment, status, calls, interface diagnostics, directional counts, parse and audit statistics
Decisive-field positive-control condition summary	12	13	model, condition, rows, accuracy, wrapped accuracy, wrapper change rate ...
Decisive-field positive-control gate summary	2	10	model, full-state accuracy, only-decisive accuracy, best control accuracy, recovery fraction ...
Decisive-field positive-control task cells	240	4	model, task family, full-state accuracy, only-decisive accuracy
Decisive-field positive-control leakage audit	5	4	audit item, evidence, status, interpretation
Control-binding wrapper scrambled summary	2	7	model, rows, raw scrambled following, wrapped scrambled following, raw expected accuracy ...
Local strict matching condition summary	2	10	condition, rows, accuracy, invalid rate, canonical-action entropy, token and latency means
Local strict matching gate summary	1	12	status, prompt match rate, completion match rate, total-token ratio, latency ratio ...
SWE-bench file-localization predictor summary	11	5	predictor, hit@3 AUC, hit@3 Pearson, no-violation AUC, sign convention
SWE-bench file-localization model summary	6	13	model, model-instance rows, localization hit@3, violation rates, binding score and execution-slice boundary columns
SWE-bench file-localization gate summary	1	21	status, rows, model-instance rows, hit@3 counts, AUC summaries, wrapper status and execution-slice boundary columns
Derived effect-size summary	6	3	evidence block, direction, stronger summary
Proxy token/compute audit summary	9	5	audit item, measurement or proxy, availability, evidence source, interpretation
Entropy calibration threshold summary	44	12	source, variant, entropy gaps, strict threshold, pass flag
Interface-diagnostic taxonomy	4	6	source, experiment, model, diagnostic type, rows, interpretation
Claim-to-evidence reproducibility map	12	7	claim, legacy export ID, result export, criterion summary, analysis entry point, SHA-256
Statistical sensitivity summary	8	10	contrast, independent unit, direction count, minimum contrast, sign test

S8. Statistical and scoring definitions

Table S64: Dataset-level statistical sensitivity summary for primary contrasts.
Exact sign tests are one-sided under a 0.5 null probability for a positive dataset direction; two-sided values are doubled and capped at one.

Contrast	Unit	Units	Positive units	Minimum contrast	Mean contrast	LODO minimum mean	Top-effect removed mean	One-sided sign p
finite behavior: structured-control-high-stochasticity composite	dataset namespace	7	7	0.832	0.886	0.875	0.875	0.0078125
reason ablation: structured-control-reason lesion B_RSI	dataset namespace	7	7	0.964	1.017	0.982	0.982	0.0078125
veto ablation: structured-control-veto lesion B_VEI	dataset namespace	7	7	0.992	0.997	0.996	0.996	0.0078125
memory behavior: structured-control-high-stochasticity B_MCI	dataset namespace	7	7	0.556	0.774	0.751	0.751	0.0078125
self-continuity behavior: structured-control-high-stochasticity B_SCI	dataset namespace	7	7	0.608	0.811	0.795	0.795	0.0078125
schema AFCI: structured full-stochastic plain	dataset namespace	7	7	0.262	0.302	0.297	0.297	0.0078125
schema AFCI: structured full-weakest control comparator	dataset namespace	7	7	0.492	0.500	0.499	0.499	0.0078125
open-weight AFCI: structured-control-high-stochasticity	dataset namespace	7	7	0.467	0.478	0.476	0.476	0.0078125

Let $\mathcal{D}$ denote the seven source corpora and $\mathcal{I}_{d,v}$ the retained generations for dataset d and variant v . Raw final-action unpredictability is reported as Shannon entropy (Shannon, 1948),

$$H_{d,v} = - \sum_{a \in \mathcal{A}_{d,v}} \hat{p}_{d,v}(a) \log_2 \hat{p}_{d,v}(a), \quad (2)$$

where $\mathcal{A}_{d,v}$ is the observed final-action support and $\hat{p}_{d,v}$ is the empirical action frequency. The baseline structured composite is

$$C_{d,v} = \frac{1}{4} (\text{SCI}_{d,v} + \text{RSI}_{d,v} + \text{VEI}_{d,v} + \text{ACI}_{d,v}). \quad (3)$$

For matched-interface controls, each scored generation has

$$\text{AFCI}_{i,v} = \frac{1}{4} (R_{i,v} + M_{i,v} + Q_{i,v} + S_{i,v}), \quad (4)$$

where R , M , Q and S denote reason-action, memory-action, veto-action and self-action alignment terms. The implementation uses emitted trace fields, first impulse, final action and predefined field-source metadata to evaluate these terms. Thus AFCI is an interface-level action-field coupling analysis.

For the finite-action-code benchmark, the canonicalizer maps the raw final action o_i to $\hat{y}_i = g(o_i) \in \mathcal{Y}$, where $\mathcal{Y} = \{\text{ACTION_A}, \text{ACTION_B}, \text{VETO}, \text{DEFER}, \text{RECALL_PRIOR}, \text{INVALID_OR_UNMAPPED}\}$. If y_i^- is the scorer-side expected action before the manipulation and y_i^+ is the scorer-side expected action after the manipulation, then

$$c_i = \mathbf{1}\{\hat{y}_i = y_i^+\}, \quad (5)$$

$$u_i = \mathbf{1}\{z_i = \text{irrelevant}\} \mathbf{1}\{\hat{y}_i \neq y_i^-\}. \quad (6)$$

The component score for component k is the target-minus-irrelevant contrast

$$B_k(v) = \widehat{\text{Pr}}(c_i = 1 \mid z_i \in \mathcal{T}_k, v) - \widehat{\text{Pr}}(u_i = 1 \mid z_i \in \mathcal{N}_k, v), \quad (7)$$

where $\mathcal{T}_k$ is the target-probe set and $\mathcal{N}_k$ is the irrelevant or placebo-probe set. The implementation also records cue specificity as the mean of the four target response rates minus the irrelevant-cue change rate. This equals the mean of the four component scores when they are all present. The reported finite-action composite averages those four scores and cue specificity, and therefore gives the same value as the four-component mean for complete records. Cue specificity is consequently a summary of the same responses, rather than an independent fifth source of evidence.

Dataset-cluster bootstrap intervals resample dataset identifiers with replacement, following the non-parametric bootstrap principle of estimating uncertainty from resampled empirical units (Efron and Tibshirani, 1993). For a metric m , comparator q , and dataset d , the paired contrast is

$$\delta_d(q; m) = \bar{m}_{d, \text{A4}} - \bar{m}_{d, q}, \quad (8)$$

with A4 replaced by A4_full for matched-interface AFCI. The reported lower endpoint of a two-sided 95% interval is the empirical 2.5th percentile of the bootstrap distribution over $|\mathcal{D}|$ -dataset resamples.

For entropy calibration, layer-specific entropy $H_{d,v}^{(\ell)}$ is computed for raw, rule-canonical, semantic-cluster and action-family representations. If $\theta = (T, p)$ denotes a candidate high-stochasticity temperature/top-p setting, the predefined calibration gap is

$$\Gamma(\theta) = \max_{\ell} \left| \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A5}(\theta)}^{(\ell)} - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} H_{d, \text{A4}}^{(\ell)} \right|. \quad (9)$$

The predefined matching criterion is $\Gamma(\theta) \leq 0.15$; calibration and holdout gaps are reported separately in Section S5.

S9. Open-weight models and additional controls

This section documents the controlled finite-action and software experiments. It covers the open-weight control matrix, strict field deletion, decisive-field positive control, local consistency guard, control-comparison criteria, forced-budget entropy and token comparisons, entropy-prior control, SWE-bench Lite localization, counterfactual transformations, API configurations and expanded human ratings. These studies complement the state-writing and execution analyses by specifying performance under their specified information and scoring conditions.

Table S65: Validation controls and interpretation status.

Evidence block	Scope	Main result	Interpretation status
Open-weight finite-action matrix	Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct and Mistral-7B-Instruct-v0.3; 126,000 scored rows per model (108,000 generation and 18,000 offline-policy records); three agent scaffolds; 20 task families; seven variants; three seeds	Structured accuracy exceeded distribution-matched offline/no-context controls in 236/240 model-scaffold-task cells, scrambled-context controls in 240/240 cells and no-fields controls in 238/240 cells; mean deltas were 0.593, 0.895 and 0.550, respectively	Supports open-weight size and family validation of the no-context control result
Independent Mistral family	Mistral-7B-Instruct-v0.3 full matrix	Core no-context controls satisfied the predefined criteria in 60/60 cells for no-fields, scrambled-context and distribution-matched controls	Supports non-Qwen model-family generalization
Strict Qwen32B target lesion	Qwen2.5-32B strict lesion slice; 28,800 scored generations	Structured accuracy exceeded strict target lesion in 60/60 cells; structured mean 0.998, strict-lesion mean 0.037, mean delta 0.961	Sensitivity to removal of the decisive task field
Decisive-field positive control	gpt-5.4-mini and qwen3-max ; 20 task families; six events per family; 1,440 fallback-free API generations	Full-state and only-decisive-field accuracy were both 1.000 in both models, while the best non-decisive control was 0.208; the only-decisive field is intentionally visible and action-code-bearing	Supports a diagnostic decisive-field readout within the finite-action protocol
Local control-binding wrapper	Same 1,440-row positive-control experiment; no second model call	Irrelevant-cue accuracy remained 1.000; raw scrambled-field following was 0.958 for gpt-5.4-mini and 0.533 for qwen3-max , reduced to 0.033 in both models after wrapping	Deterministic constraint enforcement within the finite-action interface
Formal Qwen control-audit slice	63,360 planned and observed rows; no duplicate analysis keys; parse/error rate 0.140%	Structured accuracy exceeded no-fields, scrambled-context, distribution-matched, strict-lesion and formal control variants in the relevant cells	Supports formal-control behavior in the Qwen slice
Token-prompt audit slice	17,280 planned and observed rows; no duplicate analysis keys; parse/error rate 0.069%	Prompt-token counts were aligned for the entropy-token variant, but mean total-token ratios and latency proxies remained outside the strict token/compute criterion	Prompt-token alignment with residual differences in total tokens and latency
Strict entropy criterion	Formal entropy calibration and holdout audit	Maximum entropy gaps exceed the registered threshold across model-scaffold cells	Distributional calibration under the registered criterion
Local forced-budget entropy/token audit	Qwen2.5-1.5B-Instruct CPU local inference; 20 task families; two events per family; 80 generations	Prompt and completion tokens matched exactly by event; total-token ratio 1.000; latency ratio 0.997; canonical-action entropy gap 0.020 bits; structured accuracy was 0.900 versus 0.325 for the action-prior no-state control	Exact prompt and completion matching in the local configuration
Entropy-prior no-field behavioral control	Qwen2.5-14B and Qwen2.5-32B calibration and holdout rows; 24,480 planned and observed rows; no duplicate analysis keys; parse/error rate 0.102%	The entropy-prior no-field behavioral control remained below the structured-control variant in 120/120 cells, with mean delta 0.660; the strict entropy-gap criterion was satisfied in only 1/6 model-scaffold cells	Behavioral comparison under a final-action prior; the strict entropy criterion holds in 1/6 cells

Evidence block	Scope	Main result	Interpretation status
SWE-bench Lite issue-to-file association	<code>princeton-nlp/SWE-bench_Lite</code> test split; 300 issues and six API models; 18,000 condition/repeat rows aggregate to 1,800 model–issue records	Under issue-grouped validation with training-only preprocessing, AUC is 0.737216 for the reference-free baseline and 0.924486 with reference-dependent diagnostic features; paired difference 0.187269, 95% full-refit interval [0.126941, 0.267774]	Reference-dependent association; estimator and direct held-group evaluation are defined in Supplementary Note 1 (p. 18)
SWE-bench Lite reliability calibration	Same 1,800 model–issue records; issue-grouped logistic probabilities from the reference-free baseline and reference-dependent augmentation	Reliability curves and threshold summaries characterize predicted localization, hard-constraint violations and constraint-clean localization	Retrospective calibration under reference-dependent features; reference-free evaluation is reported separately
Counterfactual transform QC	Deterministic counterfactual-flip transform-quality screen	Text–label consistency screening retained 57/240 candidate counterfactual rows as transform-quality boundary rows	Robustness analysis conditions on text–label consistency
Consistency-passing counterfactual transform	Frozen transform on Qwen2.5-14B and Qwen2.5-32B with jointly re-generated decisive context and expected action	The consistency-passing transform satisfied the predefined criterion in 120/120 scaffold–family cells; mean structured accuracy was 1.000, compared with 0.992 in the original structured rows	Robustness under the specified deterministic transformation
Expanded human ratings	1,200 shared items, three graduate-student raters	Core scores are complete; 150 of 8,400 item–dimension combinations have a 2:1 split across 68 items; mean pairwise Cohen’s kappa is 0.961, minimum individual value 0.860 and minimum dimension mean 0.897	Stratified sampling, masked items, individual scores and majority aggregation are described in Supplementary Note 5 (p. 20)
Formal API external validation	Provider-reported run-time identifiers <code>gpt-5.4-mini</code> , <code>gemini-3.1-flash-lite-preview</code> , <code>deepseek-v3.2</code> and <code>qwen3-max</code> ; 14,400 planned rows; 11,520 fallback-free remote API calls	All rows were present, duplicate analysis keys were zero, fallback calls were zero, unrecovered generations were zero and one parse/action error record remained. Structured accuracy exceeded no-fields, scrambled-context, distribution-matched and strict-lesion controls in 238/240, 240/240, 240/240 and 240/240 model–scaffold–task–family cells; all model-, scaffold- and task-family-cluster bootstrap lower endpoints of the two-sided 95% intervals were positive	Historical robustness of the explicit prompt/interface controls; not validation of latent state or interactive architectures

The decisive-field positive-control experiment is a controlled finite-action test. The only-decisive condition exposes the current event’s minimal action-relevant field and asks whether the model follows that field rather than surface context, a calibrated action prior or a decisive field from another event. The examples below specify the visible field and its associated action rule; the expected action column remains scorer-side metadata used for evaluation, whereas the visible field is the intervention-carrying state supplied to the action query.

Probe component	Example visible only-decisive field	Scorer-side target role
Reason	Decisive reason field for the event: calibrated task evidence selects <code>ACT_B</code> .	Target checks whether final action follows the current reason field.
Memory	Decisive memory field for the event: the corrected prior commitment is <code>ACT_A</code> ; stale or unrelated memory is non-decisive.	Target checks whether final action follows corrected memory rather than injected memory.
Veto/constraint	Decisive constraint/veto field for the event: a hard constraint invalidates ordinary actions and requires <code>ACT_VETO</code> .	Target checks whether final action respects the current constraint field.

The decisive-field experiment distinguishes hidden scorer metadata from the intentionally visible action-relevant field. Its field-only condition supplies the current event’s minimum decisive information.

The leakage check therefore tests whether the hidden target variable or key entered the generated prompt, while the non-decisive conditions test removal or mismatch of the current field.

Audit item	Evidence	Status	Interpretation
<code>expected_action</code> column not a prompt key	Prompt construction serializes condition, event id, task family, candidate actions and context; <code>expected_action</code> is retained in traces for scorer-side evaluation after generation.	pass	The scorer target variable is not exposed as a prompt field or JSON key.
Scorer-side target used post-generation	Across 1,440 traces, <code>expected_action_match</code> is computed after generation from canonical <code>final_action</code> versus <code>expected_action</code> .	pass	The target column is a scorer-side annotation rather than a model-output field.
Non-decisive controls do not expose current decisive state	<code>surface_context_only</code> and <code>action_prior_only</code> had <code>field_event_id=none</code> in 240/240 rows each; <code>scrambled_decisive_field</code> used a non-current field id in 240/240 rows.	pass	Surface, prior and scrambled controls separate action priors or wrong-state fields from the current decisive field.
Only-decisive action-token status	<code>only_decisive_causal_field</code> exposed a visible decisive action matching the scorer target in 240/240 rows; full-state and irrelevant-cue rows did the same by design.	scope	The positive-control claim tests following the current minimal decisive field; it is not an answer-token-free hidden-reasoning claim.
Label wording scope	The prompt contains a generic instruction not to infer hidden expected-action labels, but does not serialize the <code>expected_action</code> column or scorer target key.	pass	The generic instruction names the evaluation concept; it does not provide the hidden scorer variable.

Control criteria and repair outcomes. No-context and controlled comparisons show that finite-action performance depended on structured information under the tested rules. Entropy and total-compute matching remain subject to their observed calibration results. The separate local Qwen2.5-1.5B experiment satisfied its prompt-padding and fixed-generation-length conditions. SWE-bench Lite localization covers 300 issues; the exploratory repair evaluation contains 50 selected issues and two models. Of those 100 model-issue submissions, one resolved its issue in the official harness evaluation, and the association criterion was inconclusive. A local application and repair filter retained eight patches, all of which completed the harness with none resolving the issue. The counterfactual-transform records separately identify the screened transformations and the consistency-passing deterministic transformation used in the robustness comparison.

References

- Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. doi: 10.1177/001316446002000104.
- DeepSeek. Introducing deepseek-v4.1-flash: smarter, faster, more efficient, September 2026. URL <https://www.deepseek.com/en/news/deepseek-v4-1-flash/>. Published 10 September 2026; accessed 25 September 2026. Includes the official V4-Pro API alias migration notice.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3): 379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.